

Supplementary appendix

Joint treatment and source-discrepancy modeling with locally robust commensurate Bayesian additive regression trees

This appendix provides the joint posterior calculations, control-mean information derivation, fixed-region theoretical results, complete joint-model study, and the separate-treatment survival, simulation and application analyses. Main-text equation and theorem references are identified explicitly. The prior ESS reference and the implemented prior-scale selection procedure are described separately.

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A Posterior computation

A.1 Joint priors, tree support and conditioning

The primary likelihood includes every trial and external observation, with S_i and A_i as defined in the main text. The outcome centering constant is the sample mean across all three groups and is restored for predictions. The joint range rule uses their outcome range R_y . The tree priors are independent across trees, and the leaf priors are independent given tree structures and shared hyperparameters.

For source $s \in \{1, 2\}$, use

$$\sigma_s^2 \sim \text{IG}(v_\sigma/2, v_\sigma \lambda_{\sigma,s}/2), \quad v_\sigma = 3,$$

where $p(u) \propto u^{-a-1} \exp(-b/u)$. The scale is $\lambda_{\sigma,s} = \widehat{\sigma}_s^2 \chi_{3,0.10}^2/3$. The joint Gaussian wrapper estimates $\widehat{\sigma}_s$ by regressing centered outcomes on covariates within source; its trial fit includes both randomized arms. This empirical-Bayes preliminary fit omits the treatment indicator. When the treatment effect is nonzero, its residuals therefore contain treatment variation as well as residual outcome variation, and the resulting scale depends on the effect and allocation. We retain this choice in the reported experiment; no treatment-adjusted prior-scale sensitivity has been fitted. The resulting scales, centering and cutpoint grid are held fixed in posterior sampling. The forest scales are $\lambda_f^2 = R_y^2/(16H_f)$ and $\tau_1^2 = R_y^2/(16H_g)$.

The joint implementation assigns the following normalized weight over admissible trees. If $I(T)$ and $L(T)$ are internal and terminal nodes, $d(v)$ is node depth, $V(v)$ counts ancestor-admissible variables, and $C(v, s_v)$ counts available cuts for the selected variable, then

$$\pi_k(T) \propto \prod_{v \in I(T)} \frac{p_k\{d(v)\}}{V(v)C(v, s_v)} \prod_{\ell \in L(T)} [1 - p_k\{d(\ell)\}] \mathbf{1}\{T \text{ is admissible}\}. \quad (\text{A.1})$$

Admissibility requires ancestor-consistent cuts and at least five fitting observations per leaf: all sources for f , and all trial participants for g . Normalization is over the resulting finite tree support. Even a terminal node with no remaining cut retains its factor $1 - p_k(d)$. Its normalized marginal tree law is not the usual recursively normalized Galton–Watson law. This globally restricted weight differs from a recursive prior that assigns forced termination probability one at such nodes. The calibration tree simulator uses the latter recursion, another reason its ESS is a prior-setting reference rather than an exact characterization of the fitted tree prior.

Table S1: Primary joint LRC-BART specification.

Quantity	Specification
Mean	$f(x) + Sg(x) + A\psi$
Residual variances	One shared trial variance; one external variance
Forests	$H_f = 50, H_g = 5$; independent tree structures
Depth factors	$(\alpha_f, \beta_f) = (0.95, 2)$; $(\alpha_g, \beta_g) = (0.5, 3)$
Treatment prior	$\psi \sim \text{N}(0, 100)$, with variance 100
Mixture	$w \sim \text{Beta}(1, 1)$, updated; $v_0 = 3$ spike prior truncated below τ_1^2
Tree moves	Grow/prune/change probabilities 0.3/0.3/0.4; a stump proposes grow
Posterior sampling	Three chains; 30,000 warm-up and 12,000 retained iterations; five g sweeps per iteration
Scale calibration	Control outcomes only; requested control-mean count 100; 20 blocks; 141-point scale grid

A.2 Partial residuals

For a prognostic tree h and discrepancy tree h , respectively, define

$$\begin{aligned} r_i^f &= Y_i - f^{(-h)}(x_i) - S_i g(x_i) - A_i \psi, \quad \text{all observations,} \\ r_i^g &= Y_i - f(x_i) - g^{(-h)}(x_i) - A_i \psi, \quad S_i = 1. \end{aligned} \quad (\text{A.2})$$

The current tree is omitted from the indicated sum. For an f update both trial arms have precision $1/\sigma_1^2$ and external observations have precision $1/\sigma_2^2$. All trial participants contribute to a g update. These partial responses are held fixed during the structure proposal. In the separate-treatment variant, only the two control groups enter these calculations, with $A_i = 0$.

A.3 Marginal leaf likelihoods

Within a specified leaf, write r_i for the relevant partial residuals, suppressing their source labels only in the generic normal integral. For an f leaf these include all r_i^f , with their respective residual variances; for a g leaf they include r_i^g for $S_i = 1$.

Lemma A.1 (Normal leaf integral). *Suppose $r_i \mid \theta \sim N(\theta, \sigma_i^2)$ independently in a leaf and $\theta \sim N(0, q^2)$, with $q^2 > 0$. Let $A := \sum_i \sigma_i^{-2}$ and $B := \sum_i r_i \sigma_i^{-2}$. The marginal leaf density is*

$$p(\mathbf{r}_\ell \mid \mathcal{T}_{h^k}^k, \text{rest}) = \left[\prod_{i \in \ell} \phi(r_i; 0, \sigma_i^2) \right] (1 + q^2 A)^{-1/2} \exp \left\{ \frac{q^2 B^2}{2(1 + q^2 A)} \right\}. \quad (\text{A.3})$$

Here $k \in \{f, g\}$, $\mathcal{T}_{h^k}^k$ denotes the current component-specific tree, and θ denotes its leaf parameter. The last argument of ϕ is variance; the result also holds for an empty leaf, using $A = B = 0$ and an empty product of one.

Proof. Factor the observation-level product from the normal likelihood and integrate the remaining prior-weighted kernel:

$$\frac{1}{\sqrt{2\pi q^2}} \int \exp \left\{ -\frac{1}{2}(A + q^{-2})\theta^2 + B\theta \right\} d\theta. \quad (\text{A.4})$$

Completing the square gives mean $B/(A + q^{-2})$ and precision $A + q^{-2}$. Its normalizing integral is $\{q^2(A + q^{-2})\}^{-1/2} \exp\{B^2/[2(A + q^{-2})]\}$, which is the stated factor. $\square$

For an f leaf, let $n_{s,\ell}, S_{s,\ell}$ be the count and sum of its partial residuals from source s . In the joint model, source-1 counts include both randomized arms, unlike the main-text sample size n_1 that counts concurrent controls only. Write A and B for the leaf precision and precision-weighted residual sum. The marginal likelihood factor is

$$\begin{aligned} A &:= \frac{n_{1,\ell}}{\sigma_1^2} + \frac{n_{2,\ell}}{\sigma_2^2}, & B &:= \frac{S_{1,\ell}}{\sigma_1^2} + \frac{S_{2,\ell}}{\sigma_2^2}, \\ M_f &= (1 + \lambda_f^2 A)^{-1/2} \exp \left\{ \frac{\lambda_f^2 B^2}{2(1 + \lambda_f^2 A)} \right\}. \end{aligned} \quad (\text{A.5})$$

For a g leaf, redefine A and B using only its trial-participant count and partial-residual sum. Integrate its parameter under each component, then sum over its indicator:

$$\begin{aligned} A &:= \frac{n_{1,\ell}}{\sigma_1^2}, & B &:= \frac{S_{1,\ell}}{\sigma_1^2}, \\ M_g &= w(1 + \tau_0^2 A)^{-1/2} \exp\left\{ \frac{\tau_0^2 B^2}{2(1 + \tau_0^2 A)} \right\} \\ &\quad + (1 - w)(1 + \tau_1^2 A)^{-1/2} \exp\left\{ \frac{\tau_1^2 B^2}{2(1 + \tau_1^2 A)} \right\}. \end{aligned} \quad (\text{A.6})$$

Thus both $\theta_{h^{s\ell}}^g$ and $z_{h^{s\ell}}$ are marginalized for a discrepancy-tree proposal. The quantities w, τ_0^2, τ_1^2 are held fixed, not integrated out in this step. For an empty trial leaf, $M_g = w + (1 - w) = 1$.

A.4 Metropolis–Hastings tree updates

The marginal tree likelihood is the observation-level density product times the product of its leaf factors. The observation product is identical under both partitions, so it cancels from the Metropolis–Hastings ratio. Suppressing the tree index and component label, a proposal $\mathcal{T} \rightarrow \mathcal{T}^*$ is accepted with probability

$$\min \left\{ 1, \frac{p(\mathcal{T}^*)}{p(\mathcal{T})} \frac{q(\mathcal{T} | \mathcal{T}^*)}{q(\mathcal{T}^* | \mathcal{T})} \frac{\prod_{\ell \in \mathcal{T}^*} M_\ell}{\prod_{\ell \in \mathcal{T}} M_\ell} \right\}. \quad (\text{A.7})$$

The tree prior uses the restricted weights in (A.1); the proposal ratio accounts for the choice of move, node and rule. For a grow move replacing parent P with children L, R , the likelihood contribution reduces to $M_L M_R / M_P$. The current implementation uses grow, prune and change proposals; an inadmissible proposal is rejected.

A.5 Conditional parameter distributions

After accepting or rejecting a structure, sample node parameters for the retained tree. For an f leaf,

$$\theta_{h^f \ell}^f | \text{rest} \sim \text{N} \left(\frac{B}{A + \lambda_f^{-2}}, \frac{1}{A + \lambda_f^{-2}} \right). \quad (\text{A.8})$$

For a nonempty g leaf, let $\bar{r} = S_{1,\ell}/n_{1,\ell}$. Its spike probability is

$$P(z_{h^{s\ell}} = 1 | \text{rest}) = \frac{w \phi(\bar{r}; 0, \tau_0^2 + \sigma_1^2/n_{1,\ell})}{w \phi(\bar{r}; 0, \tau_0^2 + \sigma_1^2/n_{1,\ell}) + (1 - w) \phi(\bar{r}; 0, \tau_1^2 + \sigma_1^2/n_{1,\ell})}. \quad (\text{A.9})$$

This compares the density of the partial-residual mean under the two integrated components, including their prior weights. Given component variance τ_k^2 ($k = 0$ for $z = 1$ and $k = 1$ for $z = 0$),

$$\theta_{h^{s\ell}}^g | z_{h^{s\ell}}, \text{rest} \sim \text{N} \left(\frac{B}{A + \tau_k^{-2}}, \frac{1}{A + \tau_k^{-2}} \right). \quad (\text{A.10})$$

An empty leaf instead draws $z \sim \text{Bernoulli}(w)$ and θ from its selected prior component; division by $n_{1,\ell} = 0$ is never required.

Let L_g be the total number of discrepancy terminal nodes, $K_0 := \sum_{h^g, \ell} z_{h^g \ell}$ and $S_0 := \sum_{h^g, \ell} z_{h^g \ell} (\theta_{h^g \ell}^g)^2$. With $w \sim \text{Beta}(a_w, b_w)$, where $a_w = b_w = 1$ by default,

$$w \mid \text{rest} \sim \text{Beta}(a_w + K_0, b_w + L_g - K_0). \quad (\text{A.11})$$

When the spike variance is updated under the hierarchical prior,

$$\tau_0^2 \mid \text{rest} \sim \text{scaled-Inv } \chi^2 \left(v_0 + K_0, \frac{v_0 s_0^2 + S_0}{v_0 + K_0} \right) \text{ restricted to } (0, \tau_1^2). \quad (\text{A.12})$$

For each source, write n_s^{fit} for its fitting sample size, equal to N for source 1 and n_2 for source 2 in the joint model. Define its residual sum of squares using the full current mean, RSS_s . Then

$$\sigma_s^2 \mid \text{rest} \sim \text{IG} \left(\frac{v_\sigma + n_s^{\text{fit}}}{2}, \frac{v_\sigma \lambda_{\sigma,s} + \text{RSS}_s}{2} \right). \quad (\text{A.13})$$

The residual sums of squares use $Y_i - f(x_i) - S_i g(x_i) - A_i \psi$. The coefficient ψ is drawn from main-text equation (8); this is a full conditional, with f, g held at their current values.

A.6 Joint sampling cycle

1. Initialize both forests, their leaf parameters and indicators, ψ , the mixture parameters and the two residual variances.
2. Update each f tree using all treatment-adjusted partial responses, the marginal factor M_f , and the conditional leaf distribution.
3. Update the g trees using all trial participants' treatment-adjusted partial responses and M_g ; draw each retained tree's indicators and leaf parameters. The reported joint fits repeat this discrepancy sweep five times.
4. Update τ_0^2 and w from their conditional distributions.
5. Draw ψ from its Gaussian full conditional, then update the two residual variances.
6. Repeat and retain the joint draws after warm-up. Prediction and interval summaries average over the retained states; conditional update variances are not substituted for marginal posterior uncertainty.

The ordering above describes a composition of conditionally invariant updates. The implementation updates the treatment coefficient before the residual variances, and updates leaf parameters after every accepted or rejected tree proposal. The tree move integrates leaf parameters and discrepancy indicators while holding shared scales and w fixed. Immediate conditional draws for the retained tree restore those marginalized variables before another block uses them. Five discrepancy sweeps repeat valid updates and do not change the target distribution. These facts concern the specified transition kernel, not its mixing rate.

A.7 Separate-treatment and single-arm computation

The supplementary Gaussian/survival experiments and application use a different implementation that fits the treatment surface separately and applies the two-source model only to controls. For that variant, source-1 counts in the formulas above count concurrent controls, and no treatment term enters a control-tree residual. The prognostic forest has $H_f = 50$ in simulations and $H_f = 10$ or 50 in the application, with $H_g = 5$ throughout. Outcome ranges are calculated separately for pooled controls and treated participants.

The treatment sum of trees is fitted separately under the normal or augmented-normal likelihood. It uses $H_{\text{trt}} = 50$, $(\alpha_{\text{trt}}, \beta_{\text{trt}}) = (0.95, 2)$, independent $\theta_{h_{\text{trt}}\ell}^{\text{trt}} \sim N(0, \lambda_{\text{trt}}^2)$ with $\lambda_{\text{trt}}^2 = R_{y,\text{trt}}^2 / (16H_{\text{trt}})$, and $\sigma_{\text{trt}}^2 \sim \text{IG}(v_\sigma/2, v_\sigma \lambda_{\sigma,\text{trt}}/2)$. The treatment-specific $\lambda_{\sigma,\text{trt}}$ follows the preliminary residual rule in Section A.1.

These separate-treatment simulations use 1,000 warm-up and 1,000 retained draws per fit. Application fits use four chains with 2,000 warm-up and 2,000 retained draws per chain. Their prior-scale calibration uses 1,000 warm-up and 1,000 retained historical draws in simulation and 2,000 warm-up and 4,000 retained draws in the application.

When there are no concurrent controls, every discrepancy leaf has $M_g = 1$. In this separate-treatment variant, g is sampled from its specified prior and σ_1^2 is set equal to σ_2^2 . The reported single-arm fits fix w and $\tau_0^2 = \min(s_0^2, \tau_1^2/2)$ and permit discrepancy trees with no concurrent-control observations. The shared cutpoint specification and empirical scales are conditioned on. These prior-driven control predictions are distinct from the joint randomized-trial coefficient in main-text equation (1).

B Prior effective sample size: comparison, derivation and computation

B.1 Comparison with established prior-ESS definitions

Table S2 places the proposed definition beside the variance calibration of Neuenschwander et al. (2010), the curvature match of Morita et al. (2008), and the expected local-information-ratio (ELIR) definition of Neuenschwander et al. (2020). Throughout the table, D_H denotes the historical dataset, θ_h denotes a historical-study parameter or estimand, with h indexing historical studies when applicable, and $\theta_\star$ denotes the corresponding current-study parameter or estimand. The two Neuenschwander rows separate the complete-pooling calibration anchor from the partial-pooling MAP prior whose ESS is inferred. For Morita's method, $Y_{1:m}$ denotes a hypothetical sample of size m from the reference sampling model; write $i_\varepsilon(\theta_\star) := -\partial_{\theta_\star}^2 \log \pi_\varepsilon(\theta_\star)$ for low-information-prior curvature and $j_m(Y_{1:m}; \theta_\star) := -\partial_{\theta_\star}^2 \log \{\prod_{r=1}^m p(Y_r | \theta_\star)\}$ for observed likelihood information. For the proposed method, the target prior is the RWD-informed prior for the current-control mean, and $(\theta_h, \delta_\star, \theta_\star)$ corresponds to (μ_f, μ_g, μ) in the main text.

Table S2: Comparison of prior effective sample size definitions. Panel A summarizes each target; Panel B gives the corresponding reference or calibration and ESS.

Method/case	Target		
	Target prior	Information measure	Information quantity
Neuenschwander et al. (2010): complete pooling	$\pi_0(\theta_*) := p(\theta_* D_H, \tau = 0)$, the complete-pooling MAP prior	Inverse marginal variance, where $V_0 := \text{Var}_{\pi_0}(\theta_*)$	$\mathcal{I}_0 := 1/V_0$, the complete-pooling prior information
Neuenschwander et al. (2010): partial pooling	$\pi_{\text{target}}(\theta_*) := p(\theta_* D_H)$, the partial-pooling MAP prior	Inverse marginal variance, where $V_{\text{target}} := \text{Var}_{\pi_{\text{target}}}(\theta_*)$	$\mathcal{I}_{\text{target}} := 1/V_{\text{target}}$, the partial-pooling prior information
Morita–Thall–Müller (2008)	Supplied target prior $\pi_{\text{target}}(\theta_*)$	Negative log-prior curvature at $\bar{\theta}_* := E_{\pi_{\text{target}}}(\theta_*)$	$\mathcal{I}_{\text{target}} := -\partial_{\theta_*}^2 \log \pi_{\text{target}}(\theta_*) \Big _{\theta_* = \bar{\theta}_*}$
ELIR	Supplied target prior $\pi_{\text{target}}(\theta_*)$	Pointwise negative log-prior curvature	$\mathcal{I}_{\text{target}}(\theta_*) := -\partial_{\theta_*}^2 \log \pi_{\text{target}}(\theta_*)$
Proposed conditional variance match	RWD-informed target prior $\pi_{\text{target}}(\theta_* w, \tau_0^2)$ induced for $\theta_* := \theta_h + \delta_*$ by the historical posterior and discrepancy prior	Inverse conditional marginal variance, where $V_{\text{target}}(w, \tau_0^2) := \text{Var}_{\pi_{\text{target}}}(\theta_* w, \tau_0^2)$	$\mathcal{I}_{\text{target}}(w, \tau_0^2) := 1/V_{\text{target}}(w, \tau_0^2)$

Table S2 (continued): Panel B, reference and ESS

Method/case	Reference			ESS	
	Information source	Information measure	Information quantity	Matching or calibration	Result
Neuenschwander: complete pooling	–	–	–	Anchor $\text{ESS}_0 := N_H$, where N_H is the total historical sample size	$\text{ESS}_0 = N_H$
Neuenschwander: partial pooling	–	–	–	$\mathcal{J}_{\text{target}} = (\text{ESS}_{\text{target}}/N_H) \cdot \mathcal{J}_0$	$\text{ESS}_{\text{target}} = N_H V_0 / V_{\text{target}}$
Morita–Thall– Müller	$\pi_{\varepsilon, m}(\theta_* Y_{1:m}) \propto \pi_{\varepsilon}(\theta_*) \prod_{r=1}^m p(Y_r \theta_*)$	Expected negative log-posterior curvature at $\bar{\theta}_*$	$\mathcal{J}_{\text{ref}}(m) := i_{\varepsilon}(\bar{\theta}_*) + E_{Y_{1:m}} \{j_m(Y_{1:m}; \bar{\theta}_*)\}$	$\mathcal{J}_{\text{target}} \approx \mathcal{J}_{\text{ref}}(m)$; choose the closest match	$\text{ESS}_{\text{MTM}} := \arg \min_{m \geq 0} \mathcal{J}_{\text{target}} - \mathcal{J}_{\text{ref}}(m) $
ELIR	One observation $Y \sim p(\cdot \theta_*)$	Expected Fisher information in one observation	$\mathcal{J}_{\text{ref}}(\theta_*) := i_F(\theta_*) = E_{Y \theta_*} \{-\partial_{\theta_*}^2 \log p(Y \theta_*)\}$	$\mathcal{J}_{\text{target}}(\theta_*) = m(\theta_*) \mathcal{J}_{\text{ref}}(\theta_*)$	$\text{ESS}_{\text{ELIR}} := E_{\pi_{\text{target}}} \{m(\theta_*)\}$
Proposed method	One observation $Y \sim p(\cdot \theta_*)$	Expected Fisher information in one observation	$\mathcal{J}_{\text{ref}}(\theta_*) := i_F(\theta_*)$	$\mathcal{J}_{\text{target}}(w, \tau_0^2) = m(w, \tau_0^2; \theta_*) \mathcal{J}_{\text{ref}}(\theta_*)$	$\text{ESS}(w, s_0^2) := E_{\tau_0^2} \{m(w, \tau_0^2; \theta_*)\}$; for the normal reference, $i_F = 1/\sigma_1^2$

Neuenschwander’s complete-pooling prior supplies a known ESS anchor. Morita matches the target-prior curvature to the expected curvature of a posterior constructed from a low-information prior and hypothetical data. ELIR averages local ratios of prior curvature to one-observation Fisher information over the target prior. For the proposed method, this target prior is the RWD-informed prior; the ESS averages conditional inverse-variance matches over the calibration distribution of τ_0^2 .

B.2 Calculation of historical uncertainty

For the main-text historical component $\mu_f = N^{-1} \sum_{i=1}^N f(x_i^*)$,

$$V_f = \frac{1}{N^2} \sum_{i=1}^N \sum_{i'=1}^N \text{Cov}\{f(x_i^*), f(x_{i'}^*) \mid \mathbf{Y}_2, \mathbf{X}_2, \mathbf{X}_*\}. \quad (\text{B.1})$$

Given preliminary posterior draws $f^{(b)}$, form $\mu_f^{(b)} := N^{-1} \sum_i f^{(b)}(x_i^*)$ and estimate V_f by their sample variance. This calculation integrates tree and leaf uncertainty and retains cross-profile covariance. Averaging the pointwise posterior variances would omit the covariance terms.

B.3 Derivation of discrepancy uncertainty

Let $\mathcal{T}^g := (\mathcal{T}_{h^g}^g)_{h^g=1}^{H_g}$ and define

$$a_{h^g \ell} := \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{x_i^* \in \ell \text{ of tree } h^g\}. \quad (\text{B.2})$$

Regrouping the trial-profile average of the discrepancy trees gives

$$\mu_g = \sum_{h^g, \ell} a_{h^g \ell} \theta_{h^g \ell}^g. \quad (\text{B.3})$$

Within each tree, $a_{h^g \ell} \geq 0$ and $\sum_{\ell} a_{h^g \ell} = 1$. Conditional on the trees and indicators, the independent zero-mean leaf priors imply

$$\begin{aligned} \mathbb{E}(\mu_g \mid \mathcal{T}^g, \mathbf{z}, w, \tau_0^2) &= 0, \\ \text{Var}(\mu_g \mid \mathcal{T}^g, \mathbf{z}, w, \tau_0^2) &= \sum_{h^g, \ell} a_{h^g \ell}^2 \{z_{h^g \ell} \tau_0^2 + (1 - z_{h^g \ell}) \tau_1^2\}. \end{aligned} \quad (\text{B.4})$$

The conditional mean remains zero after integrating over the indicators, so the between-mean term in the law of total variance vanishes. Since $\mathbb{E}(z_{h^g \ell} \mid w) = w$,

$$\text{Var}(\mu_g \mid \mathcal{T}^g, w, \tau_0^2) = \{w \tau_0^2 + (1 - w) \tau_1^2\} \sum_{h^g, \ell} a_{h^g \ell}^2.$$

The same argument applies when integrating over the prior trees, giving main-text equation 11. Thus V_g integrates leaf, indicator and tree uncertainty at fixed w, τ_0^2 . The identity is exact under the stated independent zero-mean priors; Monte Carlo enters only in evaluating the tree expectation.

B.4 Proof of the prior ESS curve theorem

Under the all-spike calibration, let

$$C_g := \mathbb{E}_{\mathcal{T}^g} \left[\sum_{h^s, \ell} a_{h^s \ell}^2 \right].$$

Every finite tree has at least one occupied terminal node and $\sum_{\ell} a_{h^s \ell}^2 \leq 1$, so $0 < C_g \leq H_g$. If $U \sim \chi_{v_0}^2$, the scaled inverse-chi-square variable can be coupled over all scale values as $\tau_0^2 = v_0 s_0^2 / U$. Main-text equation 14 becomes

$$\text{ESS}_{\tau_0}(s_0^2) = \mathbb{E}_U \left[\frac{\sigma_1^2}{V_f + C_g v_0 s_0^2 / U} \right].$$

For every $U > 0$, the integrand is continuous, strictly decreasing and strictly convex in $s_0^2 > 0$, and is bounded above by σ_1^2 / V_f . Dominated convergence establishes continuity and the limits

$$\lim_{s_0^2 \downarrow 0} \text{ESS}_{\tau_0}(s_0^2) = \frac{\sigma_1^2}{V_f}, \quad \lim_{s_0^2 \rightarrow \infty} \text{ESS}_{\tau_0}(s_0^2) = 0.$$

Pointwise strict inequalities preserve strict monotonicity and strict convexity after expectation. The intermediate value theorem and strict monotonicity give a unique positive solution for each target in $(0, \sigma_1^2 / V_f)$. This proves main-text Theorem 1 without requiring $\mathbb{E}(\tau_0^2)$ to be finite.

B.5 Monte Carlo evaluation and implemented prior-scale selection

Estimate V_f from the posterior averages of f over the standardization profiles and estimate C_g by simulating discrepancy trees from the calibration prior. For independent $U_b \sim \chi_{v_0}^2$, $b = 1, \dots, B$, evaluate

$$\widehat{\text{ESS}}_{\tau_0}(s_0^2) := \frac{1}{B} \sum_{b=1}^B \frac{\widehat{\sigma}_1^2}{\widehat{V}_f + (v_0 s_0^2 / U_b) \widehat{C}_g}. \quad (\text{B.5})$$

Reusing the U_b values over the scale grid makes the Monte Carlo curve monotone. The preliminary f predictions are posterior draws conditional on the historical dataset.

Algorithm B1: full-variance ESS evaluation and target selection.

1. Specify the current-control standardization profiles, historical outcome likelihood, candidate splitting rules and fixed reference residual variance estimate.
2. Fit the historical model, average each posterior prediction draw over the standardization profiles and estimate V_f .
3. Simulate discrepancy trees from the calibration prior and estimate C_g .
4. Draw $U_b \sim \chi_{v_0}^2$ and calculate (B.5) over candidate scales.
5. For an attainable target, solve the monotone equation or select a grid approximation; record targets outside the attainable range.

The implemented selector divides the posterior draws of μ_f into $J = 20$ consecutive blocks, calculates a variance in each block and rescales the block variances so that their arithmetic mean equals the variance

across all draws. Write the rescaled values as $\widehat{V}_{f,k}$. It substitutes each $\widehat{V}_{f,k}$ into (B.5), yielding $\widehat{\text{ESS}}_k(s_0^2)$, and uses their arithmetic mean as the central curve. With $m_* := \min(m_{\text{requested}}, \widehat{\sigma}_1^2 / \widehat{V}_f)$, the selector chooses the smallest supplied scale satisfying

$$\frac{1}{J} \sum_{k=1}^J \mathbf{1}\{\widehat{\text{ESS}}_k(s_0^2) \leq m_*\} \geq 0.95. \quad (\text{B.6})$$

If no grid value satisfies the rule, it chooses the largest value. If the requested target reaches or exceeds the full-variance ceiling, the current implementation chooses the smallest grid value.

The block-based central curve does not generally have the full-variance ceiling as its zero-scale limit because

$$\lim_{s_0^2 \downarrow 0} \frac{1}{J} \sum_{k=1}^J \widehat{\text{ESS}}_k(s_0^2) = \frac{1}{J} \sum_{k=1}^J \frac{\widehat{\sigma}_1^2}{\widehat{V}_{f,k}} \geq \frac{\widehat{\sigma}_1^2}{\widehat{V}_f},$$

with equality only when the positive block variances are equal. Consecutive blocks of an autocorrelated Markov chain also do not supply a formal 95% posterior or repeated-sampling guarantee. The following figure illustrates the curves for one separate-treatment Gaussian two-arm dataset; Table S11(a) evaluates the full-variance expression in the application.

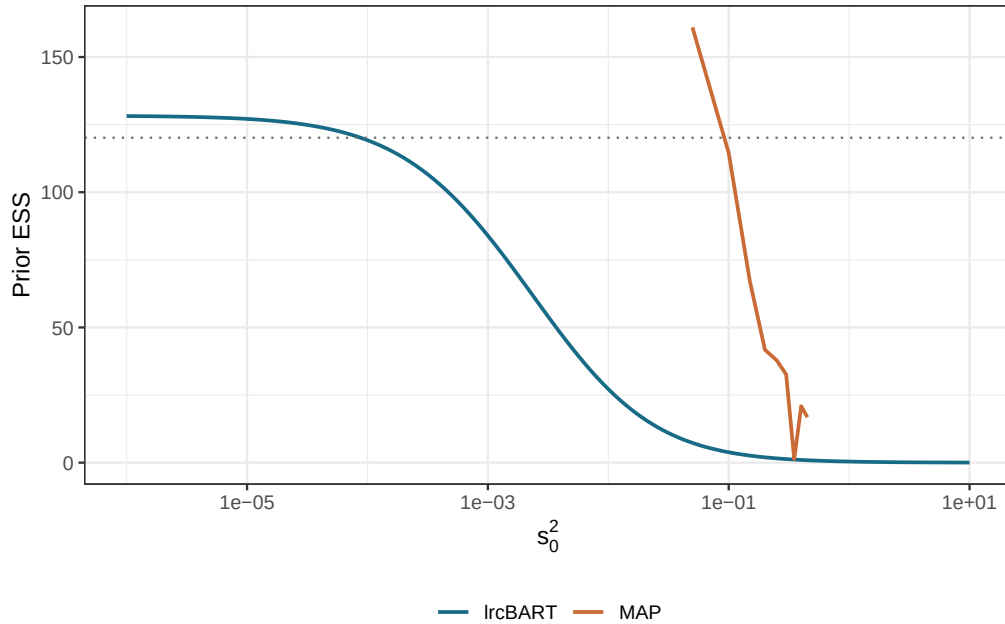


Figure S1: Illustrative prior ESS curves for the separate-treatment Gaussian two-arm Sc1, replicate 1. The lrcBART curve is the block-based central ESS summary over the variance-scale grid; the dotted horizontal line is its full-variance ceiling. The MAP overlay uses its corresponding prior ESS calculation. Because MAP and lrcBART use different prior constructions and ESS implementations, the common horizontal axis denotes each method's supplied variance-scale parameter rather than identical prior distributions. The curves use one dataset and are not averaged over replicates.

B.6 Pointwise and overall means

For one standardization profile x , define the scalar current-control target $\mu(x) := f(x) + g(x)$ and

$$V_f(x) := \text{Var}\{f(x) \mid \mathbf{Y}_2, \mathbf{X}_2, x\}.$$

One terminal node in each discrepancy tree has weight one, so

$$V_g(w, \tau_0^2; x) = H_g\{w\tau_0^2 + (1-w)\tau_1^2\}.$$

Under the all-spike reference, the pointwise ESS is therefore

$$\text{ESS}_{\tau_0}(s_0^2; x) = E_{\tau_0^2} \left[\frac{\sigma_1^2}{V_f(x) + H_g\tau_0^2} \right].$$

The matched reference observation has the same target, $Y^{\text{ref}} \mid x \sim N\{\mu(x), \sigma_1^2\}$. Thus the target and reference describe the same conditional mean parameter.

For standardization profiles $x_1, \dots, x_n$, the overall target is $\bar{\mu} = n^{-1} \sum_i \mu(x_i)$ and

$$\text{Var}(\bar{\mu}) = \frac{1}{n^2} \left[\sum_i \text{Var}\{\mu(x_i)\} + \sum_{i \neq i'} \text{Cov}\{\mu(x_i), \mu(x_{i'})\} \right]. \quad (\text{B.7})$$

The same fitted functions inform all profiles, so the covariance terms generally do not vanish. Inverse variance is also nonlinear. Pointwise ESS values therefore do not generally sum or average to the overall ESS; the overall ESS must be calculated from the variance of $\bar{\mu}$.

B.7 Reference interpretation and implementation qualifications

The theoretical reference treats σ_1^2 as fixed. Numerical calibration substitutes a fixed estimate $\hat{\sigma}_1^2$; in two-arm analyses this estimate uses randomized controls. All reported analyses use all trial profiles, including the treated profiles in a single-arm analysis, and a single-arm analysis uses the RWD-control residual variance as the reference scale under $\sigma_1^2 = \sigma_2^2$.

For survival outcomes, censoring enters the preliminary historical likelihood and therefore affects V_f . The reference information unit remains one uncensored normal observation on the log-time scale.

The preliminary historical fit, calibration tree simulation and final joint fit use different admissible split rules. The current calibration simulator limits depth to 12, while the separate-treatment two-arm fit requires at least five randomized controls per discrepancy leaf; the primary joint fit instead uses all trial participants for that restriction. Prior ESS calibration uses an untruncated spike-variance distribution; the two-arm sampler uses its upper-truncated counterpart, and single-arm fits fix the spike variance at $\min(s_0^2, \tau_1^2/2)$. The joint fitting tree weight is specified in (A.1), whereas its prior-tree calibration uses a locally forced-terminal recursion. Accordingly, the reported prior ESS corresponds to the stated calibration specification. Aligning the full-variance rule, scale law and tree prior with the final fit would require a new calibration and evaluation of the affected analyses.

B.8 Treatment information in the joint likelihood

The control-mean target and the treatment coefficient have different uncertainty. For fixed forests, indicators and scales, collect all f and g leaf parameters in β_0 , with prior covariance Λ_0 . Write $Z_0 = [Z_f, \text{diag}(S)Z_g]$ for their observation-level design: an f row reaches one leaf per f tree, and a g row is zero for an external observation. Let A be the treatment column and W the diagonal matrix of source precisions. Normal updating gives joint precision

$$Q = \begin{pmatrix} \Lambda_0^{-1} + Z_0^\top W Z_0 & Z_0^\top W A \\ A^\top W Z_0 & v_\psi^{-1} + A^\top W A \end{pmatrix}.$$

Integrating the forest coefficients yields

$$\text{Var}(\psi \mid \text{trees, indicators, scales, } \mathcal{D}) = \left[v_\psi^{-1} + A^\top W A - A^\top W Z_0 (\Lambda_0^{-1} + Z_0^\top W Z_0)^{-1} Z_0^\top W A \right]^{-1}. \quad (\text{B.8})$$

This is the inverse Schur complement of the forest block. The subtracted quadratic form is nonnegative, so the variance is at least the full-conditional variance $(v_\psi^{-1} + A^\top W A)^{-1}$ used to draw ψ . Integrating tree and scale states requires averaging the conditional variances and adding the variance of their conditional means. A control-mean precision ratio neither supplies these covariance terms nor defines an incremental ESS for treatment. Equation (B.8) is a Gaussian identity, not a new treatment-ESS definition.

B.9 Marginal ELIR for the sum of tree contributions

For a differentiable scalar prior density p_μ and normal reference variance σ_1^2 , marginal ELIR is

$$\text{ESS}_{\text{ELIR}}(p_\mu) = \sigma_1^2 \int \frac{\{p'_\mu(u)\}^2}{p_\mu(u)} du,$$

when the location information is finite and the boundary terms required for integration by parts vanish. This is the expected prior-curvature definition of Neuenschwander et al. (2020). It uses the density of the total mean, not separate leaf conditional distributions from backfitting. For independent normal leaf contributions u_j with variances v_j and fixed target weights r_j , the weighted sum has

$$\text{ESS}_{\text{ELIR}}\left(\sum_j r_j u_j\right) = \left(\sum_j \frac{r_j^2}{e_j}\right)^{-1}, \quad e_j = \sigma_1^2 / v_j.$$

The weights are target-profile proportions within a tree; at one profile each reached leaf has weight one. The inverse sum arises because the tree contributions add to the mean and their variances add under independence. Fitted contributions from different trees are generally dependent.

For a historical Gaussian fit conditional on its forest structures and scales, let Z_f be the historical leaf-incidence matrix, $\Lambda_f = \lambda_f^2 I$, and y_2 the centered historical responses. The joint posterior of the historical leaf coefficients has covariance and mean

$$\Sigma_f = (\Lambda_f^{-1} + Z_f^\top Z_f / \sigma_2^2)^{-1}, \quad m_f = \Sigma_f Z_f^\top y_2 / \sigma_2^2.$$

Let c_f and a_g contain the target proportions in the historical and discrepancy leaves, and let D_z be the diagonal

matrix of discrepancy component variances. Conditional on the joint state η , independence of the external posterior and discrepancy prior gives

$$\mu \mid \eta, \mathcal{D}_2 \sim N(M_\eta, V_\eta), \quad M_\eta = m_0 + c_f^\top m_f, \quad V_\eta = c_f^\top \Sigma_f c_f + a_g^\top D_z a_g,$$

where m_0 restores centering. A finite approximation to the mixing distribution is $p_B(u) = \sum_{b=1}^B \omega_b \phi(u; M_b, V_b)$ with positive weights summing to one. Historical forest states must be sampled jointly from their historical-data posterior, while discrepancy states come from the specified prior. They cannot be assembled by independently resampling fitted trees. In an AFT fit, the state also includes the augmented historical log times.

For this finite normal mixture, let $s_b(u) = (M_b - u)/V_b$. Its marginal score is the conditional mean of the component score given u . Total variance of the scores and integration by parts give

$$\frac{\sigma_1^2}{\text{Var}_{p_B}(\mu)} \leq \text{ESS}_{\text{ELIR}}(p_B) \leq \sigma_1^2 \sum_b \frac{\omega_b}{V_b}.$$

These bounds refer to the same mixture. The implemented calibration instead integrates tree and indicator uncertainty inside V_g , substitutes the historical marginal variance, and averages over an untruncated scale distribution. The bounds therefore do not order that calibration against the ELIR of the fitted model. The reported studies use the variance reference, not a marginal-ELIR replacement.

C Fixed-region theory and control-source benchmarks

C.1 Joint treatment and discrepancy learning

We prove main-text Theorem 2. Condition on a fixed finite partition, the group design and known, fixed positive residual variances. Collect parameters in $\vartheta = (f_1, \dots, f_J, g_1, \dots, g_J, \psi)^\top$. The design has, within each region j , rows $e_{f_j}^\top$, $(e_{f_j} + e_{g_j})^\top$ and $(e_{f_j} + e_{g_j} + e_\psi)^\top$ for external controls, concurrent controls and treated participants. Here e_k denotes a coordinate vector in $\mathbb{R}^{2J+1}$.

If all three row types have positive counts, the design has full column rank. Indeed, a vector orthogonal to every external row has each f coordinate zero; orthogonality to concurrent-control rows then makes each g coordinate zero, and treated rows determine the remaining treatment coordinate. Let X_0 contain one of each row type and let q_n be the smallest group-by-region observation precision, $n_{a,j}/\sigma_a^2$, where $\sigma_{\text{trt}}^2 = \sigma_1^2$. As all counts increase, $q_n \rightarrow \infty$ and

$$G_n = X^\top W X \succeq q_n X_0^\top X_0,$$

so $\lambda_{\min}(G_n) \rightarrow \infty$ without a relative growth-rate restriction.

Conditional on a discrepancy allocation $z \in \{0, 1\}^J$, let Q_z be the prior precision of ϑ . It is fixed and positive definite. The Gaussian posterior has

$$V_z = (Q_z + G_n)^{-1}, \quad m_z = V_z X^\top W Y.$$

At the fixed true parameter ϑ^* ,

$$\mathbb{E}_{\vartheta^*}(m_z) - \vartheta^* = -V_z Q_z \vartheta^*, \quad \text{Cov}_{\vartheta^*}(m_z) = V_z G_n V_z \preceq V_z.$$

The operator norm of V_z is at most $1/\lambda_{\min}(G_n)$, so both the squared bias and sampling variance tend to zero. It follows that

$$\mathbb{E}_{\vartheta^*} \left[\int \|\vartheta - \vartheta^*\|^2 d\Pi_z(\vartheta | Y) \right] = \text{tr}(V_z) + \mathbb{E}_{\vartheta^*} \|m_z - \vartheta^*\|^2 \rightarrow 0.$$

Markov's inequality yields posterior concentration in probability for each component. There are only 2^J components, and the mixture probability outside a neighborhood is bounded by the largest component probability, whatever their posterior weights. Norm equivalence in the fixed-dimensional parameter space gives the theorem's displayed concentration statement.

For a region with $|g_j^*| \neq c$, a draw that is on the opposite side of the tolerance boundary must satisfy $|g_j - g_j^*| \geq ||g_j^*| - c| > 0$. Concentration makes this probability vanish. A finite union establishes the simultaneous practical-agreement limits. This proof does not require consistent recovery of the spike/slab labels.

Removing all concurrent-control rows leaves the nonzero null direction $(0_J, 1_J, -1)$. It changes every g_j by the same constant and the treatment coefficient by its negative. Thus the single-arm likelihood does not identify them separately. The all-three-groups-per-region assumption is a sufficient condition for the theorem; it is not claimed to be a minimal design condition.

C.2 Control-source benchmarks

We study two properties: learning practical agreement and characterizing the estimation benefit and error induced by borrowing. The following separate control-source results use a Gaussian model on a fixed finite partition $R_1, \dots, R_J$ of the target covariate domain. Within region j , independent historical and trial controls have means f_j and $f_j + g_j$, with known positive residual variances σ_2^2 and σ_1^2 . Here g_j is the total regional source difference. Both source means are unknown and estimated jointly. Use independent priors across regions, with $f_j \sim N(0, \lambda_j^2)$ independent of

$$g_j \sim wN(0, \tau_0^2) + (1-w)N(0, \tau_1^2), \quad 0 < w < 1, \quad 0 < \tau_0^2 < \tau_1^2 < \infty.$$

All prior parameters are fixed, with $0 < \lambda_j^2 < \infty$.

Theorem C.1 (Learning local practical agreement). *Suppose the regional model is correctly specified, with true parameters f_j^*, g_j^* . As the numbers $n_{1,j}$ and $n_{2,j}$ of controls from both sources tend to infinity in every region, for every $\varepsilon > 0$,*

$$\Pi \left\{ \max_{1 \leq j \leq J} (|f_j - f_j^*| + |g_j - g_j^*|) > \varepsilon \mid \mathcal{D} \right\} \rightarrow 0$$

almost surely under the true sampling law, where $\mathcal{D}$ denotes the observed control data. Thus the current-control means $f_j + g_j$ are also consistently learned. If $|g_j^| \neq c$ in every region, the following limits hold simultaneously:*

$$B_{c,j} := \Pi(|g_j| < c \mid \mathcal{D}) \rightarrow \begin{cases} 1, & |g_j^*| < c, \\ 0, & |g_j^*| > c, \end{cases} \quad \text{almost surely.} \quad (\text{C.1})$$

For this control-source benchmark, $B_{c,j}$ is the probability based only on the two control groups. For $x \in R_j$, it is constant within the region. The posterior increasingly supports practical agreement where the true difference is small and rules it out where the difference exceeds the tolerance. This interpretation uses posterior probabilities directly, without a labeling rule.

Estimation benefit and conflict risk. For an exact risk calculation in one region, retain the discrepancy mixture but replace the normal prior on the historical mean f by the flat prior $p(f) \propto 1$. This is a separate benchmark assumption. Write $\mu_1 = f + g$, $d = \mu_1 - f$ for the true discrepancy, $v_s = \sigma_s^2/n_s > 0$, $V = v_1 + v_2$, and $D = \bar{Y}_1 - \bar{Y}_2$. Let

$$\begin{aligned} \gamma(D) &= \frac{w\phi(D; 0, V + \tau_0^2)}{w\phi(D; 0, V + \tau_0^2) + (1-w)\phi(D; 0, V + \tau_1^2)}, \\ a_k &= \frac{v_1}{V + \tau_k^2}, \quad a(D) = \gamma(D)a_0 + \{1 - \gamma(D)\}a_1, \end{aligned} \quad (\text{C.2})$$

where $\phi(\cdot; m, v)$ is a normal density with mean m and variance v .

Theorem C.2 (Regional borrowing benefit and conflict risk). *Under the flat-baseline regional benchmark, the posterior mean of the current-control mean is*

$$\hat{\mu}_1 = \{1 - a(D)\}\bar{Y}_1 + a(D)\bar{Y}_2. \quad (\text{C.3})$$

The coefficient $a(D)$ decreases strictly with $|D| > 0$ and tends to $a_1 > 0$ as $|D| \rightarrow \infty$. Put $a_* = v_1/V$. Its repeated-sampling bias and mean squared error are

$$\begin{aligned} \text{Bias}_d(\hat{\mu}_1) &= -\text{E}_d\{a(D)D\}, \\ R(d) &:= \text{E}_d(\hat{\mu}_1 - \mu_1)^2 = \frac{v_1 v_2}{V} + \text{E}_d[\{(a_* - a(D))D - a_* d\}^2], \end{aligned} \quad (\text{C.4})$$

where the expectations on the right are over $D \sim N(d, V)$. Under exact agreement, $d = 0$, the estimator is unbiased and $R(0) < v_1$, the mean squared error of the trial-only mean. For fixed sample sizes and prior parameters, as $|d| \rightarrow \infty$,

$$\frac{\text{Bias}_d(\hat{\mu}_1)}{d} \longrightarrow -a_1, \quad \frac{R(d)}{d^2} \longrightarrow a_1^2.$$

Theorem C.2 establishes an estimation benefit under agreement and quantifies the remaining error under conflict. The finite normal slab leaves residual shrinkage, so the benchmark does not have uniformly bounded borrowing bias over arbitrarily large discrepancies. The mean squared error result does not imply that every realized posterior interval is shorter.

C.3 Proof of the control-source learning theorem

Fix region j , suppress its index on $v_s := \sigma_s^2/n_{s,j}$, and let $\beta_j = (f_j, g_j)^\top$. The sufficient statistic

$$\hat{\beta}_j = \begin{pmatrix} \bar{Y}_{2,j} \\ \bar{Y}_{1,j} - \bar{Y}_{2,j} \end{pmatrix} \quad (\text{C.5})$$

has normal sampling distribution with mean $\beta_j^* = (f_j^*, g_j^*)^\top$ and covariance

$$\Sigma_j = \begin{pmatrix} v_2 & -v_2 \\ -v_2 & v_1 + v_2 \end{pmatrix}. \quad (\text{C.6})$$

As both sample sizes increase, the strong law gives $\hat{\beta}_j \rightarrow \beta_j^*$ almost surely and $\Sigma_j \rightarrow 0$.

Conditional on discrepancy component $k \in \{0, 1\}$, the prior covariance of β_j is $P_{jk} = \text{diag}(\lambda_j^2, \tau_k^2)$. Normal updating gives

$$\begin{aligned} C_{jk} &= (\Sigma_j^{-1} + P_{jk}^{-1})^{-1}, \\ m_{jk} &= C_{jk} \Sigma_j^{-1} \hat{\beta}_j = \hat{\beta}_j - C_{jk} P_{jk}^{-1} \hat{\beta}_j. \end{aligned} \quad (\text{C.7})$$

Because $0 \prec C_{jk} \preceq \Sigma_j$ in the positive-semidefinite order, $C_{jk} \rightarrow 0$. The matrix P_{jk} is fixed and positive definite, so $m_{jk} \rightarrow \beta_j^*$ almost surely. Each component posterior therefore puts probability tending to one in any neighborhood of β_j^* . The posterior mixture has the same property: its probability outside a neighborhood is at most the larger of the two component probabilities. This argument does not require the mixture allocation probability to converge to zero or one.

A finite union bound over j establishes the simultaneous posterior convergence in the theorem. The bound

$$|(f_j + g_j) - (f_j^* + g_j^*)| \leq |f_j - f_j^*| + |g_j - g_j^*|$$

proves learning of the current-control means. Finally, if $|g_j^*| \neq c$, put $\delta_j = ||g_j^*| - c| > 0$. A posterior draw for which $|g_j|$ and $|g_j^*|$ are on opposite sides of c must satisfy $|g_j - g_j^*| \geq \delta_j$. Posterior concentration makes this event's probability vanish. There are finitely many regions, so the practical-agreement limits hold almost surely in all of them simultaneously. This conclusion concerns posterior probabilities directly and introduces no decision cutoff for $B_{c,j}$.

C.4 Proof of the flat-baseline control-risk theorem

Joint posterior and borrowing coefficient. Suppress the region index and write $\mu_1 = f + g$, $v_s = \sigma_s^2/n_s$, $V = v_1 + v_2$, and $D = \bar{Y}_1 - \bar{Y}_2$. Integrating the flat-baseline likelihood over f gives

$$\int_{\mathbb{R}} \phi(\bar{Y}_1; f + g, v_1) \phi(\bar{Y}_2; f, v_2) df = \phi(D; g, V). \quad (\text{C.8})$$

Integrating this expression against the two proper discrepancy components gives the positive finite mixture $w\phi(D; 0, V + \tau_0^2) + (1 - w)\phi(D; 0, V + \tau_1^2)$. Thus the joint posterior is proper, and its spike probability is $\gamma(D)$ in equation (C.2). Under component k ,

$$g \mid D, k \sim N\left(\frac{\tau_k^2}{V + \tau_k^2} D, \frac{V \tau_k^2}{V + \tau_k^2}\right).$$

Define the precision-weighted mean and coefficient

$$T := \frac{v_2 \bar{Y}_1 + v_1 \bar{Y}_2}{V}, \quad a_* := \frac{v_1}{V}. \quad (\text{C.9})$$

Conditional on g and the observations, the posterior mean of μ_1 is $T + a_*g$. Hence its component-specific posterior mean is $\bar{Y}_1 - a_k D$, where $a_k = v_1 / (V + \tau_k^2)$. Averaging over the components proves equation (C.3).

The posterior spike odds are

$$\frac{\gamma(D)}{1 - \gamma(D)} = C \exp(-\kappa D^2), \quad C = \frac{w}{1 - w} \sqrt{\frac{V + \tau_1^2}{V + \tau_0^2}}, \quad \kappa = \frac{1}{2} \left(\frac{1}{V + \tau_0^2} - \frac{1}{V + \tau_1^2} \right) > 0. \quad (\text{C.10})$$

It follows that $\gamma(D)$ decreases strictly with $|D| > 0$ and tends to zero as $|D| \rightarrow \infty$. Because $a_0 > a_1 > 0$, the coefficient $a(D) = a_1 + (a_0 - a_1)\gamma(D)$ has the stated monotonicity and limit. Also $0 < a(D) < a_* = v_1/V$.

Exact risk and benefit under agreement. Let $d = \mu_1 - f$ denote the fixed true discrepancy in the sampling experiment. Under independent normal sampling, T and D are independent, with

$$\mathbb{E}_d(T) = \mu_1 - a_* d, \quad \text{Var}_d(T) = \frac{v_1 v_2}{V}, \quad D \sim \mathcal{N}(d, V). \quad (\text{C.11})$$

The posterior-mean estimator can be written as $\hat{\mu}_1 = T + \{a_* - a(D)\}D$. Taking its expectation or squared error proves the bias and risk expressions in equation (C.4).

At $d = 0$, $a(D)$ is even and D is symmetrically distributed, so $\mathbb{E}_0\{a(D)D\} = 0$. The estimator is therefore unbiased. Since $0 < a(D) < a_*$ and $P(D \neq 0) = 1$,

$$\begin{aligned} R(0) &= \frac{v_1 v_2}{V} + \mathbb{E}_0[\{a_* - a(D)\}^2 D^2] \\ &< \frac{v_1 v_2}{V} + a_*^2 \mathbb{E}_0(D^2) = \frac{v_1 v_2}{V} + \frac{v_1^2}{V} = v_1. \end{aligned} \quad (\text{C.12})$$

This is a strict repeated-sampling mean squared error improvement over the trial-only mean. It is not a statement about posterior interval width in every dataset. The normal-expectation formula for $R(d)$ is continuous in d , by dominated convergence on compact sets of d . Thus a neighborhood of exact agreement also has $R(d) < v_1$, although its size depends on both sampling variances and the prior parameters.

For numerical evaluation, a second risk identity follows by normal integration by parts:

$$R(d) = v_1 + \mathbb{E}_d\{a(D)^2 D^2\} - 2v_1 \mathbb{E}_d\{a(D) + D a'(D)\}. \quad (\text{C.13})$$

Indeed, $\mathbb{E}(\bar{Y}_1 - \mu_1 \mid D) = a_*(D - d)$ and $\mathbb{E}_d\{(D - d)h(D)\} = V \mathbb{E}_d\{h'(D)\}$ for $h(D) = a(D)D$. The smooth bounded coefficient and its Gaussian-decaying derivative make the integration valid. Treating $a(D)$ as a fixed coefficient would omit the derivative term and would not give the correct risk.

Severe conflict at fixed sample sizes. Write $b(D) = (a_0 - a_1)\gamma(D)$. Then $a(D) = a_1 + b(D)$ and $0 \leq b(D) \leq (a_0 - a_1)C \exp(-\kappa D^2)$. For each fixed integer $q \geq 0$, completing the square gives

$$\mathbb{E}_d\{|D|^q e^{-\kappa D^2}\} = O \left[(1 + |d|^q) \exp \left\{ -\frac{\kappa d^2}{1 + 2\kappa V} \right\} \right].$$

Consequently, the bias contribution $\mathbb{E}_d\{b(D)D\}$ tends to zero. Comparing the risk formula with the constant-coefficient estimator $\bar{Y}_1 - a_1 D$ also leaves only polynomial multiples of these vanishing expectations. There-

fore

$$\begin{aligned} \text{Bias}_d(\hat{\mu}_1) &= -a_1 d + o(1), \\ R(d) &= a_1^2 d^2 + (1 - a_1)^2 v_1 + a_1^2 v_2 + o(1), \end{aligned} \quad |d| \rightarrow \infty. \quad (\text{C.14})$$

Dividing by d and d^2 , respectively, proves the theorem's limits. A finite normal slab therefore leaves unbounded borrowing bias over arbitrarily large true discrepancies at fixed sample sizes.

Increasing trial information is a different limit. With fixed positive prior scales, fixed true d , and bounded v_2 , the inequality $a(D) \leq v_1/\tau_0^2$ gives

$$|\text{Bias}_d(\hat{\mu}_1)| \leq \frac{v_1}{\tau_0^2} \mathbb{E}_d |D| \leq \frac{v_1}{\tau_0^2} (|d| + \sqrt{v_1 + v_2}) \rightarrow 0 \quad \text{as } v_1 \downarrow 0. \quad (\text{C.15})$$

This limit does not permit the spike variance to shrink arbitrarily with the sample size.

C.5 Separate-treatment contrast benchmark

Let $p_j \geq 0$, $\sum_j p_j = 1$, be fixed target-region weights. Write $e_j = \hat{\mu}_{1,j} - \mu_{1,j}$ and $b_j = \mathbb{E}(e_j)$. The mean squared error of the standardized control estimate is

$$\mathbb{E} \left(\sum_j p_j e_j \right)^2 = \left(\sum_j p_j b_j \right)^2 + \sum_{j,k} p_j p_k \text{Cov}(e_j, e_k). \quad (\text{C.16})$$

Under independent regional data and fixed independent regional priors in the flat-baseline benchmark, the off-diagonal covariance terms vanish. If all regions have exact agreement, the regional estimators are unbiased and their aggregate risk is $\sum_j p_j^2 R_j(0) < \sum_j p_j^2 v_{1,j}$, the risk of the standardized trial-only control mean.

For this separate-treatment contrast benchmark, write the error of the same independent treatment estimator as e_{trt} , with bias b_{trt} . The contrast error is $e_{\text{trt}} - \sum_j p_j e_j$, so its mean squared error is

$$\text{Var}(e_{\text{trt}}) + \sum_{j,k} p_j p_k \text{Cov}(e_j, e_k) + \left(b_{\text{trt}} - \sum_j p_j b_j \right)^2.$$

Under exact agreement in all regions, the strict control-risk improvement therefore also improves contrast estimation relative to the same treatment estimator with trial-only controls. Under conflict, the weighted control biases must be retained; if the treatment estimator is unbiased, the contrast bias is $-\sum_j p_j b_j$.

These formulas condition on the regional weights and design. Shared estimated mixture parameters and overlapping trees can induce regional dependence, so covariance terms cannot be discarded for the fitted ensemble. Under the learning model of Theorem C.1, posterior consistency of all regional control means also implies consistency of their fixed weighted mean. Adding a consistently learned treatment mean yields consistency of the corresponding contrast. An extension from fixed regions and weights to a learned ensemble standardized over empirical trial covariates requires its own design and posterior-concentration argument.

C.6 Supporting properties of a conditional discrepancy leaf

The following two lemmas retain the original leaf-level calculations as supporting results. Fix a discrepancy-tree partition, every other tree contribution, $n \geq 1$ trial partial residuals after treatment adjustment $\mathbf{r}$ in a leaf,

$\sigma_1^2 > 0$, $0 < w < 1$, and $0 < \tau_0^2 < \tau_1^2 < \infty$. Let $\bar{r}$ be their mean, and let $z = 1$ denote the spike component. Normal marginalization gives

$$\frac{P(z = 1 \mid \mathbf{r})}{P(z = 0 \mid \mathbf{r})} = \frac{w}{1-w} \sqrt{\frac{\tau_1^2 + \sigma_1^2/n}{\tau_0^2 + \sigma_1^2/n}} \exp \left[-\frac{\bar{r}^2}{2} \left\{ \frac{1}{\tau_0^2 + \sigma_1^2/n} - \frac{1}{\tau_1^2 + \sigma_1^2/n} \right\} \right]. \quad (\text{C.17})$$

For this subsection, write these odds as $C_\ell e^{-\kappa_\ell \bar{r}^2}$, with $C_\ell, \kappa_\ell > 0$, and put $\gamma_\ell(\bar{r}) = P(z = 1 \mid \mathbf{r})$ and $\omega_k = \tau_k^2 / (\tau_k^2 + \sigma_1^2/n)$.

Lemma C.1 (Bounded additional shrinkage in a fixed leaf). *Under the stated conditional leaf model, the difference between the mixture posterior mean and the slab-only posterior mean is bounded as a function of $\bar{r}$ and tends to zero as $|\bar{r}| \rightarrow \infty$.*

Proof. The component means are $\omega_k \bar{r}$. Hence the difference is $\Delta_{\text{mix}}(\bar{r}) = -\gamma_\ell(\bar{r})(\omega_1 - \omega_0)\bar{r}$. Since $\gamma_\ell(\bar{r}) \leq C_\ell e^{-\kappa_\ell \bar{r}^2}$,

$$|\Delta_{\text{mix}}(\bar{r})| \leq (\omega_1 - \omega_0) C_\ell |\bar{r}| e^{-\kappa_\ell \bar{r}^2} \leq \frac{(\omega_1 - \omega_0) C_\ell}{\sqrt{2e\kappa_\ell}}.$$

The first bound tends to zero in both tails. Relative to the unshrunk likelihood mean $\bar{r}$, the remaining slab shrinkage is $(\omega_1 - 1)\bar{r}$, which is unbounded at fixed n and finite τ_1^2 . $\square$

Lemma C.2 (Conditional learning of a leaf discrepancy). *At fixed sample size, the posterior spike probability decreases strictly with $|\bar{r}| > 0$. If the fixed leaf receives independent $N(d, \sigma_1^2)$ partial residuals, all other quantities remain fixed, and $n \rightarrow \infty$, then for its parameter θ and $|d| \neq c$, $P(|\theta| < c \mid \mathbf{r}) \rightarrow \mathbf{1}\{|d| < c\}$ almost surely. The spike probability need not converge to zero or one.*

Proof. Monotonicity follows from equation (C.17). Almost surely $\bar{r} \rightarrow d$. Each component posterior has mean $\omega_k \bar{r} \rightarrow d$ and variance $(n/\sigma_1^2 + \tau_k^{-2})^{-1} \rightarrow 0$. Thus each component, and their posterior mixture, concentrates at d . Positive separation from the tolerance boundary proves the probability limit. Nevertheless, the spike probability converges to

$$\frac{w\phi(d; 0, \tau_0^2)}{w\phi(d; 0, \tau_0^2) + (1-w)\phi(d; 0, \tau_1^2)},$$

which is strictly between zero and one for finite d . Learning a discrepancy does not uniquely identify its overlapping prior component. $\square$

C.7 Extension to the full model

The regional theorems fix the partition and all prior parameters; the risk theorem additionally uses a flat baseline prior. In the fitted model, tree structures, other contributions, the shared mixture probability, and the spike variance are estimated and dependent, with data-dependent scale choices. The proper Gaussian baseline prior also contributes shrinkage. None of the regional risk or learning results automatically transfers to that setting. A sum of tree functions can have multiple tree decompositions, so the scientifically meaningful target is the total $g(x)$, together with $f(x)$ and $f(x) + g(x)$.

A full-model learning result would require joint posterior concentration of the historical and current-control functions over the target covariate distribution, with adequate support from both sources. Subject to separation from the tolerance boundary, such concentration could justify convergence of posterior practical-agreement probabilities averaged over that population. It would not by itself imply uniform convergence at every profile, exact recovery of tree boundaries, or an estimation-risk improvement.

Without trial controls, the separate treatment likelihood supplies no information about g . Neither a regional theorem nor an ESS calculation supplies that missing information. The regional-discrepancy figure reports finite-sample empirical behavior; Table 4 reports patient-profile prior information and is not evidence that the discrepancy is consistently learned in the single-arm application. Censored-outcome extensions likewise require additional arguments rather than treating imputed log times as fully observed independent data in these proofs.

D Survival outcomes under separate treatment modeling

D.1 Observed-data likelihood

For group $a \in \{\text{trt}, 1, 2\}$, let $T_{a,i}$ denote event time, $C_{a,i}$ censoring time, $\tilde{T}_{a,i} := \min(T_{a,i}, C_{a,i})$ and $\delta_{a,i} := \mathbf{1}(T_{a,i} \leq C_{a,i})$. Conditional noninformative censoring is assumed given the covariates and source. For a source-specific lognormal model with log-time mean $\eta_{a,i}$ and residual standard deviation σ_a , the outcome likelihood contribution is

$$\left[\frac{1}{\tilde{T}_{a,i}} \phi(\log \tilde{T}_{a,i}; \eta_{a,i}, \sigma_a^2) \right]^{\delta_{a,i}} \left[1 - \Phi \left(\frac{\log \tilde{T}_{a,i} - \eta_{a,i}}{\sigma_a} \right) \right]^{1-\delta_{a,i}}. \quad (\text{D.1})$$

The mean functions are $\eta_{\text{trt}}(x) := f_{\text{trt}}(x)$, $\eta_1(x) := f(x) + g(x)$ and $\eta_2(x) := f(x)$, with $\eta_{a,i} := \eta_a(x_{a,i})$. Each group has its residual standard deviation σ_a . Both events and censored observations contribute to parameter inference.

D.2 Data augmentation

For an event, the latent response is $Y_{a,i} = \log \tilde{T}_{a,i}$. For a censored observation, draw

$$Y_{a,i} \mid \text{rest}, \delta_{a,i} = 0 \sim \text{N}(\eta_{a,i}, \sigma_a^2) \text{ restricted to } (\log \tilde{T}_{a,i}, \infty). \quad (\text{D.2})$$

Conditional on these latent responses, use the Gaussian leaf and variance updates under the separate-treatment specification in Appendix A.7. The trial and historical censoring indicators are both used. Prior ESS retains the uncensored normal reference described in Appendix B.7.

D.3 Standardized survival curves and population medians

At each posterior draw, average the individual survival curves over the trial standardization profiles:

$$\bar{S}_a(t) := \frac{1}{N} \sum_{s \in \{\text{trt}, 1\}} \sum_{i=1}^{n_s} \Phi \left(\frac{\eta_a(x_{s,i}) - \log t}{\sigma_a} \right), \quad a \in \{\text{trt}, 1\}. \quad (\text{D.3})$$

Here $\eta_1(x) = f(x) + g(x)$, $\eta_{\text{trt}}(x) = f_{\text{trt}}(x)$; in a single-arm analysis, $\sigma_1 = \sigma_2$. For horizon $t^* > 0$, define $R_a(t^*) := \int_0^{t^*} \bar{S}_a(t) dt$. The population median M_a solves $\bar{S}_a(M_a) = 1/2$; the median-survival ratio is M_{trt}/M_1 . This median of a mixture generally differs from the mean or median of the individual conditional medians. When the true log-time treatment shift is constant and residual distributions are identical between trial arms, the true ratio is $\exp(\text{effect}) = 1.4$.

The simulation analysis uses root-finding with a lower bound 0.05 and an upper cap 30, replacing roots beyond those limits by the relevant bound. These bounds truncate extreme posterior medians. The primary lrcBART median calculations use each arm's own fitted residual standard deviation, with $\sigma_1 = \sigma_2$ under the single-arm equal-variance assumption.

D.4 Restricted mean survival time

For $T \sim \text{lognormal}(\eta, \sigma^2)$,

$$\begin{aligned} \mathbb{E}\{\min(T, t^*)\} &= \mathbb{E}\{T \mathbf{1}(T \leq t^*)\} + t^* P(T > t^*) \\ &= \exp(\eta + \sigma^2/2) \Phi\left(\frac{\log t^* - \eta - \sigma^2}{\sigma}\right) + t^* \Phi\left(\frac{\eta - \log t^*}{\sigma}\right). \end{aligned} \quad (\text{D.4})$$

The first term follows by completing the square in the truncated lognormal first moment. Averaging (D.4) over trial standardization profiles gives population RMST at each draw. The application uses this analytic expression with $t^* = 5$ years and each arm's residual standard deviation; ratio and difference intervals are obtained from their posterior draws.

The additional simulation RMST summaries use a 120-point equally spaced grid from 0.001 to the restriction horizon and multiply the average survival probability over that grid by the horizon. The horizons are 3 years for the two-arm study and 5 years for the single-arm study. The true per-replicate RMST ratio is calculated under the data-generating model at the same trial standardization profiles and horizon. Unlike the median ratio under a common log-time shift, this restricted ratio generally varies by dataset.

D.5 Supplementary simulation RMST functionals

For both the two-arm and single-arm simulation summaries, each arm's RMST is calculated using that arm's own fitted residual standard deviation. Thus the control curve at posterior draw d uses $\sigma_1^{(d)}$, while the treatment curve uses $\sigma_{\text{trt}}^{(d)}$; no residual-scale substitution is applied.

Simulation ratios replace each control RMST denominator by its maximum with 0.05, while arm-specific summaries use the unfloored values. Tables S5 and S7 report these denominator-floored ratios. Their errors are evaluated against the data-generating RMST ratio at the same profiles and horizon. These supplementary operating characteristics therefore include the effect of the denominator floor.

E Separate-treatment simulation design and results

All experiments in this appendix use independently fitted treatment and control models. They are distinct from the primary joint-model experiment in Appendix G, with different implementations and replicate counts.

In the source labels used here, group 1 includes concurrent controls only in the control likelihood. Sc4 denotes a global shift and Sc5 a regional shift throughout this appendix.

E.1 Covariates, selection and scenario definitions

Each setting has ten observed covariates. Here X_j denotes scalar predictor j , whereas the bold $\mathbf{X}_a$ in the model denotes the covariate collection for group a . For a profile x , x_j denotes coordinate j . Before source selection, the ordinary covariates are independent normals with unit variance and mean vector

$$(1, -1, 0.5, -1, 2, -2, 2, -3, 1, 0). \quad (\text{E.1})$$

With probability 0.15, X_1 is instead drawn from an equally weighted mixture of $N(-3, 0.5^2)$ and $N(3, 0.5^2)$. Separate trial and historical candidate pools are generated. In Sc1, Sc4 and Sc5, source eligibility is independent of covariates. In Sc2, the trial-eligibility probability is

$$\text{logit}^{-1} \left\{ \frac{b_D(X_5 + X_6) - q}{t_D} \right\}, \quad (\text{E.2})$$

where q is the empirical quantile of the pool's selection score at the intended trial fraction. Two-arm studies use $b_D = -1.20, t_D = 0.5$; single-arm studies use $b_D = -0.55, t_D = 1$. Thus the single-arm and two-arm selection strengths differ. Exact numbers of eligible trial and historical observations are then sampled from their respective pools. In two-arm trials, treatment is assigned independently with probability $2/3$ for Gaussian outcomes and $3/4$ for survival outcomes.

Sc3 is included only in the two-arm studies. Latent U_5, U_6 replace X_5, X_6 in both the outcome function and selection score. They are generated from the corresponding normal distributions. The observed proxies are $X_j = \rho U_j^{\text{std}} + \sqrt{1 - \rho^2} e_j$, where U_j^{std} is standardized within its candidate pool and $e_j \sim N(0, 1)$ independently. We use $\rho = -0.5, 0, 0.5$. Conditioning on these proxies generally does not remove source differences induced by latent-variable selection.

Sc4 adds $\delta \in \{1, 2\}$ to every historical Gaussian outcome or latent log event time. Sc5 adds δ only outside a compatible region: $X_5 > 2$ with $\delta \in \{0.5, 1, 2\}$, or $X_7 \leq 2$ with $\delta \in \{1, 2\}$. The true trial-minus-historical discrepancy is zero inside that region and $-\delta$ outside.

Table S3: Design-specific data-generating constants.

Design	b_0 : Sc1	b_0 : Sc2/3	$\sigma_1 = \sigma_{\text{trt}}$	σ_2	Censoring
Gaussian, two-arm	0.42	-0.08	1.5	1.8	None
Survival, two-arm	-0.20	-0.70	1.4	1.8	$C_d \sim \text{Exp}(0.02);$ $C_a \sim U(2, 3)$
Gaussian, single-arm	0.40	-0.10	1.5	1.5	None
Survival, single-arm	0.20	0.10	1.25	1.25	$C_d \sim \text{Exp}(0.09);$ $C_a \sim U(5, 5.5)$

Sc4 and Sc5 use the Sc1 intercept before adding the historical shift. Sc3 is not run in the single-arm studies. The survival constants apply on the log-time scale, with independent random dropout C_d and administrative censoring C_a . Censoring rates are generated by the outcome and censoring distributions; they are not fixed percentages.

E.2 Outcome functions and censoring

The common nonlinear mean function, before a treatment or historical shift, is

$$\begin{aligned}
 F(x) := & b_0 + \beta_1 x_1^2 + \beta_2 \exp\{-(x_2 + 1)^2/2\} + \beta_3 |x_3 - 0.5| + \beta_4 |x_4 + 1| \\
 & + \beta_5 \tanh(x_5 - 2) + \beta_6 \tanh(x_6 + 2) \\
 & + 2 \left\{ \frac{(x_5 - 2)^2}{1 + ((x_5 - 2)/1.5)^2} + \frac{(x_6 + 2)^2}{1 + ((x_6 + 2)/1.5)^2} \right\} + \sum_{j=7}^{10} \beta_j x_j,
 \end{aligned} \tag{E.3}$$

with $\beta = (-0.50, -0.75, -0.50, -0.50, -1.35, -0.80, -0.01, -0.01, -0.01, -0.01)$. In Sc3, substitute the latent variables in the fifth and sixth positions. For Gaussian outcomes add independent normal noise and a treatment effect of 1 for treated trial participants. For survival outcomes exponentiate the noisy log-time response, with treatment effect $\log(1.4)$. The effect is constant in the reported settings; no additional treatment-by-covariate interaction is generated.

Survival observations are $\min(T, C_d, C_a)$ with event indicator $\mathbf{1}(T \leq C_d, T \leq C_a)$. Two-arm administrative follow-up is generated as 3 minus a uniform entry time on $(0, 1)$; single-arm follow-up is 5.5 minus uniform entry on $(0, 0.5)$. Historical shifts are applied to latent event times before evaluating censoring. The historical candidate pool is regenerated independently for each replicate.

E.3 Truth and target population

For each dataset, true conditional means and control/treatment survival distributions are evaluated at that dataset's trial profiles. The Gaussian target is their average treatment-minus-control difference, which equals 1 under the constant-effect design. The survival primary target is the ratio of medians of the two trial-standardized population curves, equal to 1.4 under the generated common log-time shift. The RMST target uses the corresponding restricted integrals and varies with the profiles and restriction horizon. Squared errors are computed against the relevant dataset-specific truth before aggregation.

E.4 Comparator specifications and configuration coverage

The hierarchical comparator uses source-specific regression coefficients with a common estimated hypermean. Intercept and predictor-specific discrepancy-variance components have $\text{IG}(3/2, 3s_\tau^2/2)$ priors, where s_τ^2 is the scale listed in Table S4; it is not a probability or historical sample-size fraction. Hypermeans have diffuse normal priors. This comparator pools regression parameters through their prior rather than specifying a nonlinear discrepancy surface.

The MAP comparator uses the historical sample mean and its standard error in a study-level normal model, a normal location prior with standard deviation 10, and an inverse-gamma prior $\text{IG}(3/2, 3a/2)$ for its heterogeneity parameter, with a selected by the MAP prior ESS procedure. Its predictive prior is approximated by a normal mixture and updated using the trial-control sample mean and standard error. This study-level comparator does not use a BART outcome model. Its prior ESS procedure is distinct from the lrcBART definition. PSCL follows the propensity-score-integrated construction described by Chen et al. (2020).

There are 17 method/configuration combinations per Gaussian two-arm setting and 13 per survival two-arm setting. Each reported single-arm setting has eight combinations. All reported configurations have 100

Table S4: Methods and reporting configurations.

Method	Specification and included settings
LM / AFT	Parametric outcome regression. Two-arm versions fit trial controls only (-NP), pool controls (-CP), or add a source indicator (-PP). Single-arm -CP gives historical outcomes likelihood weight 1. Treated outcomes are fitted separately.
BART	Corresponding -NP, -CP, or -PP control fits; separate treated surface. Single-arm BART-CP uses historical controls without a discrepancy sum of trees.
Hierarchical LM / AFT	Source-specific intercepts and coefficients have common hypermeans and inverse-gamma hierarchical variances, with discrepancy-variance prior scale $s_{\tau}^2 = 0.25$ in two-arm studies and $s_{\tau}^2 = 0.05$ in single-arm studies.
PSCL	Gaussian two-arm only. Propensity-score estimation on the ten observed covariates, distance-based borrowing allocation with total borrowing parameter 100, and composite-likelihood outcome analysis.
MAP	Gaussian two-arm only. Normal study-level mean and standard-error summary of historical controls; MAP construction and normal-mixture approximation; requested ESS 100, 75 or 50. No covariate adjustment in this comparator.
lrcBART, two-arm	$H_f = 50, H_g = 5$, estimated w ; ESS targets 100 (primary), 75, 50, f90, f50, f25.
lrcBART, single-arm	$H_f = 50, H_g = 5$; fixed $w = 1$ with targets 100 (primary), f90 or s0min; fixed $w = 0.9$ with targets 100 or f90.

f90, f50 and f25 denote fractions of the dataset-specific calibration ceiling. s0min sets $s_0^2 = 10^{-6}$ without target selection. The single-arm target 100 is an absolute count at both survival trial sizes, including $n_{\text{trt}} = 30$.

completed replicates.

E.5 Performance measures and Monte Carlo uncertainty

Let ζ denote the estimand being evaluated: the Gaussian mean contrast, survival median ratio or RMST ratio. For point estimates $\hat{\zeta}_r$ and dataset-specific truths ζ_r , write $e_r := \hat{\zeta}_r - \zeta_r$ over $R = 100$ replicates. We calculate

$$\widehat{\text{Bias}} := R^{-1} \sum_r e_r, \quad \widehat{\text{RMSE}} := \sqrt{R^{-1} \sum_r e_r^2}. \quad (\text{E.4})$$

Gaussian point estimates use posterior means and survival ratios use posterior medians, with the corresponding point estimates used for likelihood-based methods. Coverage is the proportion of reported 95% intervals containing truth. The uncertainty column reports the mean posterior SD for Bayesian methods and the mean standard error for PSCL; interval width is averaged within each setting. These summaries differ from the empirical SD of point estimates across datasets.

For Bayesian benefit decisions, empirical power is the proportion satisfying posterior probability above 0.95 for an effect greater than 0.5 (Gaussian) or a ratio greater than 1 (survival). The PSCL decision instead requires its lower confidence limit to exceed the Gaussian threshold. These are different decision rules. Type I error was not evaluated, and the rules were not calibrated to a common demonstrated Type I error rate.

Monte Carlo standard errors are

$$\widehat{\text{SE}}(\widehat{\text{Bias}}) = \frac{s_e}{\sqrt{R}}, \quad \widehat{\text{SE}}(\widehat{\text{RMSE}}) \simeq \frac{s_{e^2}}{2\sqrt{R\widehat{\text{RMSE}}}}, \quad \widehat{\text{SE}}(\hat{p}) = \sqrt{\hat{p}(1-\hat{p})/R}. \quad (\text{E.5})$$

Here s_e and s_{e^2} are the sample standard deviations of the replicate errors and squared errors, and $\hat{p}$ is the corresponding empirical coverage or power proportion. The RMSE formula is a delta-method approximation. A plug-in proportion standard error of zero at an observed proportion of zero or one does not establish absence of uncertainty; exact binomial intervals would still have nonzero width. These errors describe each row, not a paired between-method contrast. Comparisons should account for the common datasets if formal uncertainty in performance differences is required.

E.6 Gaussian two-arm results

All 12 settings and 17 configurations are reported. Table S18 selects Sc1, Sc2, Sc3 with $\rho = 0$, Sc4 with $\delta = 1$, and Sc5 outside $X_5 > 2$ with $\delta = 1$.

Table S5: Complete Gaussian simulation results.

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
<i>Mean difference: Sc1</i>						
lrcBART: 100	0.003 (0.018)	0.174 (0.012)	0.173	0.676	0.950 (0.022)	0.890 (0.031)
lrcBART: 75	0.008 (0.017)	0.173 (0.011)	0.175	0.684	0.960 (0.020)	0.900 (0.030)
lrcBART: 50	0.001 (0.018)	0.175 (0.011)	0.177	0.694	0.950 (0.022)	0.890 (0.031)
lrcBART: f90	0.003 (0.017)	0.172 (0.011)	0.171	0.669	0.950 (0.022)	0.880 (0.032)
lrcBART: f50	0.004 (0.017)	0.174 (0.011)	0.174	0.681	0.940 (0.024)	0.900 (0.030)
lrcBART: f25	0.002 (0.018)	0.183 (0.012)	0.179	0.700	0.930 (0.026)	0.860 (0.035)
BART-NP	-0.010 (0.021)	0.209 (0.013)	0.206	0.810	0.970 (0.017)	0.730 (0.044)
BART-CP	0.009 (0.015)	0.153 (0.011)	0.146	0.570	0.950 (0.022)	0.980 (0.014)
BART-PP	-0.001 (0.020)	0.197 (0.013)	0.211	0.826	0.950 (0.022)	0.800 (0.040)
LM-NP	-0.019 (0.035)	0.348 (0.022)	0.335	1.310	0.930 (0.026)	0.390 (0.049)
LM-CP	0.015 (0.026)	0.262 (0.019)	0.241	0.946	0.940 (0.024)	0.710 (0.045)
LM-PP	-0.005 (0.035)	0.349 (0.022)	0.341	1.334	0.950 (0.022)	0.390 (0.049)
hier. LM	-0.002 (0.033)	0.329 (0.022)	0.321	1.258	0.940 (0.024)	0.450 (0.050)
PSCL	0.019 (0.029)	0.290 (0.019)	0.267	1.048	0.920 (0.027)	0.480 (0.050)
MAP (100)	0.023 (0.030)	0.302 (0.023)	0.302	1.183	0.940 (0.024)	0.530 (0.050)
MAP (75)	0.023 (0.031)	0.305 (0.022)	0.305	1.195	0.950 (0.022)	0.520 (0.050)
MAP (50)	0.019 (0.032)	0.315 (0.023)	0.315	1.234	0.950 (0.022)	0.490 (0.050)
<i>Mean difference: Sc2</i>						
lrcBART: 100	0.003 (0.019)	0.186 (0.014)	0.180	0.702	0.940 (0.024)	0.860 (0.035)
lrcBART: 75	0.007 (0.019)	0.187 (0.014)	0.180	0.706	0.940 (0.024)	0.890 (0.031)
lrcBART: 50	0.000 (0.018)	0.184 (0.014)	0.181	0.708	0.950 (0.022)	0.870 (0.034)
lrcBART: f90	0.005 (0.019)	0.188 (0.013)	0.182	0.708	0.940 (0.024)	0.860 (0.035)
lrcBART: f50	-0.000 (0.019)	0.192 (0.014)	0.185	0.725	0.950 (0.022)	0.830 (0.038)
lrcBART: f25	-0.005 (0.020)	0.198 (0.014)	0.187	0.731	0.940 (0.024)	0.820 (0.038)
BART-NP	-0.032 (0.021)	0.211 (0.016)	0.203	0.795	0.940 (0.024)	0.750 (0.043)
BART-CP	0.048 (0.018)	0.188 (0.015)	0.176	0.688	0.930 (0.026)	0.950 (0.022)
BART-PP	-0.004 (0.020)	0.200 (0.014)	0.211	0.826	0.970 (0.017)	0.740 (0.044)
LM-NP	-0.062 (0.035)	0.358 (0.025)	0.317	1.246	0.900 (0.030)	0.440 (0.050)
LM-CP	0.343 (0.031)	0.463 (0.028)	0.283	1.108	0.730 (0.044)	0.920 (0.027)
LM-PP	0.007 (0.036)	0.355 (0.027)	0.334	1.304	0.950 (0.022)	0.450 (0.050)
hier. LM	0.052 (0.036)	0.363 (0.026)	0.320	1.253	0.930 (0.026)	0.510 (0.050)

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Table S5 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
PSCL	0.007 (0.032)	0.317 (0.022)	0.292	1.145	0.910 (0.029)	0.390 (0.049)
MAP (100)	0.186 (0.042)	0.458 (0.030)	0.384	1.501	0.870 (0.034)	0.550 (0.050)
MAP (75)	0.168 (0.041)	0.444 (0.030)	0.380	1.482	0.860 (0.035)	0.530 (0.050)
MAP (50)	0.140 (0.041)	0.427 (0.029)	0.377	1.471	0.890 (0.031)	0.500 (0.050)
<i>Mean difference: Sc3, $\rho = -0.5$</i>						
lrcBART: 100	0.049 (0.027)	0.269 (0.018)	0.276	1.079	0.960 (0.020)	0.590 (0.049)
lrcBART: 75	0.054 (0.027)	0.273 (0.019)	0.274	1.070	0.950 (0.022)	0.630 (0.048)
lrcBART: 50	0.054 (0.027)	0.274 (0.018)	0.275	1.074	0.950 (0.022)	0.620 (0.049)
lrcBART: f90	0.046 (0.026)	0.267 (0.018)	0.274	1.073	0.960 (0.020)	0.620 (0.049)
lrcBART: f50	0.054 (0.027)	0.273 (0.019)	0.276	1.080	0.940 (0.024)	0.620 (0.049)
lrcBART: f25	0.057 (0.027)	0.271 (0.018)	0.275	1.066	0.960 (0.020)	0.650 (0.048)
BART-NP	0.007 (0.028)	0.276 (0.018)	0.284	1.116	0.950 (0.022)	0.530 (0.050)
BART-CP	0.844 (0.022)	0.872 (0.021)	0.205	0.800	0.010 (0.010)	1.000 (0.000)
BART-PP	0.043 (0.026)	0.260 (0.017)	0.274	1.073	0.980 (0.014)	0.650 (0.048)
LM-NP	-0.005 (0.035)	0.348 (0.023)	0.355	1.390	0.930 (0.026)	0.420 (0.049)
LM-CP	0.870 (0.029)	0.916 (0.026)	0.262	1.025	0.100 (0.030)	0.990 (0.010)
LM-PP	0.089 (0.034)	0.350 (0.022)	0.347	1.361	0.960 (0.020)	0.560 (0.050)
hier. LM	0.136 (0.035)	0.378 (0.023)	0.352	1.374	0.940 (0.024)	0.620 (0.049)
PSCL	0.687 (0.030)	0.747 (0.027)	0.280	1.097	0.320 (0.047)	0.990 (0.010)
MAP (100)	0.254 (0.040)	0.471 (0.029)	0.379	1.480	0.910 (0.029)	0.650 (0.048)
MAP (75)	0.236 (0.040)	0.459 (0.029)	0.375	1.462	0.900 (0.030)	0.640 (0.048)
MAP (50)	0.214 (0.039)	0.443 (0.028)	0.372	1.454	0.910 (0.029)	0.610 (0.049)
<i>Mean difference: Sc3, $\rho = 0$</i>						
lrcBART: 100	0.009 (0.032)	0.315 (0.025)	0.279	1.090	0.930 (0.026)	0.560 (0.050)
lrcBART: 75	0.010 (0.032)	0.315 (0.024)	0.280	1.096	0.940 (0.024)	0.550 (0.050)
lrcBART: 50	0.015 (0.032)	0.315 (0.025)	0.281	1.099	0.950 (0.022)	0.570 (0.050)
lrcBART: f90	0.007 (0.032)	0.317 (0.025)	0.279	1.087	0.950 (0.022)	0.550 (0.050)
lrcBART: f50	0.009 (0.032)	0.314 (0.025)	0.281	1.098	0.930 (0.026)	0.550 (0.050)
lrcBART: f25	0.015 (0.032)	0.316 (0.025)	0.280	1.098	0.940 (0.024)	0.530 (0.050)
BART-NP	-0.046 (0.033)	0.328 (0.025)	0.287	1.125	0.950 (0.022)	0.520 (0.050)
BART-CP	0.957 (0.023)	0.985 (0.024)	0.202	0.789	0.010 (0.010)	1.000 (0.000)
BART-PP	0.004 (0.031)	0.309 (0.024)	0.276	1.080	0.940 (0.024)	0.550 (0.050)
LM-NP	-0.057 (0.036)	0.367 (0.030)	0.355	1.387	0.930 (0.026)	0.340 (0.047)
LM-CP	0.989 (0.028)	1.027 (0.028)	0.254	0.993	0.050 (0.022)	1.000 (0.000)
LM-PP	0.047 (0.035)	0.350 (0.025)	0.350	1.366	0.940 (0.024)	0.500 (0.050)
hier. LM	0.085 (0.036)	0.373 (0.027)	0.358	1.399	0.930 (0.026)	0.510 (0.050)
PSCL	0.667 (0.028)	0.721 (0.026)	0.273	1.068	0.290 (0.045)	0.980 (0.014)

Continued on next page

Table S5 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
MAP (100)	0.202 (0.039)	0.434 (0.029)	0.376	1.464	0.910 (0.029)	0.610 (0.049)
MAP (75)	0.182 (0.038)	0.419 (0.028)	0.373	1.459	0.940 (0.024)	0.590 (0.049)
MAP (50)	0.162 (0.037)	0.401 (0.027)	0.370	1.446	0.940 (0.024)	0.590 (0.049)
<i>Mean difference: Sc3, $\rho = 0.5$</i>						
lrcBART: 100	-0.031 (0.027)	0.275 (0.019)	0.274	1.074	0.920 (0.027)	0.560 (0.050)
lrcBART: 75	-0.027 (0.027)	0.272 (0.020)	0.273	1.066	0.920 (0.027)	0.570 (0.050)
lrcBART: 50	-0.026 (0.027)	0.272 (0.020)	0.272	1.064	0.920 (0.027)	0.580 (0.049)
lrcBART: f90	-0.029 (0.027)	0.271 (0.019)	0.271	1.058	0.920 (0.027)	0.580 (0.049)
lrcBART: f50	-0.027 (0.028)	0.275 (0.020)	0.272	1.068	0.920 (0.027)	0.530 (0.050)
lrcBART: f25	-0.024 (0.027)	0.271 (0.019)	0.272	1.059	0.920 (0.027)	0.570 (0.050)
BART-NP	-0.079 (0.027)	0.280 (0.019)	0.281	1.101	0.920 (0.027)	0.470 (0.050)
BART-CP	0.772 (0.021)	0.799 (0.020)	0.205	0.803	0.050 (0.022)	1.000 (0.000)
BART-PP	-0.040 (0.027)	0.269 (0.020)	0.271	1.059	0.920 (0.027)	0.550 (0.050)
LM-NP	-0.098 (0.032)	0.331 (0.024)	0.350	1.370	0.950 (0.022)	0.310 (0.046)
LM-CP	0.806 (0.026)	0.845 (0.024)	0.265	1.037	0.160 (0.037)	1.000 (0.000)
LM-PP	-0.019 (0.033)	0.326 (0.022)	0.348	1.364	0.970 (0.017)	0.460 (0.050)
hier. LM	0.036 (0.033)	0.334 (0.020)	0.348	1.360	0.970 (0.017)	0.530 (0.050)
PSCL	0.578 (0.028)	0.641 (0.025)	0.281	1.100	0.450 (0.050)	0.970 (0.017)
MAP (100)	0.125 (0.041)	0.429 (0.025)	0.374	1.455	0.930 (0.026)	0.570 (0.050)
MAP (75)	0.106 (0.040)	0.410 (0.023)	0.371	1.445	0.940 (0.024)	0.570 (0.050)
MAP (50)	0.085 (0.039)	0.399 (0.023)	0.368	1.439	0.950 (0.022)	0.550 (0.050)
<i>Mean difference: Sc4, $\delta = 1$</i>						
lrcBART: 100	-0.007 (0.020)	0.201 (0.014)	0.192	0.751	0.940 (0.024)	0.810 (0.039)
lrcBART: 75	-0.012 (0.020)	0.197 (0.013)	0.193	0.754	0.940 (0.024)	0.790 (0.041)
lrcBART: 50	-0.014 (0.021)	0.208 (0.015)	0.193	0.751	0.930 (0.026)	0.780 (0.041)
lrcBART: f90	-0.007 (0.020)	0.198 (0.014)	0.193	0.755	0.930 (0.026)	0.820 (0.038)
lrcBART: f50	-0.013 (0.020)	0.201 (0.013)	0.192	0.749	0.930 (0.026)	0.810 (0.039)
lrcBART: f25	-0.015 (0.020)	0.201 (0.013)	0.193	0.753	0.940 (0.024)	0.780 (0.041)
BART-NP	-0.010 (0.021)	0.209 (0.013)	0.206	0.810	0.970 (0.017)	0.730 (0.044)
BART-CP	-0.689 (0.015)	0.706 (0.015)	0.147	0.576	0.000 (0.000)	0.000 (0.000)
BART-PP	-0.012 (0.020)	0.197 (0.013)	0.212	0.832	0.970 (0.017)	0.770 (0.042)
LM-NP	-0.019 (0.035)	0.348 (0.022)	0.335	1.310	0.930 (0.026)	0.390 (0.049)
LM-CP	-0.726 (0.026)	0.772 (0.026)	0.242	0.944	0.180 (0.038)	0.010 (0.010)
LM-PP	-0.054 (0.035)	0.352 (0.020)	0.341	1.334	0.960 (0.020)	0.350 (0.048)
hier. LM	-0.104 (0.034)	0.358 (0.021)	0.330	1.288	0.930 (0.026)	0.290 (0.045)
PSCL	-0.483 (0.029)	0.563 (0.026)	0.268	1.052	0.530 (0.050)	0.060 (0.024)
MAP (100)	-0.230 (0.041)	0.467 (0.025)	0.360	1.406	0.860 (0.035)	0.170 (0.038)

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Table S5 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
MAP (75)	-0.204 (0.040)	0.451 (0.026)	0.360	1.404	0.870 (0.034)	0.180 (0.038)
MAP (50)	-0.175 (0.040)	0.433 (0.025)	0.358	1.397	0.910 (0.029)	0.220 (0.041)
<i>Mean difference: Sc4, $\delta = 2$</i>						
lrcBART: 100	0.000 (0.020)	0.196 (0.012)	0.190	0.739	0.960 (0.020)	0.840 (0.037)
lrcBART: 75	0.001 (0.019)	0.194 (0.013)	0.190	0.740	0.930 (0.026)	0.840 (0.037)
lrcBART: 50	-0.005 (0.020)	0.196 (0.013)	0.191	0.745	0.950 (0.022)	0.830 (0.038)
lrcBART: f90	-0.005 (0.020)	0.197 (0.013)	0.190	0.742	0.950 (0.022)	0.820 (0.038)
lrcBART: f50	-0.005 (0.020)	0.196 (0.012)	0.191	0.743	0.960 (0.020)	0.830 (0.038)
lrcBART: f25	-0.009 (0.020)	0.197 (0.013)	0.191	0.745	0.950 (0.022)	0.820 (0.038)
BART-NP	-0.010 (0.021)	0.209 (0.013)	0.206	0.810	0.970 (0.017)	0.730 (0.044)
BART-CP	-1.393 (0.016)	1.402 (0.016)	0.153	0.597	0.000 (0.000)	0.000 (0.000)
BART-PP	-0.015 (0.019)	0.194 (0.013)	0.213	0.833	0.980 (0.014)	0.780 (0.041)
LM-NP	-0.019 (0.035)	0.348 (0.022)	0.335	1.310	0.930 (0.026)	0.390 (0.049)
LM-CP	-1.466 (0.026)	1.490 (0.026)	0.245	0.957	0.000 (0.000)	0.000 (0.000)
LM-PP	-0.097 (0.035)	0.358 (0.020)	0.342	1.333	0.940 (0.024)	0.340 (0.047)
hier. LM	-0.119 (0.036)	0.377 (0.022)	0.338	1.325	0.920 (0.027)	0.300 (0.046)
PSCL	-0.985 (0.029)	1.027 (0.028)	0.272	1.067	0.080 (0.027)	0.000 (0.000)
MAP (100)	-0.124 (0.040)	0.414 (0.025)	0.366	1.431	0.940 (0.024)	0.300 (0.046)
MAP (75)	-0.112 (0.040)	0.414 (0.025)	0.365	1.427	0.920 (0.027)	0.300 (0.046)
MAP (50)	-0.107 (0.039)	0.405 (0.025)	0.362	1.419	0.940 (0.024)	0.320 (0.047)
<i>Mean difference: Sc5, X_5, $\delta = 0.5$</i>						
lrcBART: 100	-0.048 (0.019)	0.191 (0.011)	0.181	0.705	0.940 (0.024)	0.850 (0.036)
lrcBART: 75	-0.046 (0.018)	0.186 (0.011)	0.181	0.710	0.950 (0.022)	0.810 (0.039)
lrcBART: 50	-0.037 (0.018)	0.187 (0.011)	0.183	0.714	0.960 (0.020)	0.840 (0.037)
lrcBART: f90	-0.050 (0.019)	0.193 (0.011)	0.180	0.705	0.950 (0.022)	0.810 (0.039)
lrcBART: f50	-0.041 (0.018)	0.186 (0.011)	0.182	0.710	0.980 (0.014)	0.800 (0.040)
lrcBART: f25	-0.033 (0.019)	0.188 (0.011)	0.184	0.719	0.980 (0.014)	0.810 (0.039)
BART-NP	-0.010 (0.021)	0.209 (0.013)	0.206	0.810	0.970 (0.017)	0.730 (0.044)
BART-CP	-0.165 (0.015)	0.223 (0.012)	0.146	0.570	0.800 (0.040)	0.720 (0.045)
BART-PP	-0.011 (0.019)	0.194 (0.012)	0.211	0.825	0.960 (0.020)	0.750 (0.043)
LM-NP	-0.019 (0.035)	0.348 (0.022)	0.335	1.310	0.930 (0.026)	0.390 (0.049)
LM-CP	-0.169 (0.026)	0.311 (0.022)	0.241	0.947	0.870 (0.034)	0.410 (0.049)
LM-PP	-0.017 (0.035)	0.346 (0.021)	0.344	1.347	0.950 (0.022)	0.360 (0.048)
hier. LM	-0.032 (0.033)	0.332 (0.021)	0.322	1.260	0.930 (0.026)	0.370 (0.048)
PSCL	-0.105 (0.029)	0.308 (0.020)	0.268	1.051	0.920 (0.027)	0.340 (0.047)
MAP (100)	-0.079 (0.031)	0.323 (0.023)	0.306	1.200	0.950 (0.022)	0.360 (0.048)
MAP (75)	-0.071 (0.032)	0.328 (0.024)	0.310	1.215	0.930 (0.026)	0.370 (0.048)

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Table S5 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
MAP (50)	-0.060 (0.033)	0.331 (0.024)	0.317	1.245	0.930 (0.026)	0.360 (0.048)
<i>Mean difference: Sc5, X_5, $\delta = 1$</i>						
lrcBART: 100	-0.050 (0.021)	0.214 (0.012)	0.196	0.766	0.930 (0.026)	0.690 (0.046)
lrcBART: 75	-0.051 (0.021)	0.211 (0.012)	0.195	0.759	0.920 (0.027)	0.720 (0.045)
lrcBART: 50	-0.047 (0.020)	0.207 (0.012)	0.193	0.751	0.950 (0.022)	0.750 (0.043)
lrcBART: f90	-0.055 (0.021)	0.218 (0.013)	0.196	0.764	0.940 (0.024)	0.730 (0.044)
lrcBART: f50	-0.046 (0.021)	0.210 (0.013)	0.194	0.756	0.950 (0.022)	0.720 (0.045)
lrcBART: f25	-0.044 (0.020)	0.204 (0.013)	0.192	0.751	0.950 (0.022)	0.730 (0.044)
BART-NP	-0.010 (0.021)	0.209 (0.013)	0.206	0.810	0.970 (0.017)	0.730 (0.044)
BART-CP	-0.334 (0.015)	0.366 (0.014)	0.147	0.573	0.370 (0.048)	0.350 (0.048)
BART-PP	-0.014 (0.020)	0.195 (0.012)	0.213	0.835	0.970 (0.017)	0.760 (0.043)
LM-NP	-0.019 (0.035)	0.348 (0.022)	0.335	1.310	0.930 (0.026)	0.390 (0.049)
LM-CP	-0.353 (0.026)	0.439 (0.025)	0.242	0.945	0.660 (0.047)	0.190 (0.039)
LM-PP	-0.030 (0.035)	0.348 (0.021)	0.344	1.345	0.970 (0.017)	0.370 (0.048)
hier. LM	-0.063 (0.033)	0.338 (0.021)	0.323	1.267	0.920 (0.027)	0.360 (0.048)
PSCL	-0.230 (0.029)	0.370 (0.022)	0.269	1.055	0.850 (0.036)	0.210 (0.041)
MAP (100)	-0.160 (0.034)	0.378 (0.024)	0.322	1.263	0.920 (0.027)	0.270 (0.044)
MAP (75)	-0.144 (0.034)	0.372 (0.024)	0.324	1.275	0.940 (0.024)	0.280 (0.045)
MAP (50)	-0.122 (0.035)	0.366 (0.024)	0.329	1.289	0.920 (0.027)	0.290 (0.045)
<i>Mean difference: Sc5, X_5, $\delta = 2$</i>						
lrcBART: 100	-0.013 (0.020)	0.201 (0.013)	0.190	0.741	0.940 (0.024)	0.820 (0.038)
lrcBART: 75	-0.007 (0.020)	0.198 (0.013)	0.190	0.744	0.930 (0.026)	0.840 (0.037)
lrcBART: 50	-0.013 (0.021)	0.209 (0.014)	0.192	0.751	0.950 (0.022)	0.800 (0.040)
lrcBART: f90	-0.011 (0.021)	0.205 (0.013)	0.191	0.747	0.920 (0.027)	0.800 (0.040)
lrcBART: f50	-0.015 (0.020)	0.201 (0.012)	0.190	0.744	0.960 (0.020)	0.790 (0.041)
lrcBART: f25	-0.008 (0.020)	0.200 (0.012)	0.190	0.739	0.950 (0.022)	0.830 (0.038)
BART-NP	-0.010 (0.021)	0.209 (0.013)	0.206	0.810	0.970 (0.017)	0.730 (0.044)
BART-CP	-0.679 (0.014)	0.694 (0.014)	0.150	0.585	0.010 (0.010)	0.000 (0.000)
BART-PP	-0.012 (0.020)	0.198 (0.012)	0.214	0.833	0.970 (0.017)	0.760 (0.043)
LM-NP	-0.019 (0.035)	0.348 (0.022)	0.335	1.310	0.930 (0.026)	0.390 (0.049)
LM-CP	-0.722 (0.026)	0.767 (0.026)	0.245	0.958	0.180 (0.038)	0.010 (0.010)
LM-PP	-0.049 (0.035)	0.348 (0.020)	0.350	1.366	0.950 (0.022)	0.360 (0.048)
hier. LM	-0.105 (0.034)	0.358 (0.021)	0.330	1.290	0.930 (0.026)	0.290 (0.045)
PSCL	-0.479 (0.029)	0.560 (0.026)	0.273	1.068	0.520 (0.050)	0.050 (0.022)
MAP (100)	-0.216 (0.040)	0.454 (0.025)	0.359	1.402	0.860 (0.035)	0.180 (0.038)
MAP (75)	-0.190 (0.040)	0.439 (0.025)	0.358	1.395	0.900 (0.030)	0.210 (0.041)
MAP (50)	-0.163 (0.039)	0.423 (0.025)	0.357	1.392	0.910 (0.029)	0.220 (0.041)

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Table S5 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
<i>Mean difference: $Sc5$, X_7, $\delta = 1$</i>						
lrcBART: 100	-0.047 (0.021)	0.217 (0.012)	0.196	0.762	0.920 (0.027)	0.720 (0.045)
lrcBART: 75	-0.041 (0.021)	0.213 (0.013)	0.194	0.752	0.920 (0.027)	0.710 (0.045)
lrcBART: 50	-0.040 (0.020)	0.208 (0.013)	0.194	0.753	0.940 (0.024)	0.780 (0.041)
lrcBART: f90	-0.047 (0.021)	0.219 (0.013)	0.197	0.767	0.930 (0.026)	0.720 (0.045)
lrcBART: f50	-0.037 (0.021)	0.214 (0.013)	0.194	0.757	0.930 (0.026)	0.750 (0.043)
lrcBART: f25	-0.028 (0.020)	0.202 (0.013)	0.192	0.754	0.940 (0.024)	0.790 (0.041)
BART-NP	-0.010 (0.021)	0.209 (0.013)	0.206	0.810	0.970 (0.017)	0.730 (0.044)
BART-CP	-0.336 (0.015)	0.369 (0.014)	0.147	0.574	0.370 (0.048)	0.320 (0.047)
BART-PP	-0.003 (0.020)	0.198 (0.013)	0.213	0.831	0.960 (0.020)	0.770 (0.042)
LM-NP	-0.019 (0.035)	0.348 (0.022)	0.335	1.310	0.930 (0.026)	0.390 (0.049)
LM-CP	-0.352 (0.026)	0.440 (0.025)	0.242	0.947	0.690 (0.046)	0.170 (0.038)
LM-PP	-0.025 (0.035)	0.349 (0.022)	0.342	1.342	0.940 (0.024)	0.360 (0.048)
hier. LM	-0.063 (0.033)	0.338 (0.021)	0.324	1.267	0.920 (0.027)	0.330 (0.047)
PSCL	-0.230 (0.029)	0.369 (0.023)	0.268	1.050	0.860 (0.035)	0.220 (0.041)
MAP (100)	-0.165 (0.034)	0.376 (0.024)	0.321	1.255	0.910 (0.029)	0.270 (0.044)
MAP (75)	-0.148 (0.034)	0.373 (0.024)	0.324	1.269	0.930 (0.026)	0.270 (0.044)
MAP (50)	-0.126 (0.035)	0.369 (0.024)	0.329	1.286	0.920 (0.027)	0.290 (0.045)

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Table S5 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
<i>Mean difference: Sc5, X_7, $\delta = 2$</i>						
lrcBART: 100	-0.006 (0.019)	0.193 (0.013)	0.188	0.735	0.950 (0.022)	0.830 (0.038)
lrcBART: 75	-0.002 (0.020)	0.196 (0.013)	0.189	0.738	0.930 (0.026)	0.860 (0.035)
lrcBART: 50	-0.007 (0.020)	0.198 (0.013)	0.190	0.743	0.930 (0.026)	0.830 (0.038)
lrcBART: f90	-0.003 (0.020)	0.194 (0.013)	0.189	0.739	0.950 (0.022)	0.830 (0.038)
lrcBART: f50	-0.003 (0.020)	0.194 (0.013)	0.190	0.743	0.940 (0.024)	0.840 (0.037)
lrcBART: f25	0.002 (0.020)	0.201 (0.014)	0.191	0.743	0.920 (0.027)	0.840 (0.037)
BART-NP	-0.010 (0.021)	0.209 (0.013)	0.206	0.810	0.970 (0.017)	0.730 (0.044)
BART-CP	-0.683 (0.016)	0.700 (0.015)	0.150	0.587	0.010 (0.010)	0.000 (0.000)
BART-PP	-0.006 (0.020)	0.196 (0.012)	0.214	0.835	0.970 (0.017)	0.770 (0.042)
LM-NP	-0.019 (0.035)	0.348 (0.022)	0.335	1.310	0.930 (0.026)	0.390 (0.049)
LM-CP	-0.718 (0.027)	0.766 (0.027)	0.245	0.955	0.190 (0.039)	0.000 (0.000)
LM-PP	-0.046 (0.035)	0.352 (0.022)	0.349	1.363	0.960 (0.020)	0.350 (0.048)
hier. LM	-0.103 (0.034)	0.357 (0.021)	0.330	1.292	0.910 (0.029)	0.300 (0.046)
PSCL	-0.478 (0.029)	0.559 (0.026)	0.270	1.059	0.560 (0.050)	0.070 (0.026)
MAP (100)	-0.220 (0.040)	0.457 (0.025)	0.360	1.405	0.860 (0.035)	0.170 (0.038)
MAP (75)	-0.202 (0.040)	0.445 (0.024)	0.358	1.395	0.870 (0.034)	0.190 (0.039)
MAP (50)	-0.171 (0.039)	0.428 (0.025)	0.358	1.395	0.900 (0.030)	0.210 (0.041)

Entries with parentheses are estimates (Monte Carlo standard errors). Post. SD is the mean posterior standard deviation, or the estimated standard error for PSCL; width is the mean 95% interval width. Coverage and empirical power are proportions; Bayesian decisions require posterior probability of benefit above 0.95, whereas PSCL requires its lower 95% confidence limit to exceed the benefit threshold. Each row has 100 replicates. Gaussian power uses a benefit threshold of 0.5; survival power uses a ratio threshold of 1. Targets 100, 75 and 50 are absolute counts; f90, f50 and f25 are fractions of the calibration ceiling. s0min fixes $s_0^2 = 10^{-6}$ without target selection. NP, CP and PP denote trial-only, completely pooled and source-adjusted control fits. In single-arm studies, CP uses all historical controls without a discrepancy term. The hierarchical discrepancy variance has prior $IG(3/2, 3s_\tau^2/2)$. The hierarchical comparator uses discrepancy-variance prior scale $s_\tau^2 = 0.25$.

E.7 Survival two-arm results

The primary median-survival ratio is followed by the supplementary three-year RMST ratio, using the modified functional in Appendix D.5.

Table S6: Complete Survival simulation results.

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
<i>Median-survival ratio: Sc1</i>						
lrcBART: 100	0.014 (0.029)	0.290 (0.037)	0.308	1.199	0.970 (0.017)	0.420 (0.049)
lrcBART: 75	0.018 (0.030)	0.295 (0.036)	0.309	1.207	0.980 (0.014)	0.440 (0.050)
lrcBART: 50	0.023 (0.031)	0.310 (0.037)	0.314	1.220	0.970 (0.017)	0.430 (0.050)
lrcBART: f90	0.014 (0.029)	0.289 (0.041)	0.300	1.175	0.970 (0.017)	0.450 (0.050)
lrcBART: f50	0.013 (0.029)	0.292 (0.038)	0.311	1.209	0.960 (0.020)	0.430 (0.050)
lrcBART: f25	0.017 (0.031)	0.308 (0.038)	0.315	1.227	0.970 (0.017)	0.410 (0.049)
BART-NP	0.048 (0.039)	0.392 (0.045)	0.435	1.695	0.980 (0.014)	0.330 (0.047)
BART-CP	0.010 (0.022)	0.218 (0.016)	0.228	0.889	0.950 (0.022)	0.660 (0.047)
BART-PP	-0.034 (0.032)	0.316 (0.032)	0.357	1.387	0.980 (0.014)	0.300 (0.046)
AFT-NP	-0.105 (0.044)	0.455 (0.053)	0.563	2.169	0.970 (0.017)	0.070 (0.026)
AFT-CP	0.027 (0.031)	0.307 (0.021)	0.393	1.522	1.000 (0.000)	0.330 (0.047)
AFT-PP	0.022 (0.042)	0.415 (0.038)	0.620	2.389	0.990 (0.010)	0.100 (0.030)
hier. AFT	0.050 (0.044)	0.443 (0.060)	0.600	2.314	0.990 (0.010)	0.140 (0.035)
<i>Median-survival ratio: Sc2</i>						
lrcBART: 100	0.058 (0.034)	0.341 (0.030)	0.325	1.261	0.960 (0.020)	0.480 (0.050)
lrcBART: 75	0.061 (0.034)	0.346 (0.032)	0.326	1.268	0.940 (0.024)	0.430 (0.050)
lrcBART: 50	0.057 (0.034)	0.347 (0.034)	0.324	1.265	0.960 (0.020)	0.470 (0.050)
lrcBART: f90	0.058 (0.034)	0.345 (0.031)	0.326	1.267	0.950 (0.022)	0.480 (0.050)
lrcBART: f50	0.047 (0.035)	0.348 (0.031)	0.329	1.279	0.960 (0.020)	0.460 (0.050)
lrcBART: f25	0.042 (0.036)	0.357 (0.035)	0.331	1.287	0.960 (0.020)	0.420 (0.049)
BART-NP	0.117 (0.043)	0.440 (0.060)	0.444	1.725	0.960 (0.020)	0.320 (0.047)
BART-CP	0.130 (0.028)	0.310 (0.028)	0.330	1.288	0.970 (0.017)	0.630 (0.048)
BART-PP	-0.019 (0.035)	0.349 (0.025)	0.370	1.431	0.970 (0.017)	0.280 (0.045)
AFT-NP	0.018 (0.053)	0.529 (0.049)	0.593	2.280	0.960 (0.020)	0.160 (0.037)
AFT-CP	1.202 (0.079)	1.435 (0.093)	0.906	3.518	0.570 (0.050)	0.920 (0.027)
AFT-PP	0.403 (0.071)	0.817 (0.083)	0.785	3.014	0.920 (0.027)	0.300 (0.046)
hier. AFT	0.417 (0.070)	0.815 (0.078)	0.765	2.944	0.930 (0.026)	0.390 (0.049)

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Table S6 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
<i>Median-survival ratio: Sc3, $p = -0.5$</i>						
lrcBART: 100	0.041 (0.053)	0.531 (0.074)	0.488	1.890	0.920 (0.027)	0.290 (0.045)
lrcBART: 75	0.044 (0.054)	0.535 (0.076)	0.486	1.873	0.920 (0.027)	0.330 (0.047)
lrcBART: 50	0.047 (0.053)	0.532 (0.077)	0.488	1.895	0.930 (0.026)	0.290 (0.045)
lrcBART: f90	0.043 (0.054)	0.542 (0.082)	0.489	1.885	0.910 (0.029)	0.280 (0.045)
lrcBART: f50	0.042 (0.053)	0.526 (0.075)	0.491	1.883	0.930 (0.026)	0.290 (0.045)
lrcBART: f25	0.044 (0.053)	0.524 (0.073)	0.489	1.890	0.930 (0.026)	0.310 (0.046)
BART-NP	0.036 (0.057)	0.566 (0.075)	0.548	2.114	0.930 (0.026)	0.210 (0.041)
BART-CP	2.120 (0.069)	2.228 (0.071)	0.790	3.069	0.000 (0.000)	1.000 (0.000)
BART-PP	0.029 (0.047)	0.472 (0.059)	0.476	1.836	0.930 (0.026)	0.240 (0.043)
AFT-NP	-0.028 (0.060)	0.601 (0.073)	0.633	2.433	0.940 (0.024)	0.150 (0.036)
AFT-CP	2.647 (0.107)	2.852 (0.116)	1.231	4.752	0.050 (0.022)	1.000 (0.000)
AFT-PP	0.463 (0.077)	0.894 (0.092)	0.831	3.188	0.920 (0.027)	0.390 (0.049)
hier. AFT	0.445 (0.082)	0.933 (0.102)	0.839	3.217	0.910 (0.029)	0.370 (0.048)
<i>Median-survival ratio: Sc3, $p = 0$</i>						
lrcBART: 100	0.076 (0.047)	0.472 (0.037)	0.521	2.012	0.970 (0.017)	0.290 (0.045)
lrcBART: 75	0.083 (0.048)	0.482 (0.038)	0.520	2.010	0.960 (0.020)	0.310 (0.046)
lrcBART: 50	0.081 (0.047)	0.479 (0.038)	0.522	2.018	0.970 (0.017)	0.290 (0.045)
lrcBART: f90	0.071 (0.047)	0.471 (0.038)	0.521	2.019	0.930 (0.026)	0.300 (0.046)
lrcBART: f50	0.079 (0.047)	0.471 (0.037)	0.521	2.003	0.970 (0.017)	0.290 (0.045)
lrcBART: f25	0.080 (0.047)	0.475 (0.038)	0.521	2.015	0.960 (0.020)	0.280 (0.045)
BART-NP	0.014 (0.050)	0.502 (0.037)	0.569	2.190	0.960 (0.020)	0.180 (0.038)
BART-CP	2.642 (0.094)	2.802 (0.104)	0.885	3.439	0.000 (0.000)	1.000 (0.000)
BART-PP	0.075 (0.044)	0.446 (0.033)	0.503	1.940	0.960 (0.020)	0.300 (0.046)
AFT-NP	-0.090 (0.053)	0.531 (0.034)	0.632	2.410	0.970 (0.017)	0.110 (0.031)
AFT-CP	3.187 (0.127)	3.429 (0.141)	1.331	5.143	0.010 (0.010)	1.000 (0.000)
AFT-PP	0.477 (0.069)	0.839 (0.064)	0.844	3.220	0.930 (0.026)	0.470 (0.050)
hier. AFT	0.415 (0.073)	0.836 (0.066)	0.870	3.330	0.920 (0.027)	0.350 (0.048)
<i>Median-survival ratio: Sc3, $p = 0.5$</i>						
lrcBART: 100	0.004 (0.047)	0.468 (0.040)	0.483	1.863	0.930 (0.026)	0.240 (0.043)
lrcBART: 75	0.014 (0.048)	0.475 (0.040)	0.477	1.853	0.930 (0.026)	0.260 (0.044)
lrcBART: 50	0.016 (0.048)	0.477 (0.040)	0.483	1.864	0.930 (0.026)	0.250 (0.043)

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Table S6 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
lrcBART: f90	0.009 (0.047)	0.469 (0.041)	0.481	1.860	0.930 (0.026)	0.270 (0.044)
lrcBART: f50	0.012 (0.047)	0.469 (0.040)	0.484	1.865	0.930 (0.026)	0.260 (0.044)
lrcBART: f25	0.012 (0.047)	0.469 (0.039)	0.481	1.860	0.930 (0.026)	0.290 (0.045)
BART-NP	-0.004 (0.052)	0.514 (0.036)	0.540	2.090	0.940 (0.024)	0.200 (0.040)
BART-CP	2.083 (0.079)	2.225 (0.084)	0.775	3.013	0.030 (0.017)	1.000 (0.000)
BART-PP	0.006 (0.045)	0.445 (0.034)	0.470	1.815	0.940 (0.024)	0.250 (0.043)
AFT-NP	-0.083 (0.054)	0.544 (0.037)	0.616	2.352	0.910 (0.029)	0.120 (0.032)
AFT-CP	2.601 (0.124)	2.878 (0.133)	1.220	4.716	0.070 (0.026)	1.000 (0.000)
AFT-PP	0.422 (0.072)	0.830 (0.065)	0.827	3.164	0.950 (0.022)	0.400 (0.049)
hier. AFT	0.391 (0.076)	0.849 (0.063)	0.830	3.184	0.930 (0.026)	0.350 (0.048)
<i>Median-survival ratio: Sc4, $\delta = 1$</i>						
lrcBART: 100	-0.007 (0.033)	0.332 (0.040)	0.329	1.277	0.940 (0.024)	0.370 (0.048)
lrcBART: 75	-0.005 (0.033)	0.332 (0.037)	0.330	1.279	0.960 (0.020)	0.370 (0.048)
lrcBART: 50	-0.013 (0.033)	0.331 (0.041)	0.326	1.264	0.950 (0.022)	0.380 (0.049)
lrcBART: f90	-0.006 (0.034)	0.335 (0.040)	0.327	1.271	0.950 (0.022)	0.400 (0.049)
lrcBART: f50	-0.007 (0.034)	0.335 (0.041)	0.328	1.279	0.940 (0.024)	0.370 (0.048)
lrcBART: f25	-0.013 (0.033)	0.332 (0.039)	0.331	1.287	0.940 (0.024)	0.360 (0.048)
BART-NP	0.048 (0.039)	0.392 (0.045)	0.435	1.695	0.980 (0.014)	0.330 (0.047)
BART-CP	-0.729 (0.010)	0.736 (0.009)	0.112	0.436	0.000 (0.000)	0.000 (0.000)
BART-PP	-0.080 (0.030)	0.311 (0.029)	0.346	1.342	0.960 (0.020)	0.230 (0.042)
AFT-NP	-0.105 (0.044)	0.455 (0.053)	0.563	2.169	0.970 (0.017)	0.070 (0.026)
AFT-CP	-0.768 (0.013)	0.780 (0.013)	0.180	0.697	0.080 (0.027)	0.000 (0.000)
AFT-PP	-0.212 (0.035)	0.405 (0.024)	0.526	2.015	0.960 (0.020)	0.030 (0.017)
hier. AFT	-0.159 (0.043)	0.459 (0.051)	0.545	2.094	0.970 (0.017)	0.060 (0.024)
<i>Median-survival ratio: Sc4, $\delta = 2$</i>						
lrcBART: 100	-0.005 (0.033)	0.325 (0.041)	0.326	1.272	0.960 (0.020)	0.340 (0.047)
lrcBART: 75	-0.006 (0.033)	0.327 (0.042)	0.326	1.265	0.960 (0.020)	0.350 (0.048)
lrcBART: 50	-0.002 (0.033)	0.330 (0.040)	0.329	1.281	0.970 (0.017)	0.400 (0.049)
lrcBART: f90	-0.007 (0.033)	0.327 (0.040)	0.329	1.277	0.970 (0.017)	0.360 (0.048)
lrcBART: f50	-0.006 (0.033)	0.331 (0.042)	0.328	1.274	0.960 (0.020)	0.380 (0.049)
lrcBART: f25	-0.002 (0.033)	0.327 (0.042)	0.329	1.282	0.980 (0.014)	0.370 (0.048)
BART-NP	0.048 (0.039)	0.392 (0.045)	0.435	1.695	0.980 (0.014)	0.330 (0.047)

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Table S6 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
BART-CP	-1.075 (0.005)	1.076 (0.005)	0.058	0.227	0.000 (0.000)	0.000 (0.000)
BART-PP	-0.098 (0.030)	0.317 (0.024)	0.340	1.315	0.960 (0.020)	0.230 (0.042)
AFT-NP	-0.105 (0.044)	0.455 (0.053)	0.563	2.169	0.970 (0.017)	0.070 (0.026)
AFT-CP	-1.123 (0.006)	1.124 (0.006)	0.083	0.322	0.000 (0.000)	0.000 (0.000)
AFT-PP	-0.434 (0.030)	0.524 (0.023)	0.430	1.643	0.890 (0.031)	0.010 (0.010)
hier. AFT	-0.187 (0.047)	0.502 (0.050)	0.560	2.145	0.950 (0.022)	0.050 (0.022)
<i>Median-survival ratio: Sc5, X_5, $\delta = 0.5$</i>						
lrcBART: 100	-0.033 (0.031)	0.313 (0.040)	0.312	1.210	0.950 (0.022)	0.310 (0.046)
lrcBART: 75	-0.020 (0.032)	0.316 (0.042)	0.311	1.202	0.960 (0.020)	0.370 (0.048)
lrcBART: 50	-0.019 (0.032)	0.320 (0.047)	0.316	1.228	0.940 (0.024)	0.330 (0.047)
lrcBART: f90	-0.043 (0.030)	0.306 (0.039)	0.304	1.179	0.960 (0.020)	0.350 (0.048)
lrcBART: f50	-0.026 (0.031)	0.306 (0.039)	0.308	1.193	0.950 (0.022)	0.350 (0.048)
lrcBART: f25	-0.017 (0.031)	0.310 (0.037)	0.314	1.221	0.970 (0.017)	0.390 (0.049)
BART-NP	0.048 (0.039)	0.392 (0.045)	0.435	1.695	0.980 (0.014)	0.330 (0.047)
BART-CP	-0.209 (0.019)	0.279 (0.013)	0.194	0.756	0.840 (0.037)	0.260 (0.044)
BART-PP	-0.050 (0.031)	0.310 (0.030)	0.352	1.370	0.970 (0.017)	0.290 (0.045)
AFT-NP	-0.105 (0.044)	0.455 (0.053)	0.563	2.169	0.970 (0.017)	0.070 (0.026)
AFT-CP	-0.228 (0.025)	0.338 (0.018)	0.326	1.260	0.890 (0.031)	0.110 (0.031)
AFT-PP	-0.040 (0.040)	0.402 (0.033)	0.597	2.288	1.000 (0.000)	0.100 (0.030)
hier. AFT	-0.013 (0.044)	0.434 (0.059)	0.575	2.206	0.980 (0.014)	0.100 (0.030)
<i>Median-survival ratio: Sc5, X_5, $\delta = 1$</i>						
lrcBART: 100	-0.054 (0.033)	0.331 (0.039)	0.320	1.245	0.930 (0.026)	0.330 (0.047)
lrcBART: 75	-0.044 (0.034)	0.339 (0.043)	0.321	1.244	0.940 (0.024)	0.330 (0.047)
lrcBART: 50	-0.036 (0.033)	0.329 (0.040)	0.322	1.247	0.950 (0.022)	0.330 (0.047)
lrcBART: f90	-0.057 (0.034)	0.341 (0.040)	0.321	1.247	0.930 (0.026)	0.330 (0.047)
lrcBART: f50	-0.051 (0.032)	0.327 (0.041)	0.320	1.241	0.940 (0.024)	0.320 (0.047)
lrcBART: f25	-0.034 (0.033)	0.335 (0.039)	0.324	1.259	0.950 (0.022)	0.330 (0.047)
BART-NP	0.048 (0.039)	0.392 (0.045)	0.435	1.695	0.980 (0.014)	0.330 (0.047)
BART-CP	-0.391 (0.016)	0.422 (0.014)	0.165	0.645	0.420 (0.049)	0.070 (0.026)
BART-PP	-0.067 (0.031)	0.316 (0.031)	0.352	1.367	0.960 (0.020)	0.240 (0.043)
AFT-NP	-0.105 (0.044)	0.455 (0.053)	0.563	2.169	0.970 (0.017)	0.070 (0.026)
AFT-CP	-0.441 (0.020)	0.484 (0.018)	0.269	1.040	0.700 (0.046)	0.000 (0.000)

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Table S6 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
AFT-PP	-0.116 (0.037)	0.385 (0.026)	0.573	2.193	1.000 (0.000)	0.060 (0.024)
hier. AFT	-0.074 (0.043)	0.433 (0.056)	0.562	2.160	0.980 (0.014)	0.080 (0.027)
<i>Median-survival ratio: Sc5, X₅, $\delta = 2$</i>						
lrcBART: 100	-0.034 (0.035)	0.348 (0.045)	0.333	1.293	0.970 (0.017)	0.340 (0.047)
lrcBART: 75	-0.035 (0.035)	0.348 (0.044)	0.334	1.294	0.950 (0.022)	0.340 (0.047)
lrcBART: 50	-0.036 (0.035)	0.347 (0.048)	0.332	1.296	0.950 (0.022)	0.330 (0.047)
lrcBART: f90	-0.039 (0.035)	0.347 (0.040)	0.333	1.294	0.950 (0.022)	0.310 (0.046)
lrcBART: f50	-0.043 (0.034)	0.342 (0.045)	0.331	1.288	0.960 (0.020)	0.340 (0.047)
lrcBART: f25	-0.042 (0.034)	0.344 (0.042)	0.328	1.277	0.940 (0.024)	0.330 (0.047)
BART-NP	0.048 (0.039)	0.392 (0.045)	0.435	1.695	0.980 (0.014)	0.330 (0.047)
BART-CP	-0.659 (0.012)	0.670 (0.011)	0.126	0.494	0.010 (0.010)	0.000 (0.000)
BART-PP	-0.086 (0.032)	0.326 (0.032)	0.349	1.357	0.970 (0.017)	0.230 (0.042)
AFT-NP	-0.105 (0.044)	0.455 (0.053)	0.563	2.169	0.970 (0.017)	0.070 (0.026)
AFT-CP	-0.741 (0.014)	0.754 (0.014)	0.190	0.738	0.170 (0.038)	0.000 (0.000)
AFT-PP	-0.244 (0.033)	0.413 (0.023)	0.521	1.993	0.950 (0.022)	0.040 (0.020)
hier. AFT	-0.156 (0.043)	0.457 (0.050)	0.545	2.084	0.970 (0.017)	0.050 (0.022)
<i>Median-survival ratio: Sc5, X₇, $\delta = 1$</i>						
lrcBART: 100	-0.041 (0.034)	0.338 (0.037)	0.323	1.256	0.940 (0.024)	0.300 (0.046)
lrcBART: 75	-0.028 (0.035)	0.345 (0.038)	0.327	1.264	0.940 (0.024)	0.330 (0.047)
lrcBART: 50	-0.031 (0.033)	0.333 (0.039)	0.325	1.262	0.960 (0.020)	0.330 (0.047)
lrcBART: f90	-0.052 (0.034)	0.340 (0.032)	0.325	1.262	0.940 (0.024)	0.300 (0.046)
lrcBART: f50	-0.034 (0.034)	0.339 (0.037)	0.326	1.269	0.920 (0.027)	0.310 (0.046)
lrcBART: f25	-0.031 (0.033)	0.331 (0.036)	0.322	1.255	0.970 (0.017)	0.330 (0.047)
BART-NP	0.048 (0.039)	0.392 (0.045)	0.435	1.695	0.980 (0.014)	0.330 (0.047)
BART-CP	-0.428 (0.014)	0.451 (0.012)	0.160	0.623	0.340 (0.047)	0.030 (0.017)
BART-PP	-0.064 (0.032)	0.320 (0.032)	0.353	1.370	0.970 (0.017)	0.260 (0.044)
AFT-NP	-0.105 (0.044)	0.455 (0.053)	0.563	2.169	0.970 (0.017)	0.070 (0.026)
AFT-CP	-0.454 (0.020)	0.495 (0.018)	0.265	1.029	0.700 (0.046)	0.000 (0.000)
AFT-PP	-0.101 (0.038)	0.395 (0.029)	0.571	2.188	1.000 (0.000)	0.080 (0.027)
hier. AFT	-0.073 (0.043)	0.430 (0.055)	0.559	2.146	0.980 (0.014)	0.100 (0.030)
<i>Median-survival ratio: Sc5, X₇, $\delta = 2$</i>						
lrcBART: 100	-0.016 (0.033)	0.331 (0.033)	0.332	1.288	0.930 (0.026)	0.340 (0.047)

Continued on next page

Table S6 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
lrcBART: 75	-0.018 (0.033)	0.333 (0.038)	0.330	1.281	0.960 (0.020)	0.350 (0.048)
lrcBART: 50	-0.017 (0.033)	0.330 (0.036)	0.331	1.288	0.950 (0.022)	0.350 (0.048)
lrcBART: f90	-0.010 (0.034)	0.339 (0.039)	0.330	1.285	0.950 (0.022)	0.370 (0.048)
lrcBART: f50	-0.018 (0.034)	0.338 (0.041)	0.330	1.280	0.940 (0.024)	0.350 (0.048)
lrcBART: f25	-0.020 (0.035)	0.346 (0.041)	0.330	1.282	0.970 (0.017)	0.340 (0.047)
BART-NP	0.048 (0.039)	0.392 (0.045)	0.435	1.695	0.980 (0.014)	0.330 (0.047)
BART-CP	-0.718 (0.010)	0.725 (0.010)	0.115	0.450	0.000 (0.000)	0.000 (0.000)
BART-PP	-0.083 (0.031)	0.318 (0.028)	0.352	1.371	0.970 (0.017)	0.240 (0.043)
AFT-NP	-0.105 (0.044)	0.455 (0.053)	0.563	2.169	0.970 (0.017)	0.070 (0.026)
AFT-CP	-0.769 (0.013)	0.780 (0.013)	0.182	0.705	0.110 (0.031)	0.000 (0.000)
AFT-PP	-0.234 (0.035)	0.424 (0.025)	0.521	1.995	0.940 (0.024)	0.030 (0.017)
hier. AFT	-0.158 (0.044)	0.461 (0.052)	0.547	2.098	0.960 (0.020)	0.060 (0.024)
<i>Modified RMST ratio: Sc1</i>						
lrcBART: 100	0.003 (0.009)	0.087 (0.007)	0.097	0.378	0.960 (0.020)	0.340 (0.047)
lrcBART: 75	0.004 (0.009)	0.088 (0.007)	0.097	0.379	0.970 (0.017)	0.360 (0.048)
lrcBART: 50	0.005 (0.009)	0.092 (0.007)	0.098	0.384	0.960 (0.020)	0.370 (0.048)
lrcBART: f90	0.003 (0.009)	0.087 (0.008)	0.094	0.371	0.970 (0.017)	0.390 (0.049)
lrcBART: f50	0.001 (0.009)	0.088 (0.008)	0.097	0.379	0.960 (0.020)	0.340 (0.047)
lrcBART: f25	0.003 (0.009)	0.091 (0.008)	0.099	0.386	0.960 (0.020)	0.350 (0.048)
BART-NP	0.082 (0.014)	0.159 (0.013)	0.169	0.661	0.950 (0.022)	0.460 (0.050)
BART-CP	-0.004 (0.007)	0.071 (0.005)	0.072	0.280	0.950 (0.022)	0.520 (0.050)
BART-PP	-0.016 (0.010)	0.096 (0.007)	0.111	0.432	0.980 (0.014)	0.240 (0.043)
AFT-NP	-0.052 (0.011)	0.124 (0.009)	0.138	0.538	0.960 (0.020)	0.080 (0.027)
AFT-CP	-0.011 (0.008)	0.078 (0.005)	0.090	0.352	0.970 (0.017)	0.350 (0.048)
AFT-PP	-0.015 (0.010)	0.103 (0.007)	0.140	0.548	1.000 (0.000)	0.100 (0.030)
hier. AFT	-0.002 (0.011)	0.110 (0.010)	0.143	0.562	0.990 (0.010)	0.130 (0.034)
<i>Modified RMST ratio: Sc2</i>						
lrcBART: 100	0.026 (0.010)	0.107 (0.010)	0.097	0.378	0.930 (0.026)	0.470 (0.050)
lrcBART: 75	0.027 (0.011)	0.108 (0.010)	0.097	0.377	0.940 (0.024)	0.460 (0.050)
lrcBART: 50	0.026 (0.011)	0.110 (0.011)	0.096	0.379	0.940 (0.024)	0.470 (0.050)
lrcBART: f90	0.026 (0.010)	0.107 (0.009)	0.097	0.377	0.930 (0.026)	0.500 (0.050)
lrcBART: f50	0.023 (0.011)	0.109 (0.010)	0.097	0.380	0.930 (0.026)	0.450 (0.050)

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Table S6 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
lrcBART: f25	0.020 (0.011)	0.109 (0.011)	0.097	0.382	0.940 (0.024)	0.410 (0.049)
BART-NP	0.093 (0.015)	0.174 (0.018)	0.153	0.598	0.910 (0.029)	0.460 (0.050)
BART-CP	0.033 (0.009)	0.094 (0.009)	0.096	0.374	0.960 (0.020)	0.550 (0.050)
BART-PP	-0.005 (0.011)	0.109 (0.008)	0.107	0.419	0.940 (0.024)	0.280 (0.045)
AFT-NP	-0.012 (0.012)	0.122 (0.009)	0.132	0.516	0.980 (0.014)	0.170 (0.038)
AFT-CP	0.278 (0.014)	0.313 (0.016)	0.157	0.614	0.460 (0.050)	0.970 (0.017)
AFT-PP	0.101 (0.015)	0.184 (0.016)	0.161	0.626	0.880 (0.032)	0.440 (0.050)
hier. AFT	0.084 (0.015)	0.168 (0.015)	0.153	0.597	0.930 (0.026)	0.410 (0.049)
<i>Modified RMST ratio: Sc3, $\rho = -0.5$</i>						
lrcBART: 100	0.014 (0.015)	0.152 (0.019)	0.135	0.528	0.930 (0.026)	0.250 (0.043)
lrcBART: 75	0.014 (0.015)	0.151 (0.018)	0.134	0.521	0.920 (0.027)	0.300 (0.046)
lrcBART: 50	0.015 (0.015)	0.152 (0.020)	0.135	0.527	0.920 (0.027)	0.250 (0.043)
lrcBART: f90	0.014 (0.015)	0.152 (0.019)	0.135	0.527	0.920 (0.027)	0.250 (0.043)
lrcBART: f50	0.014 (0.015)	0.149 (0.019)	0.135	0.527	0.940 (0.024)	0.270 (0.044)
lrcBART: f25	0.015 (0.015)	0.149 (0.017)	0.135	0.527	0.940 (0.024)	0.260 (0.044)
BART-NP	0.013 (0.016)	0.157 (0.015)	0.152	0.595	0.950 (0.022)	0.230 (0.042)
BART-CP	0.490 (0.014)	0.509 (0.015)	0.146	0.572	0.000 (0.000)	1.000 (0.000)
BART-PP	0.006 (0.013)	0.131 (0.014)	0.128	0.501	0.930 (0.026)	0.260 (0.044)
AFT-NP	-0.035 (0.013)	0.135 (0.011)	0.135	0.527	0.920 (0.027)	0.130 (0.034)
AFT-CP	0.480 (0.016)	0.504 (0.016)	0.156	0.609	0.060 (0.024)	1.000 (0.000)
AFT-PP	0.092 (0.016)	0.186 (0.016)	0.155	0.607	0.930 (0.026)	0.450 (0.050)
hier. AFT	0.078 (0.016)	0.180 (0.016)	0.160	0.624	0.940 (0.024)	0.370 (0.048)
<i>Modified RMST ratio: Sc3, $\rho = 0$</i>						
lrcBART: 100	0.021 (0.013)	0.136 (0.009)	0.138	0.538	0.960 (0.020)	0.290 (0.045)
lrcBART: 75	0.024 (0.014)	0.139 (0.009)	0.138	0.541	0.960 (0.020)	0.300 (0.046)
lrcBART: 50	0.023 (0.014)	0.138 (0.009)	0.138	0.540	0.970 (0.017)	0.300 (0.046)
lrcBART: f90	0.020 (0.014)	0.136 (0.009)	0.137	0.537	0.970 (0.017)	0.280 (0.045)
lrcBART: f50	0.022 (0.013)	0.136 (0.009)	0.138	0.537	0.960 (0.020)	0.290 (0.045)
lrcBART: f25	0.023 (0.014)	0.137 (0.009)	0.138	0.539	0.960 (0.020)	0.290 (0.045)
BART-NP	0.006 (0.015)	0.146 (0.009)	0.155	0.606	0.980 (0.014)	0.180 (0.038)
BART-CP	0.583 (0.016)	0.604 (0.016)	0.148	0.579	0.000 (0.000)	1.000 (0.000)
BART-PP	0.019 (0.012)	0.122 (0.007)	0.131	0.508	0.980 (0.014)	0.320 (0.047)

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Table S6 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
AFT-NP	-0.054 (0.012)	0.134 (0.007)	0.132	0.516	0.950 (0.022)	0.110 (0.031)
AFT-CP	0.549 (0.016)	0.570 (0.016)	0.153	0.598	0.010 (0.010)	1.000 (0.000)
AFT-PP	0.092 (0.014)	0.166 (0.011)	0.155	0.604	0.930 (0.026)	0.510 (0.050)
hier. AFT	0.062 (0.015)	0.161 (0.011)	0.158	0.614	0.940 (0.024)	0.360 (0.048)
<i>Modified RMST ratio: Sc3, $\rho = 0.5$</i>						
lrcBART: 100	-0.001 (0.014)	0.134 (0.011)	0.133	0.518	0.920 (0.027)	0.240 (0.043)
lrcBART: 75	0.003 (0.014)	0.136 (0.010)	0.132	0.516	0.930 (0.026)	0.260 (0.044)
lrcBART: 50	0.003 (0.014)	0.137 (0.010)	0.133	0.518	0.920 (0.027)	0.290 (0.045)
lrcBART: f90	0.002 (0.014)	0.135 (0.011)	0.133	0.517	0.930 (0.026)	0.260 (0.044)
lrcBART: f50	0.002 (0.014)	0.135 (0.010)	0.133	0.517	0.920 (0.027)	0.240 (0.043)
lrcBART: f25	0.002 (0.014)	0.135 (0.010)	0.133	0.519	0.940 (0.024)	0.270 (0.044)
BART-NP	0.004 (0.016)	0.155 (0.012)	0.153	0.601	0.970 (0.017)	0.200 (0.040)
BART-CP	0.478 (0.014)	0.499 (0.014)	0.143	0.563	0.030 (0.017)	1.000 (0.000)
BART-PP	-0.004 (0.012)	0.124 (0.008)	0.127	0.496	0.940 (0.024)	0.250 (0.043)
AFT-NP	-0.052 (0.013)	0.141 (0.010)	0.132	0.517	0.900 (0.030)	0.130 (0.034)
AFT-CP	0.458 (0.016)	0.486 (0.016)	0.153	0.597	0.070 (0.026)	1.000 (0.000)
AFT-PP	0.076 (0.015)	0.166 (0.011)	0.154	0.600	0.940 (0.024)	0.440 (0.050)
hier. AFT	0.060 (0.016)	0.172 (0.013)	0.157	0.615	0.940 (0.024)	0.380 (0.049)
<i>Modified RMST ratio: Sc4, $\delta = 1$</i>						
lrcBART: 100	-0.001 (0.010)	0.101 (0.008)	0.105	0.410	0.960 (0.020)	0.320 (0.047)
lrcBART: 75	-0.001 (0.010)	0.099 (0.008)	0.105	0.411	0.960 (0.020)	0.300 (0.046)
lrcBART: 50	-0.003 (0.010)	0.100 (0.008)	0.105	0.410	0.980 (0.014)	0.290 (0.045)
lrcBART: f90	-0.001 (0.010)	0.101 (0.008)	0.105	0.410	0.970 (0.017)	0.310 (0.046)
lrcBART: f50	-0.001 (0.010)	0.100 (0.008)	0.105	0.411	0.970 (0.017)	0.290 (0.045)
lrcBART: f25	-0.002 (0.010)	0.101 (0.008)	0.106	0.414	0.960 (0.020)	0.300 (0.046)
BART-NP	0.082 (0.014)	0.159 (0.013)	0.169	0.661	0.950 (0.022)	0.460 (0.050)
BART-CP	-0.259 (0.005)	0.263 (0.005)	0.051	0.200	0.000 (0.000)	0.000 (0.000)
BART-PP	-0.023 (0.009)	0.096 (0.007)	0.112	0.437	0.980 (0.014)	0.180 (0.038)
AFT-NP	-0.052 (0.011)	0.124 (0.009)	0.138	0.538	0.960 (0.020)	0.080 (0.027)
AFT-CP	-0.249 (0.006)	0.255 (0.006)	0.068	0.266	0.030 (0.017)	0.000 (0.000)
AFT-PP	-0.076 (0.009)	0.120 (0.007)	0.129	0.506	0.940 (0.024)	0.040 (0.020)
hier. AFT	-0.059 (0.011)	0.128 (0.009)	0.139	0.545	0.970 (0.017)	0.060 (0.024)

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Table S6 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
<i>Modified RMST ratio: Sc4, $\delta = 2$</i>						
lrcBART: 100	0.008 (0.010)	0.100 (0.009)	0.108	0.422	0.960 (0.020)	0.300 (0.046)
lrcBART: 75	0.008 (0.010)	0.102 (0.009)	0.108	0.417	0.960 (0.020)	0.300 (0.046)
lrcBART: 50	0.009 (0.010)	0.102 (0.008)	0.108	0.423	0.970 (0.017)	0.320 (0.047)
lrcBART: f90	0.007 (0.010)	0.101 (0.009)	0.108	0.420	0.950 (0.022)	0.320 (0.047)
lrcBART: f50	0.008 (0.010)	0.102 (0.009)	0.108	0.422	0.950 (0.022)	0.310 (0.046)
lrcBART: f25	0.009 (0.010)	0.100 (0.009)	0.108	0.422	0.970 (0.017)	0.330 (0.047)
BART-NP	0.082 (0.014)	0.159 (0.013)	0.169	0.661	0.950 (0.022)	0.460 (0.050)
BART-CP	-0.427 (0.004)	0.429 (0.004)	0.040	0.156	0.000 (0.000)	0.000 (0.000)
BART-PP	-0.017 (0.010)	0.099 (0.007)	0.115	0.451	0.980 (0.014)	0.220 (0.041)
AFT-NP	-0.052 (0.011)	0.124 (0.009)	0.138	0.538	0.960 (0.020)	0.080 (0.027)
AFT-CP	-0.407 (0.005)	0.410 (0.005)	0.055	0.216	0.000 (0.000)	0.000 (0.000)
AFT-PP	-0.140 (0.009)	0.166 (0.008)	0.119	0.463	0.850 (0.036)	0.010 (0.010)
hier. AFT	-0.069 (0.012)	0.141 (0.009)	0.145	0.564	0.930 (0.026)	0.060 (0.024)
<i>Modified RMST ratio: Sc5, X_5, $\delta = 0.5$</i>						
lrcBART: 100	-0.023 (0.009)	0.092 (0.007)	0.095	0.372	0.960 (0.020)	0.240 (0.043)
lrcBART: 75	-0.019 (0.009)	0.091 (0.007)	0.095	0.370	0.960 (0.020)	0.270 (0.044)
lrcBART: 50	-0.020 (0.009)	0.092 (0.008)	0.096	0.375	0.950 (0.022)	0.270 (0.044)
lrcBART: f90	-0.027 (0.009)	0.091 (0.007)	0.093	0.364	0.940 (0.024)	0.230 (0.042)
lrcBART: f50	-0.021 (0.009)	0.089 (0.007)	0.094	0.369	0.970 (0.017)	0.260 (0.044)
lrcBART: f25	-0.019 (0.009)	0.092 (0.007)	0.096	0.377	0.970 (0.017)	0.280 (0.045)
BART-NP	0.082 (0.014)	0.159 (0.013)	0.169	0.661	0.950 (0.022)	0.460 (0.050)
BART-CP	-0.080 (0.006)	0.102 (0.005)	0.065	0.252	0.750 (0.043)	0.180 (0.038)
BART-PP	-0.029 (0.009)	0.094 (0.006)	0.107	0.418	0.960 (0.020)	0.210 (0.041)
AFT-NP	-0.052 (0.011)	0.124 (0.009)	0.138	0.538	0.960 (0.020)	0.080 (0.027)
AFT-CP	-0.082 (0.007)	0.108 (0.006)	0.083	0.324	0.860 (0.035)	0.110 (0.031)
AFT-PP	-0.036 (0.010)	0.105 (0.006)	0.135	0.529	1.000 (0.000)	0.090 (0.029)
hier. AFT	-0.020 (0.011)	0.111 (0.010)	0.140	0.549	0.990 (0.010)	0.110 (0.031)
<i>Modified RMST ratio: Sc5, X_5, $\delta = 1$</i>						
lrcBART: 100	-0.041 (0.009)	0.100 (0.006)	0.096	0.374	0.900 (0.030)	0.200 (0.040)
lrcBART: 75	-0.039 (0.009)	0.100 (0.007)	0.096	0.373	0.920 (0.027)	0.190 (0.039)
lrcBART: 50	-0.038 (0.009)	0.098 (0.006)	0.096	0.374	0.940 (0.024)	0.190 (0.039)

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Table S6 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
lrcBART: f90	-0.043 (0.009)	0.103 (0.007)	0.096	0.374	0.900 (0.030)	0.180 (0.038)
lrcBART: f50	-0.041 (0.009)	0.098 (0.006)	0.095	0.372	0.930 (0.026)	0.190 (0.039)
lrcBART: f25	-0.035 (0.009)	0.099 (0.006)	0.096	0.375	0.940 (0.024)	0.190 (0.039)
BART-NP	0.082 (0.014)	0.159 (0.013)	0.169	0.661	0.950 (0.022)	0.460 (0.050)
BART-CP	-0.146 (0.006)	0.157 (0.005)	0.058	0.228	0.340 (0.047)	0.030 (0.017)
BART-PP	-0.042 (0.009)	0.098 (0.007)	0.105	0.412	0.950 (0.022)	0.130 (0.034)
AFT-NP	-0.052 (0.011)	0.124 (0.009)	0.138	0.538	0.960 (0.020)	0.080 (0.027)
AFT-CP	-0.145 (0.006)	0.159 (0.006)	0.077	0.299	0.570 (0.050)	0.000 (0.000)
AFT-PP	-0.060 (0.009)	0.110 (0.007)	0.131	0.510	0.990 (0.010)	0.070 (0.026)
hier. AFT	-0.037 (0.011)	0.115 (0.009)	0.139	0.543	0.990 (0.010)	0.080 (0.027)
<i>Modified RMST ratio: Sc5, X_5, $\delta = 2$</i>						
lrcBART: 100	-0.042 (0.010)	0.105 (0.007)	0.099	0.386	0.950 (0.022)	0.120 (0.032)
lrcBART: 75	-0.044 (0.009)	0.104 (0.007)	0.099	0.387	0.920 (0.027)	0.160 (0.037)
lrcBART: 50	-0.043 (0.009)	0.104 (0.007)	0.099	0.387	0.940 (0.024)	0.180 (0.038)
lrcBART: f90	-0.045 (0.009)	0.104 (0.007)	0.099	0.384	0.930 (0.026)	0.140 (0.035)
lrcBART: f50	-0.047 (0.009)	0.105 (0.007)	0.099	0.385	0.920 (0.027)	0.160 (0.037)
lrcBART: f25	-0.044 (0.009)	0.104 (0.007)	0.098	0.382	0.910 (0.029)	0.160 (0.037)
BART-NP	0.082 (0.014)	0.159 (0.013)	0.169	0.661	0.950 (0.022)	0.460 (0.050)
BART-CP	-0.247 (0.005)	0.252 (0.005)	0.051	0.197	0.000 (0.000)	0.000 (0.000)
BART-PP	-0.052 (0.009)	0.102 (0.007)	0.105	0.411	0.950 (0.022)	0.110 (0.031)
AFT-NP	-0.052 (0.011)	0.124 (0.009)	0.138	0.538	0.960 (0.020)	0.080 (0.027)
AFT-CP	-0.244 (0.006)	0.250 (0.006)	0.068	0.266	0.020 (0.014)	0.000 (0.000)
AFT-PP	-0.102 (0.009)	0.134 (0.007)	0.121	0.473	0.920 (0.027)	0.030 (0.017)
hier. AFT	-0.060 (0.011)	0.128 (0.009)	0.138	0.538	0.970 (0.017)	0.060 (0.024)
<i>Modified RMST ratio: Sc5, X_7, $\delta = 1$</i>						
lrcBART: 100	-0.019 (0.010)	0.100 (0.007)	0.102	0.398	0.940 (0.024)	0.250 (0.043)
lrcBART: 75	-0.015 (0.010)	0.102 (0.008)	0.102	0.397	0.930 (0.026)	0.240 (0.043)
lrcBART: 50	-0.015 (0.010)	0.099 (0.007)	0.102	0.398	0.930 (0.026)	0.250 (0.043)
lrcBART: f90	-0.022 (0.010)	0.104 (0.007)	0.102	0.399	0.930 (0.026)	0.260 (0.044)
lrcBART: f50	-0.016 (0.010)	0.102 (0.007)	0.102	0.397	0.940 (0.024)	0.270 (0.044)
lrcBART: f25	-0.016 (0.010)	0.099 (0.007)	0.101	0.395	0.940 (0.024)	0.260 (0.044)
BART-NP	0.082 (0.014)	0.159 (0.013)	0.169	0.661	0.950 (0.022)	0.460 (0.050)

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Table S6 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
BART-CP	-0.148 (0.005)	0.157 (0.005)	0.060	0.234	0.360 (0.048)	0.010 (0.010)
BART-PP	-0.026 (0.009)	0.098 (0.007)	0.110	0.432	0.960 (0.020)	0.210 (0.041)
AFT-NP	-0.052 (0.011)	0.124 (0.009)	0.138	0.538	0.960 (0.020)	0.080 (0.027)
AFT-CP	-0.145 (0.006)	0.158 (0.006)	0.077	0.302	0.620 (0.049)	0.000 (0.000)
AFT-PP	-0.048 (0.010)	0.110 (0.007)	0.134	0.525	0.980 (0.014)	0.080 (0.027)
hier. AFT	-0.035 (0.011)	0.115 (0.010)	0.139	0.543	0.980 (0.014)	0.090 (0.029)
<i>Modified RMST ratio: Sc5, X_7, $\delta = 2$</i>						
lrcBART: 100	-0.014 (0.010)	0.100 (0.007)	0.103	0.404	0.950 (0.022)	0.290 (0.045)
lrcBART: 75	-0.016 (0.010)	0.098 (0.008)	0.103	0.402	0.970 (0.017)	0.260 (0.044)
lrcBART: 50	-0.016 (0.010)	0.097 (0.007)	0.103	0.402	0.960 (0.020)	0.270 (0.044)
lrcBART: f90	-0.014 (0.010)	0.099 (0.008)	0.103	0.401	0.950 (0.022)	0.310 (0.046)
lrcBART: f50	-0.017 (0.010)	0.100 (0.008)	0.103	0.402	0.940 (0.024)	0.270 (0.044)
lrcBART: f25	-0.018 (0.010)	0.102 (0.008)	0.103	0.402	0.950 (0.022)	0.260 (0.044)
BART-NP	0.082 (0.014)	0.159 (0.013)	0.169	0.661	0.950 (0.022)	0.460 (0.050)
BART-CP	-0.259 (0.005)	0.264 (0.005)	0.051	0.200	0.000 (0.000)	0.000 (0.000)
BART-PP	-0.031 (0.009)	0.098 (0.007)	0.112	0.438	0.950 (0.022)	0.190 (0.039)
AFT-NP	-0.052 (0.011)	0.124 (0.009)	0.138	0.538	0.960 (0.020)	0.080 (0.027)
AFT-CP	-0.250 (0.006)	0.257 (0.006)	0.068	0.266	0.040 (0.020)	0.000 (0.000)
AFT-PP	-0.089 (0.010)	0.130 (0.008)	0.127	0.493	0.920 (0.027)	0.050 (0.022)
hier. AFT	-0.059 (0.011)	0.128 (0.009)	0.140	0.545	0.960 (0.020)	0.070 (0.026)

Entries with parentheses are estimates (Monte Carlo standard errors). Post. SD is the mean posterior standard deviation, or the estimated standard error for PSCL; width is the mean 95% interval width. Coverage and empirical power are proportions; Bayesian decisions require posterior probability of benefit above 0.95, whereas PSCL requires its lower 95% confidence limit to exceed the benefit threshold. Each row has 100 replicates. Gaussian power uses a benefit threshold of 0.5; survival power uses a ratio threshold of 1. Targets 100, 75 and 50 are absolute counts; f90, f50 and f25 are fractions of the calibration ceiling. s0min fixes $s_0^2 = 10^{-6}$ without target selection. NP, CP and PP denote trial-only, completely pooled and source-adjusted control fits. In single-arm studies, CP uses all historical controls without a discrepancy term. The hierarchical discrepancy variance has prior $IG(3/2, 3s_T^2/2)$. The hierarchical comparator uses discrepancy-variance prior scale $s_T^2 = 0.25$.

E.8 Gaussian single-arm results

Sc1 and Sc2 at $n_{\text{trt}} = 200$ include all five lrcBART priors and all three comparator settings.

Table S7: Complete Gaussian single arm simulation results.

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
<i>Mean difference: $n_{\text{trt}} = 200$, Sc1</i>						
lrcBART: 100	-0.015 (0.016)	0.156 (0.012)	0.176	0.689	0.950 (0.022)	0.890 (0.031)
lrcBART: f90	-0.012 (0.015)	0.153 (0.011)	0.163	0.636	0.960 (0.020)	0.920 (0.027)
lrcBART: s0min	-0.015 (0.016)	0.157 (0.012)	0.151	0.591	0.910 (0.029)	0.920 (0.027)
lrcBART: $w = 0.9$, 100	-0.010 (0.016)	0.155 (0.011)	1.289	6.036	1.000 (0.000)	0.000 (0.000)
lrcBART: $w = 0.9$, f90	-0.007 (0.016)	0.159 (0.010)	1.292	6.067	1.000 (0.000)	0.000 (0.000)
BART-CP	-0.014 (0.015)	0.153 (0.011)	0.150	0.588	0.940 (0.024)	0.940 (0.024)
LM-CP	-0.029 (0.027)	0.270 (0.017)	0.252	0.983	0.960 (0.020)	0.600 (0.049)
hier. LM	-0.025 (0.027)	0.272 (0.017)	0.576	2.210	1.000 (0.000)	0.090 (0.029)
<i>Mean difference: $n_{\text{trt}} = 200$, Sc2</i>						
lrcBART: 100	0.062 (0.017)	0.177 (0.010)	0.179	0.697	0.980 (0.014)	0.910 (0.029)
lrcBART: f90	0.064 (0.017)	0.183 (0.010)	0.184	0.714	0.970 (0.017)	0.910 (0.029)
lrcBART: s0min	0.061 (0.017)	0.177 (0.010)	0.168	0.653	0.980 (0.014)	0.970 (0.017)
lrcBART: $w = 0.9$, 100	0.065 (0.017)	0.183 (0.010)	1.262	5.920	1.000 (0.000)	0.000 (0.000)
lrcBART: $w = 0.9$, f90	0.065 (0.018)	0.189 (0.010)	1.268	5.950	1.000 (0.000)	0.000 (0.000)
BART-CP	0.056 (0.017)	0.176 (0.010)	0.168	0.657	0.930 (0.026)	0.950 (0.022)
LM-CP	0.290 (0.027)	0.395 (0.024)	0.273	1.067	0.850 (0.036)	0.910 (0.029)
hier. LM	0.357 (0.027)	0.444 (0.024)	0.588	2.243	1.000 (0.000)	0.500 (0.050)

Entries with parentheses are estimates (Monte Carlo standard errors). Post. SD is the mean posterior standard deviation, or the estimated standard error for PSCL; width is the mean 95% interval width. Coverage and empirical power are proportions; Bayesian decisions require posterior probability of benefit above 0.95, whereas PSCL requires its lower 95% confidence limit to exceed the benefit threshold. Each row has 100 replicates. Gaussian power uses a benefit threshold of 0.5; survival power uses a ratio threshold of 1. Targets 100, 75 and 50 are absolute counts; f90, f50 and f25 are fractions of the calibration ceiling. s0min fixes $s_0^2 = 10^{-6}$ without target selection. NP, CP and PP denote trial-only, completely pooled and source-adjusted control fits. In single-arm studies, CP uses all historical controls without a discrepancy term. The hierarchical discrepancy variance has prior $\text{IG}(3/2, 3s_\tau^2/2)$. The single-arm hierarchical comparator uses discrepancy-variance prior scale $s_\tau^2 = 0.05$.

E.9 Survival single-arm results

Both trial sizes and both scenarios are reported, first for median-survival ratios and then for five-year RMST ratios with floored control denominators (Appendix D.5).

Table S8: Complete Survival single arm simulation results.

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
<i>Median-survival ratio: $n_{\text{trt}} = 30$, Sc1</i>						
lrcBART: 100	0.007 (0.044)	0.440 (0.046)	0.877	3.177	0.980 (0.014)	0.050 (0.022)
lrcBART: f90	0.011 (0.044)	0.439 (0.047)	0.906	3.271	1.000 (0.000)	0.050 (0.022)
lrcBART: s0min	0.007 (0.044)	0.440 (0.046)	0.877	3.177	0.980 (0.014)	0.050 (0.022)
lrcBART: $w = 0.9$, 100	0.016 (0.044)	0.442 (0.047)	3.323	10.974	1.000 (0.000)	0.000 (0.000)
lrcBART: $w = 0.9$, f90	0.016 (0.045)	0.443 (0.049)	3.220	10.974	1.000 (0.000)	0.000 (0.000)
BART-CP	0.044 (0.043)	0.430 (0.045)	0.894	3.292	0.990 (0.010)	0.060 (0.024)
AFT-CP	0.127 (0.065)	0.662 (0.074)	0.876	3.282	0.950 (0.022)	0.150 (0.036)
hier. AFT	0.177 (0.068)	0.697 (0.083)	1.862	5.683	1.000 (0.000)	0.040 (0.020)
<i>Median-survival ratio: $n_{\text{trt}} = 30$, Sc2</i>						
lrcBART: 100	0.023 (0.050)	0.496 (0.064)	1.008	3.672	0.990 (0.010)	0.070 (0.026)
lrcBART: f90	0.026 (0.050)	0.497 (0.067)	1.047	3.763	0.990 (0.010)	0.050 (0.022)
lrcBART: s0min	0.023 (0.050)	0.496 (0.064)	1.008	3.672	0.990 (0.010)	0.070 (0.026)
lrcBART: $w = 0.9$, 100	0.019 (0.050)	0.496 (0.064)	4.526	12.499	1.000 (0.000)	0.000 (0.000)
lrcBART: $w = 0.9$, f90	0.032 (0.050)	0.500 (0.067)	4.571	12.498	1.000 (0.000)	0.000 (0.000)
BART-CP	0.046 (0.048)	0.484 (0.057)	1.041	3.720	1.000 (0.000)	0.060 (0.024)
AFT-CP	0.326 (0.081)	0.867 (0.186)	1.053	3.918	0.980 (0.014)	0.210 (0.041)
hier. AFT	0.530 (0.092)	1.060 (0.218)	2.632	7.275	0.990 (0.010)	0.090 (0.029)
<i>Median-survival ratio: $n_{\text{trt}} = 200$, Sc1</i>						
lrcBART: 100	-0.020 (0.018)	0.178 (0.013)	0.240	0.934	0.990 (0.010)	0.590 (0.049)
lrcBART: f90	-0.022 (0.018)	0.183 (0.013)	0.229	0.892	0.990 (0.010)	0.650 (0.048)
lrcBART: s0min	-0.021 (0.018)	0.178 (0.013)	0.215	0.838	0.970 (0.017)	0.680 (0.047)
lrcBART: $w = 0.9$, 100	-0.022 (0.018)	0.178 (0.013)	2.431	10.097	1.000 (0.000)	0.000 (0.000)
lrcBART: $w = 0.9$, f90	-0.021 (0.018)	0.183 (0.013)	2.420	10.166	1.000 (0.000)	0.000 (0.000)
BART-CP	-0.011 (0.018)	0.180 (0.013)	0.217	0.849	0.970 (0.017)	0.730 (0.044)
AFT-CP	0.055 (0.037)	0.373 (0.027)	0.412	1.603	0.970 (0.017)	0.370 (0.048)
hier. AFT	0.097 (0.038)	0.388 (0.028)	1.386	4.191	1.000 (0.000)	0.020 (0.014)

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Table S8 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
<i>Median-survival ratio: $n_{\text{trt}} = 200$, $Sc2$</i>						
lrcBART: 100	0.070 (0.021)	0.216 (0.017)	0.257	1.003	0.980 (0.014)	0.730 (0.044)
lrcBART: f90	0.079 (0.021)	0.227 (0.016)	0.278	1.087	0.990 (0.010)	0.740 (0.044)
lrcBART: s0min	0.070 (0.021)	0.216 (0.017)	0.255	0.994	0.980 (0.014)	0.740 (0.044)
lrcBART: $w = 0.9$, 100	0.071 (0.020)	0.216 (0.017)	3.026	11.630	1.000 (0.000)	0.000 (0.000)
lrcBART: $w = 0.9$, f90	0.079 (0.021)	0.227 (0.016)	3.040	11.782	1.000 (0.000)	0.000 (0.000)
BART-CP	0.069 (0.021)	0.216 (0.015)	0.253	0.986	0.990 (0.010)	0.770 (0.042)
AFT-CP	0.676 (0.053)	0.859 (0.058)	0.668	2.583	0.820 (0.038)	0.810 (0.039)
hier. AFT	0.872 (0.057)	1.042 (0.061)	2.117	6.476	1.000 (0.000)	0.370 (0.048)
<i>Modified RMST ratio: $n_{\text{trt}} = 30$, $Sc1$</i>						
lrcBART: 100	0.007 (0.016)	0.161 (0.011)	0.211	0.825	0.990 (0.010)	0.140 (0.035)
lrcBART: f90	0.006 (0.016)	0.161 (0.011)	0.219	0.851	0.990 (0.010)	0.110 (0.031)
lrcBART: s0min	0.007 (0.016)	0.161 (0.011)	0.211	0.825	0.990 (0.010)	0.140 (0.035)
lrcBART: $w = 0.9$, 100	0.007 (0.016)	0.160 (0.011)	1.023	2.748	1.000 (0.000)	0.000 (0.000)
lrcBART: $w = 0.9$, f90	0.004 (0.016)	0.160 (0.011)	1.039	2.810	1.000 (0.000)	0.000 (0.000)
BART-CP	0.011 (0.016)	0.161 (0.011)	0.213	0.830	0.980 (0.014)	0.140 (0.035)
AFT-CP	-0.034 (0.018)	0.178 (0.013)	0.175	0.684	0.960 (0.020)	0.130 (0.034)
hier. AFT	-0.024 (0.018)	0.178 (0.013)	0.367	1.079	0.990 (0.010)	0.040 (0.020)
<i>Modified RMST ratio: $n_{\text{trt}} = 30$, $Sc2$</i>						
lrcBART: 100	-0.004 (0.012)	0.117 (0.008)	0.166	0.646	1.000 (0.000)	0.060 (0.024)
lrcBART: f90	-0.004 (0.012)	0.116 (0.009)	0.171	0.667	0.990 (0.010)	0.060 (0.024)
lrcBART: s0min	-0.004 (0.012)	0.117 (0.008)	0.166	0.646	1.000 (0.000)	0.060 (0.024)
lrcBART: $w = 0.9$, 100	-0.006 (0.012)	0.116 (0.008)	0.755	1.985	1.000 (0.000)	0.000 (0.000)
lrcBART: $w = 0.9$, f90	-0.004 (0.012)	0.115 (0.008)	0.779	2.023	1.000 (0.000)	0.000 (0.000)
BART-CP	-0.006 (0.011)	0.114 (0.009)	0.164	0.641	1.000 (0.000)	0.050 (0.022)
AFT-CP	0.050 (0.014)	0.148 (0.013)	0.160	0.627	0.970 (0.017)	0.220 (0.041)
hier. AFT	0.085 (0.014)	0.166 (0.015)	0.486	1.057	0.990 (0.010)	0.100 (0.030)
<i>Modified RMST ratio: $n_{\text{trt}} = 200$, $Sc1$</i>						
lrcBART: 100	-0.008 (0.006)	0.061 (0.004)	0.083	0.323	0.990 (0.010)	0.490 (0.050)
lrcBART: f90	-0.008 (0.006)	0.064 (0.005)	0.079	0.309	0.980 (0.014)	0.530 (0.050)
lrcBART: s0min	-0.008 (0.006)	0.061 (0.004)	0.075	0.295	0.980 (0.014)	0.570 (0.050)
lrcBART: $w = 0.9$, 100	-0.009 (0.006)	0.062 (0.005)	0.816	2.370	1.000 (0.000)	0.000 (0.000)

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Table S8 (continued)

Method / setting	Bias (MCSE)	RMSE (MCSE)	Post. SD	Width	Coverage (MCSE)	Power (MCSE)
lrcBART: $w = 0.9$, f90	-0.009 (0.006)	0.064 (0.005)	0.815	2.370	1.000 (0.000)	0.000 (0.000)
BART-CP	-0.005 (0.006)	0.062 (0.005)	0.075	0.295	0.980 (0.014)	0.550 (0.050)
AFT-CP	-0.013 (0.010)	0.096 (0.006)	0.105	0.410	0.970 (0.017)	0.330 (0.047)
hier. AFT	-0.004 (0.010)	0.096 (0.006)	0.411	0.944	1.000 (0.000)	0.020 (0.014)
<i>Modified RMST ratio: $n_{\text{trt}} = 200$, Sc2</i>						
lrcBART: 100	0.029 (0.007)	0.074 (0.005)	0.079	0.307	0.960 (0.020)	0.650 (0.048)
lrcBART: f90	0.032 (0.007)	0.076 (0.005)	0.084	0.328	0.960 (0.020)	0.620 (0.049)
lrcBART: s0min	0.029 (0.007)	0.074 (0.005)	0.078	0.305	0.960 (0.020)	0.660 (0.047)
lrcBART: $w = 0.9$, 100	0.029 (0.007)	0.073 (0.005)	0.853	2.334	1.000 (0.000)	0.000 (0.000)
lrcBART: $w = 0.9$, f90	0.032 (0.007)	0.076 (0.005)	0.847	2.353	1.000 (0.000)	0.000 (0.000)
BART-CP	0.029 (0.007)	0.075 (0.005)	0.077	0.303	0.950 (0.022)	0.700 (0.046)
AFT-CP	0.155 (0.010)	0.186 (0.010)	0.129	0.503	0.810 (0.039)	0.830 (0.038)
hier. AFT	0.185 (0.010)	0.211 (0.010)	0.465	1.122	1.000 (0.000)	0.370 (0.048)

Entries with parentheses are estimates (Monte Carlo standard errors). Post. SD is the mean posterior standard deviation, or the estimated standard error for PSCL; width is the mean 95% interval width. Coverage and empirical power are proportions; Bayesian decisions require posterior probability of benefit above 0.95, whereas PSCL requires its lower 95% confidence limit to exceed the benefit threshold. Each row has 100 replicates. Gaussian power uses a benefit threshold of 0.5; survival power uses a ratio threshold of 1. Targets 100, 75 and 50 are absolute counts; f90, f50 and f25 are fractions of the calibration ceiling. s0min fixes $s_0^2 = 10^{-6}$ without target selection. NP, CP and PP denote trial-only, completely pooled and source-adjusted control fits. In single-arm studies, CP uses all historical controls without a discrepancy term. The hierarchical discrepancy variance has prior $\text{IG}(3/2, 3s_\tau^2/2)$. The single-arm hierarchical comparator uses discrepancy-variance prior scale $s_\tau^2 = 0.05$.

E.10 Prior ESS sensitivity and regional summaries

Table S9 distinguishes the requested target, the target after ceiling capping, the block-based curve value and the selected prior scale. These summarize per-dataset prior ESS calibration. The regional-discrepancy figure summarizes posterior probabilities of a small discrepancy and posterior mean discrepancies averaged over trial-control profiles in half-unit bins on X_5 . It uses the first 40 Gaussian replicates for three regional shifts of 0.5, 1 and 2 (Sc5). For each bin, profile-specific posterior summaries are pooled across replicates and averaged; replicates contribute in proportion to their trial-control counts within that bin. The figure shows binned means, not credible intervals or replicate-variability intervals.

Table S9: Replicate-averaged lrcBART prior ESS calibration summaries.

Setting	Requested	Capped target	Curve ESS	Ceiling	Cap fraction	s_0^2
<i>gaussian, Sc1</i>						
lrcBART: 100	100.0	96.4	82.5	134.5	0.21	0.001626
lrcBART: 75	75.0	74.4	62.4	134.5	0.05	0.003083
lrcBART: 50	50.0	50.0	42.6	134.5	0.00	0.006187
lrcBART: f90	121.1	121.1	96.0	134.5	0.00	0.000967
lrcBART: f50	67.3	67.3	56.6	134.5	0.00	0.003845
lrcBART: f25	33.6	33.6	29.6	134.5	0.00	0.011093
<i>gaussian, Sc2</i>						
lrcBART: 100	100.0	22.8	24.2	22.8	1.00	0.000001
lrcBART: 75	75.0	22.8	24.2	22.8	1.00	0.000001
lrcBART: 50	50.0	22.8	24.0	22.8	0.99	0.000036
lrcBART: f90	20.5	20.5	16.0	22.8	0.00	0.005771
lrcBART: f50	11.4	11.4	9.5	22.8	0.00	0.021554
lrcBART: f25	5.7	5.7	5.0	22.8	0.00	0.060643
<i>gaussian, Sc3, $\rho = -0.5$</i>						
lrcBART: 100	100.0	99.3	82.6	150.1	0.06	0.003895
lrcBART: 75	75.0	75.0	63.1	150.1	0.00	0.006945
lrcBART: 50	50.0	50.0	43.3	150.1	0.00	0.013433
lrcBART: f90	135.1	135.1	108.1	150.1	0.00	0.001726
lrcBART: f50	75.0	75.0	63.3	150.1	0.00	0.007055
lrcBART: f25	37.5	37.5	33.1	150.1	0.00	0.020388
<i>gaussian, Sc3, $\rho = 0$</i>						
lrcBART: 100	100.0	100.0	84.4	220.1	0.00	0.005907
lrcBART: 75	75.0	75.0	64.6	220.1	0.00	0.009315
lrcBART: 50	50.0	50.0	44.0	220.1	0.00	0.016348
lrcBART: f90	198.1	198.1	156.5	220.1	0.00	0.001284
lrcBART: f50	110.1	110.1	91.8	220.1	0.00	0.005124
lrcBART: f25	55.0	55.0	48.6	220.1	0.00	0.014482
<i>gaussian, Sc3, $\rho = 0.5$</i>						
lrcBART: 100	100.0	99.3	82.6	148.4	0.06	0.003844
lrcBART: 75	75.0	75.0	63.0	148.4	0.00	0.006947
lrcBART: 50	50.0	50.0	43.4	148.4	0.00	0.013417
lrcBART: f90	133.6	133.6	106.8	148.4	0.00	0.001772
lrcBART: f50	74.2	74.2	62.5	148.4	0.00	0.007267
lrcBART: f25	37.1	37.1	32.8	148.4	0.00	0.020901

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Table S9 (continued)

Setting	Requested	Capped target	Curve ESS	Ceiling	Cap fraction	s_0^2
<i>gaussian, Sc4, $\delta = 1$</i>						
lrcBART: 100	100.0	96.4	82.5	134.5	0.21	0.001626
lrcBART: 75	75.0	74.4	62.4	134.5	0.05	0.003083
lrcBART: 50	50.0	50.0	42.6	134.5	0.00	0.006187
lrcBART: f90	121.1	121.1	96.0	134.5	0.00	0.000967
lrcBART: f50	67.3	67.3	56.6	134.5	0.00	0.003845
lrcBART: f25	33.6	33.6	29.6	134.5	0.00	0.011093
<i>gaussian, Sc4, $\delta = 2$</i>						
lrcBART: 100	100.0	96.4	82.5	134.5	0.21	0.001626
lrcBART: 75	75.0	74.4	62.4	134.5	0.05	0.003083
lrcBART: 50	50.0	50.0	42.6	134.5	0.00	0.006187
lrcBART: f90	121.1	121.1	96.0	134.5	0.00	0.000967
lrcBART: f50	67.3	67.3	56.6	134.5	0.00	0.003845
lrcBART: f25	33.6	33.6	29.6	134.5	0.00	0.011093
<i>gaussian, Sc5, X_5, $\delta = 0.5$</i>						
lrcBART: 100	100.0	96.2	83.0	132.6	0.24	0.001592
lrcBART: 75	75.0	74.4	62.0	132.6	0.04	0.003091
lrcBART: 50	50.0	50.0	42.6	132.6	0.02	0.006189
lrcBART: f90	119.3	119.3	94.3	132.6	0.00	0.001020
lrcBART: f50	66.3	66.3	55.5	132.6	0.00	0.003963
lrcBART: f25	33.1	33.1	29.1	132.6	0.00	0.011303
<i>gaussian, Sc5, X_5, $\delta = 1$</i>						
lrcBART: 100	100.0	96.0	83.0	131.7	0.26	0.001577
lrcBART: 75	75.0	74.4	62.4	131.7	0.06	0.003036
lrcBART: 50	50.0	50.0	42.4	131.7	0.00	0.006162
lrcBART: f90	118.6	118.6	93.7	131.7	0.00	0.001014
lrcBART: f50	65.9	65.9	55.1	131.7	0.00	0.003970
lrcBART: f25	32.9	32.9	28.8	131.7	0.00	0.011422
<i>gaussian, Sc5, X_5, $\delta = 2$</i>						
lrcBART: 100	100.0	95.6	81.7	130.1	0.22	0.001570
lrcBART: 75	75.0	74.3	62.0	130.1	0.05	0.003030
lrcBART: 50	50.0	49.9	42.7	130.1	0.01	0.006075
lrcBART: f90	117.1	117.1	92.8	130.1	0.00	0.001005
lrcBART: f50	65.0	65.0	54.4	130.1	0.00	0.004028
lrcBART: f25	32.5	32.5	28.7	130.1	0.00	0.011446
<i>gaussian, Sc5, X_7, $\delta = 1$</i>						
lrcBART: 100	100.0	95.3	82.7	130.6	0.25	0.001542
lrcBART: 75	75.0	74.1	62.3	130.6	0.08	0.003018
lrcBART: 50	50.0	50.0	42.6	130.6	0.01	0.006072
lrcBART: f90	117.6	117.6	93.1	130.6	0.00	0.001030
lrcBART: f50	65.3	65.3	54.7	130.6	0.00	0.004042
lrcBART: f25	32.7	32.7	28.8	130.6	0.00	0.011468
<i>gaussian, Sc5, X_7, $\delta = 2$</i>						
lrcBART: 100	100.0	95.4	82.4	129.9	0.24	0.001544
lrcBART: 75	75.0	74.2	62.2	129.9	0.06	0.002990
lrcBART: 50	50.0	50.0	42.6	129.9	0.02	0.006085

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Table S9 (continued)

Setting	Requested	Capped target	Curve ESS	Ceiling	Cap fraction	s_0^2
lrcBART: f90	117.0	117.0	92.5	129.9	0.00	0.001021
lrcBART: f50	65.0	65.0	54.4	129.9	0.00	0.004030
lrcBART: f25	32.5	32.5	28.5	129.9	0.00	0.011577
<i>survival, Sc1</i>						
lrcBART: 100	100.0	99.3	83.2	202.3	0.05	0.004399
lrcBART: 75	75.0	74.9	63.7	202.3	0.01	0.007015
lrcBART: 50	50.0	50.0	43.6	202.3	0.00	0.012328
lrcBART: f90	182.1	182.1	142.5	202.3	0.00	0.001139
lrcBART: f50	101.2	101.2	84.0	202.3	0.00	0.004398
lrcBART: f25	50.6	50.6	44.3	202.3	0.00	0.012413
<i>survival, Sc2</i>						
lrcBART: 100	100.0	36.2	38.3	36.2	1.00	0.000001
lrcBART: 75	75.0	36.2	38.1	36.2	0.99	0.000029
lrcBART: 50	50.0	35.2	35.3	36.2	0.87	0.000756
lrcBART: f90	32.6	32.6	25.5	36.2	0.00	0.005652
lrcBART: f50	18.1	18.1	15.0	36.2	0.00	0.021673
lrcBART: f25	9.1	9.1	7.9	36.2	0.00	0.061216
<i>survival, Sc3, $\rho = -0.5$</i>						
lrcBART: 100	100.0	99.9	82.9	193.6	0.01	0.006562
lrcBART: 75	75.0	75.0	63.7	193.6	0.00	0.010583
lrcBART: 50	50.0	50.0	43.5	193.6	0.00	0.019003
lrcBART: f90	174.3	174.3	137.1	193.6	0.00	0.001825
lrcBART: f50	96.8	96.8	80.8	193.6	0.00	0.007121
lrcBART: f25	48.4	48.4	42.4	193.6	0.00	0.020292
<i>survival, Sc3, $\rho = 0$</i>						
lrcBART: 100	100.0	100.0	86.2	302.7	0.00	0.009624
lrcBART: 75	75.0	75.0	66.1	302.7	0.00	0.014262
lrcBART: 50	50.0	50.0	44.7	302.7	0.00	0.024000
lrcBART: f90	272.4	272.4	217.8	302.7	0.00	0.001224
lrcBART: f50	151.3	151.3	127.5	302.7	0.00	0.004992
lrcBART: f25	75.7	75.7	66.7	302.7	0.00	0.014404
<i>survival, Sc3, $\rho = 0.5$</i>						
lrcBART: 100	100.0	99.8	84.5	201.3	0.01	0.006783
lrcBART: 75	75.0	75.0	64.4	201.3	0.00	0.011035
lrcBART: 50	50.0	50.0	44.1	201.3	0.00	0.019587
lrcBART: f90	181.2	181.2	144.8	201.3	0.00	0.001706
lrcBART: f50	100.7	100.7	84.9	201.3	0.00	0.006934
lrcBART: f25	50.3	50.3	44.3	201.3	0.00	0.020040
<i>survival, Sc4, $\delta = 1$</i>						
lrcBART: 100	100.0	98.4	82.1	177.0	0.09	0.003993
lrcBART: 75	75.0	74.8	63.0	177.0	0.03	0.006510
lrcBART: 50	50.0	50.0	42.8	177.0	0.00	0.011885
lrcBART: f90	159.3	159.3	123.8	177.0	0.00	0.001388
lrcBART: f50	88.5	88.5	73.3	177.0	0.00	0.005147
lrcBART: f25	44.2	44.2	38.6	177.0	0.00	0.014486

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Table S9 (continued)

Setting	Requested	Capped target	Curve ESS	Ceiling	Cap fraction	s_0^2
<i>survival, Sc4, $\delta = 2$</i>						
lrcBART: 100	100.0	95.1	81.2	141.1	0.25	0.003228
lrcBART: 75	75.0	74.1	61.0	141.1	0.06	0.005752
lrcBART: 50	50.0	50.0	42.0	141.1	0.01	0.010976
lrcBART: f90	127.0	127.0	97.4	141.1	0.00	0.001905
lrcBART: f50	70.5	70.5	57.8	141.1	0.00	0.006715
lrcBART: f25	35.3	35.3	30.7	141.1	0.00	0.018381
<i>survival, Sc5, X_5, $\delta = 0.5$</i>						
lrcBART: 100	100.0	98.9	82.7	191.6	0.04	0.004175
lrcBART: 75	75.0	74.8	63.4	191.6	0.02	0.006801
lrcBART: 50	50.0	50.0	43.4	191.6	0.00	0.012144
lrcBART: f90	172.4	172.4	136.3	191.6	0.00	0.001182
lrcBART: f50	95.8	95.8	80.1	191.6	0.00	0.004644
lrcBART: f25	47.9	47.9	42.1	191.6	0.00	0.013207
<i>survival, Sc5, X_5, $\delta = 1$</i>						
lrcBART: 100	100.0	99.0	82.8	181.8	0.08	0.004062
lrcBART: 75	75.0	74.8	63.1	181.8	0.02	0.006621
lrcBART: 50	50.0	50.0	43.2	181.8	0.00	0.011961
lrcBART: f90	163.6	163.6	128.1	181.8	0.00	0.001291
lrcBART: f50	90.9	90.9	75.6	181.8	0.00	0.004925
lrcBART: f25	45.5	45.5	39.8	181.8	0.00	0.013909
<i>survival, Sc5, X_5, $\delta = 2$</i>						
lrcBART: 100	100.0	97.5	82.4	159.4	0.18	0.003622
lrcBART: 75	75.0	74.5	61.8	159.4	0.03	0.006196
lrcBART: 50	50.0	50.0	42.6	159.4	0.01	0.011458
lrcBART: f90	143.5	143.5	110.5	159.4	0.00	0.001598
lrcBART: f50	79.7	79.7	65.5	159.4	0.00	0.005816
lrcBART: f25	39.9	39.9	34.6	159.4	0.00	0.016210
<i>survival, Sc5, X_7, $\delta = 1$</i>						
lrcBART: 100	100.0	98.9	82.8	186.2	0.07	0.004149
lrcBART: 75	75.0	74.8	63.3	186.2	0.02	0.006717
lrcBART: 50	50.0	50.0	43.3	186.2	0.00	0.012069
lrcBART: f90	167.6	167.6	131.6	186.2	0.00	0.001245
lrcBART: f50	93.1	93.1	77.7	186.2	0.00	0.004752
lrcBART: f25	46.6	46.6	40.9	186.2	0.00	0.013513
<i>survival, Sc5, X_7, $\delta = 2$</i>						
lrcBART: 100	100.0	98.3	82.5	167.2	0.12	0.003742
lrcBART: 75	75.0	74.7	62.1	167.2	0.02	0.006364
lrcBART: 50	50.0	50.0	42.9	167.2	0.00	0.011595
lrcBART: f90	150.5	150.5	117.1	167.2	0.00	0.001461
lrcBART: f50	83.6	83.6	69.2	167.2	0.00	0.005431
lrcBART: f25	41.8	41.8	36.5	167.2	0.00	0.015186
<i>gaussian single arm, $n_{\text{trt}} = 200$, Sc1</i>						
lrcBART: 100	100.0	100.0	81.4	152.8	0.00	0.001838
lrcBART: f90	137.6	137.6	108.5	152.8	0.00	0.000815
lrcBART: s0min	–	–	160.6	152.8	0.00	0.000001

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Table S9 (continued)

Setting	Requested	Capped target	Curve ESS	Ceiling	Cap fraction	s_0^2
lrcBART: $w = 0.9$, 100	100.0	100.0	81.4	152.8	0.00	0.001838
lrcBART: $w = 0.9$, f90	137.6	137.6	108.5	152.8	0.00	0.000815
<i>gaussian single arm, $n_{\text{trt}} = 200$, Sc2</i>						
lrcBART: 100	100.0	97.9	84.2	109.0	0.29	0.000891
lrcBART: f90	98.1	98.1	77.0	109.0	0.00	0.001173
lrcBART: s0min	–	–	114.6	109.0	0.00	0.000001
lrcBART: $w = 0.9$, 100	100.0	97.9	84.2	109.0	0.29	0.000891
lrcBART: $w = 0.9$, f90	98.1	98.1	77.0	109.0	0.00	0.001173
<i>survival single arm, $n_{\text{trt}} = 30$, Sc1</i>						
lrcBART: 100	100.0	55.9	60.5	55.9	1.00	0.000001
lrcBART: f90	50.3	50.3	38.3	55.9	0.00	0.002037
lrcBART: s0min	–	–	60.5	55.9	0.00	0.000001
lrcBART: $w = 0.9$, 100	100.0	55.9	60.5	55.9	1.00	0.000001
lrcBART: $w = 0.9$, f90	50.3	50.3	38.3	55.9	0.00	0.002037
<i>survival single arm, $n_{\text{trt}} = 30$, Sc2</i>						
lrcBART: 100	100.0	41.9	45.6	41.9	1.00	0.000001
lrcBART: f90	37.7	37.7	28.6	41.9	0.00	0.002817
lrcBART: s0min	–	–	45.6	41.9	0.00	0.000001
lrcBART: $w = 0.9$, 100	100.0	41.9	45.6	41.9	1.00	0.000001
lrcBART: $w = 0.9$, f90	37.7	37.7	28.6	41.9	0.00	0.002817
<i>survival single arm, $n_{\text{trt}} = 200$, Sc1</i>						
lrcBART: 100	100.0	99.9	79.9	133.4	0.03	0.001190
lrcBART: f90	120.0	120.0	93.1	133.4	0.00	0.000755
lrcBART: s0min	–	–	141.8	133.4	0.00	0.000001
lrcBART: $w = 0.9$, 100	100.0	99.9	79.9	133.4	0.03	0.001190
lrcBART: $w = 0.9$, f90	120.0	120.0	93.1	133.4	0.00	0.000755
<i>survival single arm, $n_{\text{trt}} = 200$, Sc2</i>						
lrcBART: 100	100.0	83.5	85.7	84.7	0.88	0.000099
lrcBART: f90	76.2	76.2	59.3	84.7	0.00	0.001206
lrcBART: s0min	–	–	90.2	84.7	0.00	0.000001
lrcBART: $w = 0.9$, 100	100.0	83.5	85.7	84.7	0.88	0.000099
lrcBART: $w = 0.9$, f90	76.2	76.2	59.3	84.7	0.00	0.001206

All quantities are means over 100 datasets, except cap fraction, which is the proportion flagged as capped. Curve ESS is the mean across block-specific ESS calculations at the selected scale. The ceiling uses the variance across all historical posterior draws. These are numerical summaries of distinct rules (Appendix B.5); neither is a posterior borrowing count. s0min bypasses target selection. A dash denotes an unavailable or inapplicable quantity.

F Separate-treatment application details and results

F.1 Analysis extract and harmonized variables

The available merged analysis extract has 30 EloKRd and 200 triplet-treated UCMM observations. Starting from 797 UCMM records, the preparation code excludes negative OS/PFS times (790 remain), requires a recorded induction start (303), excludes E-Rd/Elo-Rd (302), and retains verified three-agent regimens (200). The EloKRd analysis uses the available subset of the published study (Derman et al., 2022).

Triplet-only C3 selection. Figure S2 shows the sequential exclusions. C2 contains 302 patients after excluding E-Rd/Elo-Rd. The triplet rule then excludes 102 patients with doublets ($n = 66$), single-agent treatment ($n = 4$), four or more treatment agents ($n = 19$), or other/unclear treatment records ($n = 13$). C3 therefore contains 200 UCMM controls, or 230 observations when merged with the 30 EloKRd patients. Classification uses the recorded induction components rather than the source triplet flag. Steroids count as treatment agents; zoledronic acid is supportive therapy. VRd plus vorinostat is excluded because it contains four treatment agents, despite a source triplet flag.

The retained regimen labels and counts are VRd (107), KRd (38), CyBorD (22), CRd (5), Cy-Kd (4), IRd (4), KPd (4), VTD (4), PDd (3), VPd (2), and one each of BRd, Dara-Rd, Elo-Vd, O-Rd, VAD, VMP and VRd-Zoledronic. In this extract, PDd records bortezomib, doxorubicin and dexamethasone; O-Rd records oprozomib, lenalidomide and dexamethasone. The 13 other/unclear records are not assumed to be triplets. Covariate harmonization retains all 200 analysis controls.

The EloKRd extract supplies times from first treatment (C1D1); the UCMM code constructs times from induction start. OS ends at death or last contact. UCMM PFS ends at the first progression or death, using the recorded first-progression date with a recurrence-date fallback; otherwise it ends at last contact. Zero UCMM PFS times are floored at 0.1 month before logarithmic analysis. EloKRd PFS status 2 (death without prior progression) is recoded as an event. Differences in source-specific endpoint ascertainment may remain despite this harmonization.

Variable selection first retains variables complete in EloKRd, then drops variables with the greatest UCMM missingness until all 200 triplet-treated UCMM records are complete on the retained set. The final numeric model predictors have no missing values. This data-dependent availability rule should be distinguished from a clinically prespecified adjustment set.

F.2 Analysis and sensitivity configurations

All adjusted methods use the same harmonized predictors and standardize control and treated predictions to the 30 EloKRd profiles. KM remains unadjusted. Table S11 reports the hypothetical EloKRd control for adjusted methods and the observed UCMM cohort RMST for KM. Ratios and differences use those respective controls. The primary lrcBART analysis uses $H_f = 10$, $H_g = 5$, fixed $w = 1$ and requested ESS 30. Sensitivities cross $H_f \in \{10, 50\}$ with fixed $w \in \{1, 0.9\}$. Each of those four settings uses six requested targets: 30, 22.5, 15 and fractions 0.90, 0.50, 0.25 of its calibration ceiling. The target labels 100, 75 and 50 correspond to percentages of the 30-patient trial here; in simulations those labels are absolute counts.

The five comparator settings are KM, AFT-CP, BART-CP and hierarchical AFT at discrepancy-variance prior scales $s_\tau^2 = 0.05$ and $s_\tau^2 = 0.5$. Together with 24 lrcBART settings this gives 29 rows per endpoint.

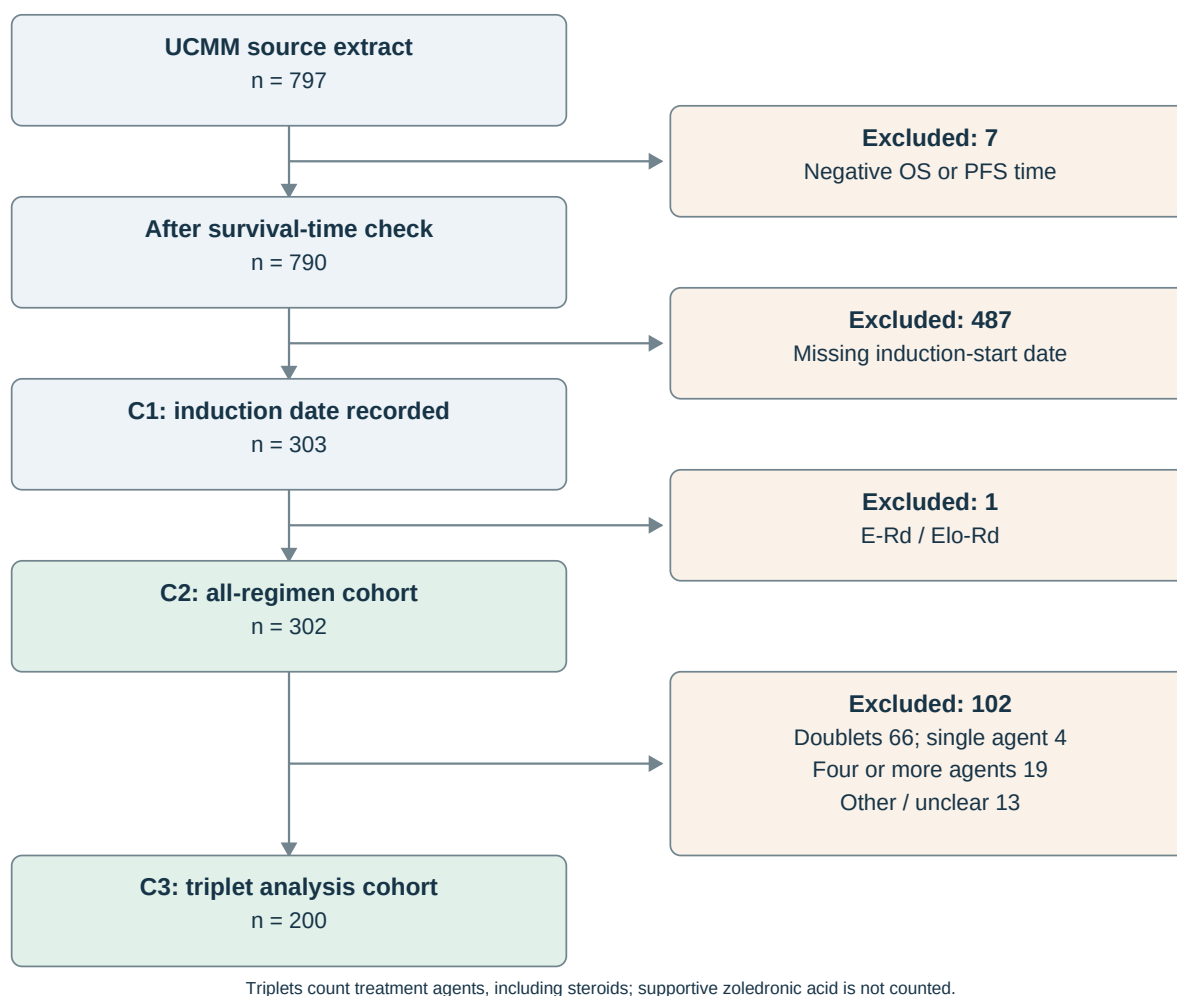


Figure S2: UCMM inclusion and exclusion flow through C2 and the triplet-only analysis cohort. Counts refer to sequential, mutually exclusive exclusions.

Table S10: Harmonized model variables in the myeloma application.

Dimension	Numeric model representation and coding
Age	Continuous age in years: enrollment age for EloKRd; age at the identification date for UCMM, with diagnosis-age fallback. Table 4 groups rows into < 50 , $[50, 60)$, $[60, 70)$ and ≥ 70 years for display, shows age to one decimal year, and uses exact age in every calculation.
Sex	Male indicator: 1 for male, 0 for female; source strings beginning with M or F are harmonized.
Race	Black and Other indicators, with White as reference. White/Caucasian and Black/African labels are combined. Other comprises remaining harmonized categories, including Other/Unknown; it is not an imputation of a known race.
Hispanic/Latino	1 for “Hispanic or Latino”; 0 for the non-Hispanic labels in the extract. Ethnicity remains a separate dimension from race.
Cytogenetic risk	EloKRd uses its recorded 0/1 high-risk flag. UCMM uses HRCA ≥ 1 when recorded; otherwise any positive $t(4;14)$, $t(14;16)$, $t(14;20)$, $\text{del}(17p)$ or $\text{gain}(1q)$ indicator gives 1. In that fallback, missing component indicators are ignored and no positive entry gives 0.
ASCT	1 for transplantation, 0 otherwise, using the cleaned transplantation indicators. In EloKRd the source variable records off-protocol transplantation. This is a postinduction characteristic.
Outcome records	PFS and OS times converted from months to years; event indicator 1 and right-censoring indicator 0.

Seven numeric predictors represent six covariate dimensions. The final extract has no missing numeric model predictors. The harmonized variables can retain source-specific measurement differences.

The main table reports $s^2_{\tau} = 0.05$; both hierarchical settings remain in the complete appendix table. RMST contrasts use the draw-by-draw calculations in Appendix D, with posterior medians and equal-tailed 95% intervals.

F.3 Complete application results

All 58 PFS and OS results are included. Similarity among lrcBART estimates does not resolve prior dependence for unobserved trial-control outcomes.

Table S11: Complete application results, including all lrcBART sensitivity settings.

Method	Setting	Ratio (95% interval)	Difference (95% interval)	EloKRd treatment RMST	EloKRd hypothetical control RMST	UCMM control RMST
<i>PFS</i>						
KM	—	1.040 (0.862, 1.254)	0.143 (-0.548, 0.834)	3.705	—	3.562
lrcBART	default, 100	0.997 (0.799, 1.229)	-0.009 (-0.768, 0.714)	3.521	3.535	—
lrcBART	default, 75	0.998 (0.792, 1.250)	-0.009 (-0.806, 0.760)	3.521	3.534	—
lrcBART	default, 50	0.997 (0.779, 1.311)	-0.011 (-0.883, 0.893)	3.521	3.536	—
lrcBART	default, f90	0.997 (0.801, 1.224)	-0.012 (-0.759, 0.702)	3.521	3.534	—
lrcBART	default, f50	0.998 (0.786, 1.280)	-0.008 (-0.847, 0.820)	3.521	3.534	—
lrcBART	default, f25	0.997 (0.760, 1.419)	-0.011 (-1.000, 1.085)	3.521	3.529	—
lrcBART	Hf50, 100	0.987 (0.798, 1.220)	-0.046 (-0.775, 0.684)	3.521	3.572	—
lrcBART	Hf50, 75	0.987 (0.792, 1.247)	-0.049 (-0.804, 0.752)	3.521	3.572	—
lrcBART	Hf50, 50	0.988 (0.782, 1.290)	-0.044 (-0.875, 0.849)	3.521	3.571	—
lrcBART	Hf50, f90	0.987 (0.798, 1.222)	-0.046 (-0.775, 0.689)	3.521	3.572	—
lrcBART	Hf50, f50	0.987 (0.783, 1.281)	-0.047 (-0.865, 0.829)	3.521	3.570	—
lrcBART	Hf50, f25	0.991 (0.754, 1.436)	-0.032 (-1.021, 1.131)	3.521	3.562	—
lrcBART	w0.9, 100	0.996 (0.738, 1.754)	-0.015 (-1.131, 1.526)	3.521	3.539	—
lrcBART	w0.9, 75	0.995 (0.736, 1.753)	-0.017 (-1.145, 1.543)	3.521	3.537	—
lrcBART	w0.9, 50	0.996 (0.730, 1.783)	-0.014 (-1.176, 1.578)	3.521	3.538	—
lrcBART	w0.9, f90	0.995 (0.740, 1.750)	-0.018 (-1.129, 1.535)	3.521	3.539	—
lrcBART	w0.9, f50	0.996 (0.733, 1.767)	-0.014 (-1.163, 1.552)	3.521	3.537	—
lrcBART	w0.9, f25	0.997 (0.725, 1.842)	-0.012 (-1.223, 1.655)	3.521	3.537	—
lrcBART	Hf50, w0.9, 100	0.984 (0.734, 1.735)	-0.057 (-1.161, 1.545)	3.521	3.582	—
lrcBART	Hf50, w0.9, 75	0.985 (0.731, 1.737)	-0.053 (-1.170, 1.545)	3.521	3.584	—
lrcBART	Hf50, w0.9, 50	0.986 (0.730, 1.765)	-0.053 (-1.179, 1.563)	3.521	3.581	—
lrcBART	Hf50, w0.9, f90	0.985 (0.734, 1.733)	-0.057 (-1.160, 1.543)	3.521	3.582	—
lrcBART	Hf50, w0.9, f50	0.985 (0.730, 1.765)	-0.054 (-1.180, 1.561)	3.521	3.583	—
lrcBART	Hf50, w0.9, f25	0.987 (0.720, 1.844)	-0.048 (-1.244, 1.673)	3.521	3.570	—
BART-CP	—	0.988 (0.796, 1.215)	-0.042 (-0.760, 0.690)	3.527	3.580	—
AFT-CP	—	0.901 (0.708, 1.105)	-0.338 (-1.080, 0.335)	3.069	3.408	—
hier. AFT	—	1.062 (0.789, 1.467)	0.192 (-0.834, 1.138)	3.355	3.148	—
hier. AFT	—	1.209 (0.731, 2.136)	0.567 (-1.204, 1.784)	3.355	2.715	—

Continued on next page

Table S11 (continued)

Method	Setting	Ratio (95% interval)	Difference (95% interval)	EloKRd treatment RMST	EloKRd hypothetical control RMST	UCMM control RMST
<i>OS</i>						
KM	—	0.891 (0.776, 1.023)	-0.505 (-1.077, 0.068)	4.122	—	4.627
lrcBART	default, 100	0.898 (0.765, 1.024)	-0.453 (-1.065, 0.101)	3.995	4.456	—
lrcBART	default, 75	0.898 (0.764, 1.034)	-0.454 (-1.073, 0.142)	3.995	4.453	—
lrcBART	default, 50	0.899 (0.763, 1.051)	-0.451 (-1.090, 0.207)	3.995	4.452	—
lrcBART	default, f90	0.898 (0.766, 1.030)	-0.454 (-1.068, 0.123)	3.995	4.455	—
lrcBART	default, f50	0.899 (0.763, 1.054)	-0.452 (-1.096, 0.216)	3.995	4.452	—
lrcBART	default, f25	0.900 (0.756, 1.120)	-0.444 (-1.149, 0.440)	3.995	4.446	—
lrcBART	Hf50, 100	0.901 (0.771, 1.024)	-0.439 (-1.039, 0.100)	3.995	4.434	—
lrcBART	Hf50, 75	0.902 (0.768, 1.037)	-0.437 (-1.052, 0.154)	3.995	4.432	—
lrcBART	Hf50, 50	0.902 (0.766, 1.057)	-0.432 (-1.064, 0.229)	3.995	4.428	—
lrcBART	Hf50, f90	0.901 (0.769, 1.032)	-0.436 (-1.043, 0.132)	3.995	4.432	—
lrcBART	Hf50, f50	0.902 (0.766, 1.061)	-0.431 (-1.068, 0.242)	3.995	4.427	—
lrcBART	Hf50, f25	0.906 (0.759, 1.141)	-0.415 (-1.121, 0.513)	3.995	4.417	—
lrcBART	w0.9, 100	0.897 (0.763, 1.067)	-0.458 (-1.098, 0.261)	3.995	4.457	—
lrcBART	w0.9, 75	0.897 (0.762, 1.072)	-0.458 (-1.110, 0.282)	3.995	4.457	—
lrcBART	w0.9, 50	0.897 (0.759, 1.087)	-0.456 (-1.122, 0.334)	3.995	4.455	—
lrcBART	w0.9, f90	0.897 (0.762, 1.067)	-0.459 (-1.104, 0.268)	3.995	4.458	—
lrcBART	w0.9, f50	0.897 (0.759, 1.090)	-0.457 (-1.123, 0.348)	3.995	4.454	—
lrcBART	w0.9, f25	0.899 (0.754, 1.156)	-0.454 (-1.154, 0.558)	3.995	4.452	—
lrcBART	Hf50, w0.9, 100	0.902 (0.766, 1.068)	-0.432 (-1.080, 0.265)	3.995	4.434	—
lrcBART	Hf50, w0.9, 75	0.903 (0.765, 1.082)	-0.432 (-1.072, 0.318)	3.995	4.432	—
lrcBART	Hf50, w0.9, 50	0.904 (0.765, 1.099)	-0.425 (-1.088, 0.371)	3.995	4.428	—
lrcBART	Hf50, w0.9, f90	0.903 (0.766, 1.076)	-0.432 (-1.067, 0.297)	3.995	4.433	—
lrcBART	Hf50, w0.9, f50	0.904 (0.765, 1.103)	-0.424 (-1.091, 0.389)	3.995	4.428	—
lrcBART	Hf50, w0.9, f25	0.907 (0.758, 1.174)	-0.413 (-1.136, 0.610)	3.995	4.418	—
BART-CP	—	0.898 (0.769, 1.017)	-0.448 (-1.048, 0.073)	3.985	4.439	—
AFT-CP	—	0.809 (0.654, 0.951)	-0.814 (-1.521, -0.203)	3.449	4.268	—
hier. AFT	—	0.961 (0.761, 1.387)	-0.153 (-1.085, 1.073)	3.807	3.953	—
hier. AFT	—	1.208 (0.777, 1.710)	0.646 (-1.057, 1.684)	3.807	3.117	—

All 58 method/endpoint results are included. Default: $H_f = 10$, $H_g = 5$, $w = 1$. Hf50 changes H_f to 50; w0.9 fixes $w = 0.9$. Application labels 100, 75 and 50 request ESS 30, 22.5 and 15, respectively; f90, f50 and f25 request the corresponding fraction of the calibration ceiling. Intervals, units and arm-column definitions follow Table 6. Contrasts use the EloKRd hypothetical control for adjusted methods and the observed UCMM control for KM. Within each endpoint, the first and second hier. AFT rows use hierarchical discrepancy-variance prior scales $s_\tau^2 = 0.05$ and $s_\tau^2 = 0.5$, respectively.

F.4 Construction of the patient-profile PFS table

Table 4 contains one row for each of the 30 EloKRd patients. Rows are ordered by age group (< 50 , $50\text{--}59$, $60\text{--}69$ and ≥ 70 years) and exact age. The age group and UCMM count are displayed separately on every row. A row retains that patient's complete model profile: exact age, sex, race, Hispanic/Latino status, cytogenetic risk and ASCT. Age is rounded to one decimal year in the display, while all calculations use its exact value. The other predictors are not set to common reference values.

The preliminary historical model is the primary PFS specification with $H_f = 10$. Its 4,000 retained posterior predictions at the EloKRd profiles give $V_f(x_i)$ directly as the sample variance across draws for each patient. For a single profile, one leaf from each discrepancy tree has weight one, so the all-spike discrepancy variance is $H_g \tau_0^2$. The expectation uses 100,000 chi-square draws with $\nu_0 = 3$ and the globally selected $s_0^2 = 0.00398107$.

For each row, the UCMM cohort count is obtained by matching historical patients on age group, sex, race, Hispanic/Latino status, cytogenetic risk and ASCT. Counts repeat when EloKRd rows share a displayed combination and should not be summed down the table. A count is descriptive and does not restrict the historical fit used for that row's ESS. Every profile uses the same historical residual-variance reference and scale law. Table 4 evaluates the full-variance expectation-of-ratios definition using the preliminary historical fit. ESS is reported to one decimal place, ranges from 3.8 to 24.5, and is expressed in uncensored normal-reference observations for the conditional mean log-PFS at the same profile. Only PFS appears in this table; OS remains in the outcome tables.

F.5 Prior ESS calibration and MCMC diagnostics

The preliminary application calibration uses 2,000 warm-up and 4,000 retained draws, 2,000 prior trees, 4,000 scale draws and the 141-value scale grid. Table S11(a) reports requested/capped targets, selected scales, the block-based curve and the full-variance ESS evaluated with the estimated prior-tree weight expectation. The latter uses 100,000 deterministic chi-square quantiles for numerical integration. Differences reflect block averaging and numerical integration error; the reciprocal-variance ordering in Appendix B applies with common scale draws.

Each lrcBART outcome fit uses four chains with 2,000 warm-up and 2,000 retained draws per chain. Table S11(b) reports $\hat{R}$, bulk ESS and tail ESS for the log RMST ratio. Prior ESS and MCMC ESS have different meanings: the former concerns equivalent reference information about a specified current-control mean, whereas the latter concerns simulation efficiency. These scalar diagnostics do not establish exploration of all tree structures.

Table S11(a): Application prior ESS calibration.

Target label	Requested	Capped target	s_0^2	Block-based ESS	Full-variance ESS	Ceiling
<i>PFS, $H_f = 10$</i>						
100	30.00	30.00	0.003981	24.70	24.43	37.03
75	22.50	22.50	0.007943	19.40	19.28	37.03
50	15.00	15.00	0.017783	13.07	13.08	37.03
f90	33.33	33.33	0.002818	27.07	26.71	37.03
f50	18.51	18.51	0.012589	15.72	15.69	37.03
f25	9.26	9.26	0.035481	8.43	8.47	37.03
<i>PFS, $H_f = 50$</i>						
100	30.00	30.00	0.001585	27.22	27.09	32.97
75	22.50	22.50	0.005623	20.28	20.25	32.97
50	15.00	15.00	0.014125	13.90	13.94	32.97
f90	29.67	29.67	0.001778	26.73	26.60	32.97
f50	16.48	16.48	0.012589	14.70	14.73	32.97
f25	8.24	8.24	0.039811	7.47	7.52	32.97
<i>OS, $H_f = 10$</i>						
100	30.00	29.14	0.000001	30.04	29.13	29.14
75	22.50	22.50	0.007079	18.93	18.66	29.14
50	15.00	15.00	0.017783	13.21	13.13	29.14
f90	26.22	26.22	0.003981	22.06	21.63	29.14
f50	14.57	14.57	0.019953	12.49	12.43	29.14
f25	7.28	7.28	0.056234	6.71	6.73	29.14
<i>OS, $H_f = 50$</i>						
100	30.00	28.21	0.000001	28.92	28.21	28.21
75	22.50	22.50	0.007079	18.55	18.33	28.21
50	15.00	15.00	0.017783	13.06	13.01	28.21
f90	25.39	25.39	0.004467	20.96	20.64	28.21
f50	14.11	14.11	0.019953	12.36	12.32	28.21
f25	7.05	7.05	0.063096	6.19	6.22	28.21

Full-variance ESS evaluates the expectation-of-ratios definition with the variance across all historical posterior draws and tree-weight estimate, using 100,000 chi-square quantiles. Block-based ESS is the block average used in the existing scale selector, with 4,000 random scale draws. Differences also reflect numerical integration error; the reciprocal-variance ordering applies when the same integration draws are used. The same calibration is used at $w = 1$ and $w = 0.9$; it remains an all-spike reference.

Table S11(b): MCMC diagnostics for the application log RMST ratio.

Setting	$\hat{R}$	Bulk MCMC ESS	Tail MCMC ESS
<i>PFS</i>			
default, 100	1.0004	7121	7347
default, 75	1.0002	7263	7405
default, 50	1.0003	7478	8092
default, f90	1.0004	7070	7279
default, f50	1.0002	7380	7791
default, f25	1.0002	7677	8093
Hf50, 100	1.0000	6977	7164
Hf50, 75	1.0000	7068	7202
Hf50, 50	1.0001	7234	7730
Hf50, f90	1.0000	6982	7199
Hf50, f50	1.0000	7207	7652
Hf50, f25	1.0000	7537	7848
w0.9, 100	1.0002	7424	7843
w0.9, 75	1.0003	7408	7686
w0.9, 50	1.0005	7396	7826
w0.9, f90	1.0002	7419	7822
w0.9, f50	1.0004	7404	7703
w0.9, f25	1.0006	7407	7716
Hf50_w0.9, 100	0.9999	7562	7559
Hf50_w0.9, 75	0.9999	7553	7673
Hf50_w0.9, 50	0.9998	7572	7687
Hf50_w0.9, f90	0.9999	7561	7559
Hf50_w0.9, f50	0.9998	7567	7650
Hf50_w0.9, f25	0.9999	7638	7928
<i>OS</i>			
default, 100	1.0000	6657	7221
default, 75	1.0000	6812	7697
default, 50	1.0000	6944	7846
default, f90	1.0000	6763	7638
default, f50	1.0000	6966	7886
default, f25	1.0001	7216	7889
Hf50, 100	1.0001	7200	7689
Hf50, 75	1.0003	7283	7849
Hf50, 50	1.0004	7386	7848
Hf50, f90	1.0002	7251	7808
Hf50, f50	1.0004	7405	7769
Hf50, f25	1.0003	7668	7690
w0.9, 100	1.0002	7045	7650
w0.9, 75	1.0001	7109	7849
w0.9, 50	1.0000	7199	7716
w0.9, f90	1.0001	7080	7809
w0.9, f50	1.0000	7213	7500
w0.9, f25	0.9999	7418	7255
Hf50_w0.9, 100	0.9999	7076	7610
Hf50_w0.9, 75	1.0000	7113	7441
Hf50_w0.9, 50	1.0001	7185	7297
Hf50_w0.9, f90	1.0000	7097	7536
Hf50_w0.9, f50	1.0000	7198	7403
Hf50_w0.9, f25	0.9998	7407	7444

Four chains, each with 2,000 warm-up and 2,000 retained draws. These effective sample sizes measure MCMC efficiency and are unrelated to prior ESS. These scalar diagnostics alone do not establish adequate exploration of all tree structures or parameters.

G Joint treatment and control validation

This appendix documents the Gaussian validation for the joint primary model. The model is

$$\begin{aligned} y_i &= f(x_i) + S_i g(x_i) + A_i \psi + \varepsilon_i, \\ v_i &= \begin{cases} \sigma_1^2, & S_i = 1, \\ \sigma_2^2, & S_i = 0, \end{cases} \quad \varepsilon_i \sim N(0, v_i), \\ \psi &\sim N(0, v_\psi), \quad v_\psi = 100. \end{aligned} \tag{G.1}$$

where $S_i = 1$ for both randomized-trial arms and $S_i = 0$ for external controls. The treatment indicator is $A_i = 1$ for treated trial subjects and zero otherwise. Thus the randomized-control mean is $f + g$, the treated mean is $f + g + \psi$, and the external-control mean is f . The standardized trial contrast is the average of the treated and control potential-mean difference over the trial profiles, and equals ψ under the common-effect model. The Heterogeneous condition deliberately violates that restriction and is scored against its realized trial-profile ATE.

G.1 Posterior computation

The prognostic function f is a sum of $H_f = 50$ trees and the discrepancy function g is a sum of $H_g = 5$ trees. Let

$$p_k(d) = \alpha_k (1 + d)^{-\beta_k}, \quad (\alpha_f, \beta_f) = (0.95, 2), \quad (\alpha_g, \beta_g) = (0.5, 3).$$

The primary tree prior uses these depth weights in a globally restricted admissible-tree distribution. For a valid tree T ,

$$\tilde{\pi}_k(T) \propto \prod_{v \in I(T)} \frac{p_k(d(v))}{V(v)C(v, s_v)} \prod_{\ell \in L(T)} \{1 - p_k(d(\ell))\} 1\{T \text{ is admissible}\}, \tag{G.2}$$

where $I(T)$ and $L(T)$ are its internal nodes and leaves, $V(v)$ is the number of ancestor-admissible splitting variables at v , and $C(v, s_v)$ is the number of available cutpoints for the selected variable. The normalizing constant is over the finite valid-tree support. Admissibility requires ancestor-consistent cuts and at least five fitting observations per leaf: all sources for f and randomized-trial observations for g . A terminal node with no remaining cut retains the factor $1 - p_k(d)$ under this globally restricted target. This is a depth-weighted restricted tree prior with a single normalizing constant over valid trees.

The minimum leaf size is five and the grow, prune and change move probabilities are 0.3, 0.3 and 0.4. The f terminal parameters have a normal prior with variance

$$\lambda_f^2 = \left\{ \frac{R_y}{2k\sqrt{H_f}} \right\}^2, \quad k = 2,$$

where R_y is the range of the centered observed outcome passed to the final fit. The discrepancy terminal parameters use

$$z_{h\ell} \mid w \sim \text{Bernoulli}(w), \quad \theta_{h\ell}^g \mid z_{h\ell} = 1 \sim N(0, \tau_0^2), \quad \theta_{h\ell}^g \mid z_{h\ell} = 0 \sim N(0, \tau_1^2).$$

Here $w \sim \text{Beta}(1, 1)$, $v_0 = 3$, τ_0^2 has a scaled inverse-chi-square prior with scale s_0^2 truncated to $(0, \tau_1^2)$, and $\tau_1^2 = R_y^2 / (4k^2 H_g)$ is fixed. The discrepancy scale is updated through the spike indicators; the slab scale is not updated. The source-specific residual variance priors have three degrees of freedom and scales obtained from source-specific residual-scale estimates and the factor $\chi_{3,0.1}^2 / 3$. The randomized-trial variance σ_1^2 is shared by treated and control trial observations; the external-control variance is σ_2^2 .

After centering the observed outcomes, the treatment-adjusted partial residual is

$$r_i = y_i - f_{-h}(x_i) - S_i g(x_i) - A_i \psi.$$

An f -tree update uses both sources with their respective precisions. A g -tree update uses randomized observations, including treated observations after subtracting $A_i \psi$, and the spike/slab leaf indicator is integrated out for structure proposals. Conditional on the forests and variances, the treatment update is

$$\begin{aligned} V_\psi &= \left(v_\psi^{-1} + \sum_{i:A_i=1} 1/v_i \right)^{-1}, \\ m_\psi &= V_\psi \sum_{i:A_i=1} \frac{y_i - f(x_i) - S_i g(x_i)}{v_i}, \\ \psi \mid - &\sim N(m_\psi, V_\psi). \end{aligned} \tag{G.3}$$

The same $A_i \psi$ term is removed before each forest update and is included in residual-variance updates. Each iteration updates the 50 f trees once, performs five sweeps over the five g trees, then updates the discrepancy hyperparameters, ψ and the two residual variances.

G.2 Training-only calibration and data-dependent scales

The calibration selects s_0^2 for the discrepancy prior; it is not an ESS calculation for the joint treatment coefficient. Stage 1 fits plain BART to the 300 external-control outcomes and evaluates its posterior mean surface at the 300 trial profiles. A separate plain BART fit to randomized-control outcomes supplies the variance estimate used as σ_1^2 in calibration. The calibration uses 20 consecutive posterior blocks, $q = 0.95$, $H_g = 5$, $v_0 = 3$, $\alpha_g = 0.5$, $\beta_g = 3$, 4,000 inverse-chi-square draws, 2,000 prior-tree draws and requested target $N_{\text{target}} = 100$. It uses 1,000 burn-in iterations and 1,000 draws for each calibration fit. The tree-grid calculation uses the combined external and trial profile covariates. Treated outcomes are excluded from this scale-selection stage.

The final joint fit independently calculates its outcome range and source residual-scale priors from the 600 outcomes supplied to that fit, including treated and control trial outcomes and external controls. This distinction is part of the fitted formulation: calibration is training-only, while the final range and variance rules are fit-specific. The selected scale is not described as supplying 100 effective observations to the treatment effect. The existing ESS quantity remains a control-mean prior reference.

G.3 Data-generating conditions and reproducibility

The study has five conditions, 40 datasets per condition, 300 trial subjects, 300 external controls and 10 covariates. Covariates and the baseline outcome surface F follow the constructions in Appendix E.1–E.2.

The outcome intercept is $b_0 = 0.42$. A common candidate pool receives independent source indicators with probability $1/2$; exactly 300 trial and 300 external observations are then sampled from their respective source groups. Within the trial, $P(A_i = 1 \mid S_i = 1) = 2/3$, giving the expected 2:1 treatment-to-control allocation. Gaussian trial and external residual standard deviations are 1.5 and 1.8.

The Compatible condition has no external shift. One-variable adds a shift of two outside $X_5 > 2$. Two-variable adds a shift of two outside the checkerboard agreement region

$$\{X_5 > 2, X_7 > 2\} \cup \{X_5 \leq 2, X_7 \leq 2\}.$$

Thus the shift is applied when exactly one of the two covariates exceeds 2. Heterogeneous uses the Two-variable source shift and treatment effect

$$\tau_i = 1 + 0.5(X_{5i} - \bar{X}_{5,\text{trial}}),$$

whose trial-profile mean is one. Null uses the same source shift and constant treatment effect $\psi = 0$. The condition names distinguish this 40-replicate study from the separate Sc1–Sc5 simulation record.

Compatible, One-variable and Two-variable share dataset seeds 92026001–92026040 and share trial data, treatment assignments and noise within each replicate. Heterogeneous uses 92126001–92126040 and Null uses 92226001–92226040. For a dataset seed s , calibration seeds are $s + 100001$ for the randomized-control stage, $s + 200001$ for the external-control stage and $s + 300001$ for the calibration calculation. Validation fit seeds are recorded in 09-method-exploration/validation/jobs.csv; the long-chain run used 30,000 burn-in iterations, 12,000 retained draws and three chains for every method-condition-replicate combination.

The reproducibility package retains the relative validation directory 09-method-exploration/validation/ and its files validation_core.R, run_validation.R and summarize_validation.R, together with the joint-package R and C++ sources. Their SHA-256 values and the output manifest are retained in the package. The original validation archive contains the raw chain files. The accompanying source package contains pooled method–dataset rows, seed and configuration fields, and checksums; it does not redistribute the raw chains.

G.4 Complete numerical results

Table S15 gives the pooled ATE bias, mean interval width, RMSE, Monte Carlo standard error of the RMSE, 95% coverage count and ATE convergence flags. The RMSE standard errors quantify finite-replicate uncertainty in the displayed RMSE; the flags report method–dataset groups and are separate from posterior uncertainty. ATE flags use $\hat{R} > 1.05$ or bulk effective sample size below 400. The paired block-bootstrap comparison uses only Compatible, One-variable and Two-variable, with shared replicate seeds and common trial data; it does not resample Heterogeneous or Null. Its interval is a sampling-uncertainty statement for the paired empirical comparison, not a posterior interval and not a correction for failed chains.

Across the 200 method-condition groups, the ATE convergence summary for Joint LRC has median/max $\hat{R} = 1.004/1.058$ and minimum/median bulk ESS 35.2/1289.5. The corresponding Joint source-BART values are 1.004/1.057 and 36.8/1438.5, and the separated values are 1.003/1.059 and 31.9/2464.0. Discrepancy-contrast diagnostics are materially worse: the maximum $\hat{R}$ values are 2.004, 2.164 and 1.987, and the minimum bulk ESS values are 4.1, 3.8 and 4.1 for Joint LRC, Joint source-BART and Separated source-

Table S14: Comparator specifications. The methods differ in mean model, variance treatment, range inputs and calibration as well as in borrowing structure.

Method	Mean model	Trial variance	Range inputs	Calibration
Joint LRC	$f(x) + Sg(x) + A\psi$; 50 f trees and 5 g trees	One σ_1^2 for both trial arms; σ_2^2 external	All 600 outcomes for R_y ; combined trial and external profiles for tree grid	Training-only control fits; $N_{\text{target}} = 100$, selected s_0^2
Joint source-BART	Source-indicator mean with common $A\psi$; no separate g ensemble	One trial variance and one external variance	All 600 outcomes; same profile inputs	None: $H_g = 0$, so the supplied discrepancy scale is unused
Separated source-BART	Separate treated mean; source-indicator model for controls	Separate treated fit; source-specific control variances	Treated fit and pooled-control fit use their own outcomes	None: $H_g = 0$; no common ψ

Table S15: Complete joint validation metrics. Bias, interval width and RMSE are pooled over the 40 datasets in each condition. Coverage is the number of central 95% intervals containing the truth out of 40. RMSE SE is the Monte Carlo standard error of the RMSE. ATE flags count method–dataset groups with $\hat{R} > 1.05$ or bulk effective sample size below 400.

Condition	Method	Bias	Width	RMSE	RMSE SE	Coverage	Flags
Compatible	Joint LRC	-0.057	0.683	0.175	0.017	38/40	0
Compatible	Joint source-BART	-0.047	0.775	0.213	0.020	39/40	1
Compatible	Separated source-BART	-0.037	0.744	0.206	0.021	37/40	1
One-variable	Joint LRC	-0.055	0.762	0.207	0.018	38/40	3
One-variable	Joint source-BART	-0.052	0.782	0.216	0.020	38/40	2
One-variable	Separated source-BART	-0.045	0.747	0.211	0.021	37/40	0
Two-variable	Joint LRC	-0.046	0.803	0.215	0.018	38/40	3
Two-variable	Joint source-BART	-0.052	0.784	0.215	0.020	38/40	2
Two-variable	Separated source-BART	-0.040	0.761	0.211	0.021	37/40	7
Heterogeneous	Joint LRC	-0.047	0.811	0.209	0.021	38/40	8
Heterogeneous	Joint source-BART	-0.039	0.789	0.211	0.021	38/40	4
Heterogeneous	Separated source-BART	-0.034	0.763	0.192	0.020	39/40	5
Null	Joint LRC	-0.024	0.810	0.230	0.029	36/40	4
Null	Joint source-BART	-0.015	0.789	0.215	0.025	38/40	3
Null	Separated source-BART	-0.018	0.769	0.210	0.022	38/40	8

BART, respectively. These diagnostics support a qualified treatment-effect comparison and do not support a claim of reliable regional discrepancy recovery.

Across Compatible, One-variable and Two-variable, pooled Joint LRC RMSE is 0.199690, compared with 0.209416 for Separated source-BART and 0.214491 for Joint source-BART. The paired MSE differences (Joint LRC minus comparator) are -0.003979 (95% block-bootstrap CI $[-0.011089, 0.002817]$) and -0.006130 (CI $[-0.010451, -0.002245]$). The corresponding RMSE reductions are 4.64% and 6.9%, below the prespecified 10% improvement criterion. The validation therefore records a modest, comparator-specific empirical pattern. It does not establish general superiority, recovery of regional discrepancies or transfer to survival or single-arm analyses.

G.5 Saved-chain sensitivity

Each chain-index summary retains all 120 datasets in Compatible, One-variable and Two-variable. Table S16 computes RMSE from the posterior mean within each saved chain, using the same datasets for every method. The percentage reduction is $100(1 - \text{RMSE}_{\text{LRC}}/\text{RMSE}_{\text{comparator}})$. No diagnostic-based exclusions are applied. The pooled direction is similar across chain indices, but this descriptive check does not establish adequate mixing or replace the three-chain diagnostics.

Table S16: Saved-chain sensitivity across 120 datasets per chain index.

Chain	RMSE			LRC reduction (%)	
	Joint LRC	Joint source	Separated source	vs separated	vs joint source
1	0.199403	0.214421	0.209069	4.623	7.004
2	0.199121	0.214920	0.209618	5.008	7.351
3	0.201897	0.215474	0.210701	4.178	6.301

H Selected displays for the separate-treatment variant

These displays summarize the separate-treatment experiments in Appendix E. They do not report the joint treatment model. All numerical entries and regional curves use the same 100-replicate simulation record described there; the regional display uses its first 40 replicates.

Table S17: Separate-treatment simulation study designs.

Design	Trial N	Historical n_2	$P(Z = 1)$	Replicates	Settings
Gaussian, two-arm	300	300	2/3	100	Sc1–Sc5; 12 settings
Survival, two-arm	300	300	3/4	100	Sc1–Sc5; 12 settings
Gaussian, single-arm	200	300	1	100	Sc1–Sc2
Survival, single-arm	200	300	1	100	Sc1–Sc2

Ten predictors are generated in each design. $Z = 1$ denotes trial treatment; two-arm allocation is Bernoulli, so realized arm sizes vary. Historical controls are regenerated for each replicate. Sc1: common covariate distributions; Sc2: selection on observed prognostic variables; Sc3: latent-variable selection with proxy correlations $-0.5, 0, 0.5$; Sc4: global historical shift of 1 or 2; Sc5: regional shifts outside $X_5 > 2$ (0.5, 1, 2) or $X_7 \leq 2$ (1, 2). Gaussian treatment effects are 1; survival effects are $\log(1.4)$ on the log-time scale.

Table S18: Gaussian mean-difference performance in representative two-arm scenarios.

Method	Bias	RMSE	Coverage	Power
<i>Sc1</i>				
lrcBART	0.003	0.174	0.950	0.890
BART-NP	-0.010	0.209	0.970	0.730
BART-CP	0.009	0.153	0.950	0.980
BART-PP	-0.001	0.197	0.950	0.800
LM-NP	-0.019	0.348	0.930	0.390
LM-CP	0.015	0.262	0.940	0.710
LM-PP	-0.005	0.349	0.950	0.390
hier. LM	-0.002	0.329	0.940	0.450
PSCL	0.019	0.290	0.920	0.480
MAP	0.023	0.302	0.940	0.530
<i>Sc2</i>				
lrcBART	0.003	0.186	0.940	0.860
BART-NP	-0.032	0.211	0.940	0.750
BART-CP	0.048	0.188	0.930	0.950
BART-PP	-0.004	0.200	0.970	0.740
LM-NP	-0.062	0.358	0.900	0.440
LM-CP	0.343	0.463	0.730	0.920
LM-PP	0.007	0.355	0.950	0.450
hier. LM	0.052	0.363	0.930	0.510
PSCL	0.007	0.317	0.910	0.390
MAP	0.186	0.458	0.870	0.550
<i>Sc3, $\rho = 0$</i>				
lrcBART	0.009	0.315	0.930	0.560
BART-NP	-0.046	0.328	0.950	0.520
BART-CP	0.957	0.985	0.010	1.000
BART-PP	0.004	0.309	0.940	0.550
LM-NP	-0.057	0.367	0.930	0.340
LM-CP	0.989	1.027	0.050	1.000
LM-PP	0.047	0.350	0.940	0.500
hier. LM	0.085	0.373	0.930	0.510
PSCL	0.667	0.721	0.290	0.980
MAP	0.202	0.434	0.910	0.610

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Table S18 (continued)

Method	Bias	RMSE	Coverage	Power
<i>Sc4</i> , $\delta = 1$				
lrcBART	-0.007	0.201	0.940	0.810
BART-NP	-0.010	0.209	0.970	0.730
BART-CP	-0.689	0.706	0.000	0.000
BART-PP	-0.012	0.197	0.970	0.770
LM-NP	-0.019	0.348	0.930	0.390
LM-CP	-0.726	0.772	0.180	0.010
LM-PP	-0.054	0.352	0.960	0.350
hier. LM	-0.104	0.358	0.930	0.290
PSCL	-0.483	0.563	0.530	0.060
MAP	-0.230	0.467	0.860	0.170
<i>Sc5</i> , X_5 , $\delta = 1$				
lrcBART	-0.050	0.214	0.930	0.690
BART-NP	-0.010	0.209	0.970	0.730
BART-CP	-0.334	0.366	0.370	0.350
BART-PP	-0.014	0.195	0.970	0.760
LM-NP	-0.019	0.348	0.930	0.390
LM-CP	-0.353	0.439	0.660	0.190
LM-PP	-0.030	0.348	0.970	0.370
hier. LM	-0.063	0.338	0.920	0.360
PSCL	-0.230	0.370	0.850	0.210
MAP	-0.160	0.378	0.920	0.270

Each row summarizes 100 alternative-hypothesis replicates. Coverage is for a 95% interval. Empirical power is the frequency of posterior probability of benefit exceeding 0.95 (PSCL: lower 95% confidence limit above the threshold). Gaussian decisions use a threshold of 0.5 and survival decisions use a ratio threshold of 1. RMSE is the square root of the mean squared error. NP denotes no historical pooling (trial controls only); CP denotes complete pooling; PP denotes source-adjusted pooling using a source indicator. PP is distinct from the hierarchical comparator. lrcBART requests ESS 100; the MAP comparator in Table 2 also requests 100. Full configurations and Monte Carlo standard errors appear in the appendix. The hierarchical comparator uses discrepancy-variance prior scale $s_{\tau}^2 = 0.25$.

Table S19: Median-survival-ratio performance in representative two-arm scenarios.

Method	Bias	RMSE	Coverage	Power
<i>Sc1</i>				
lrcBART	0.014	0.290	0.970	0.420
BART-NP	0.048	0.392	0.980	0.330
BART-CP	0.010	0.218	0.950	0.660
BART-PP	-0.034	0.316	0.980	0.300
AFT-NP	-0.105	0.455	0.970	0.070
AFT-CP	0.027	0.307	1.000	0.330
AFT-PP	0.022	0.415	0.990	0.100
hier. AFT	0.050	0.443	0.990	0.140
<i>Sc2</i>				
lrcBART	0.058	0.341	0.960	0.480
BART-NP	0.117	0.440	0.960	0.320
BART-CP	0.130	0.310	0.970	0.630
BART-PP	-0.019	0.349	0.970	0.280
AFT-NP	0.018	0.529	0.960	0.160
AFT-CP	1.202	1.435	0.570	0.920
AFT-PP	0.403	0.817	0.920	0.300
hier. AFT	0.417	0.815	0.930	0.390
<i>Sc3, $\rho = 0$</i>				
lrcBART	0.076	0.472	0.970	0.290
BART-NP	0.014	0.502	0.960	0.180
BART-CP	2.642	2.802	0.000	1.000
BART-PP	0.075	0.446	0.960	0.300
AFT-NP	-0.090	0.531	0.970	0.110
AFT-CP	3.187	3.429	0.010	1.000
AFT-PP	0.477	0.839	0.930	0.470
hier. AFT	0.415	0.836	0.920	0.350

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Table S19 (continued)

Method	Bias	RMSE	Coverage	Power
<i>Sc4</i> , $\delta = 1$				
lrcBART	-0.007	0.332	0.940	0.370
BART-NP	0.048	0.392	0.980	0.330
BART-CP	-0.729	0.736	0.000	0.000
BART-PP	-0.080	0.311	0.960	0.230
AFT-NP	-0.105	0.455	0.970	0.070
AFT-CP	-0.768	0.780	0.080	0.000
AFT-PP	-0.212	0.405	0.960	0.030
hier. AFT	-0.159	0.459	0.970	0.060
<i>Sc5</i> , X_5 , $\delta = 1$				
lrcBART	-0.054	0.331	0.930	0.330
BART-NP	0.048	0.392	0.980	0.330
BART-CP	-0.391	0.422	0.420	0.070
BART-PP	-0.067	0.316	0.960	0.240
AFT-NP	-0.105	0.455	0.970	0.070
AFT-CP	-0.441	0.484	0.700	0.000
AFT-PP	-0.116	0.385	1.000	0.060
hier. AFT	-0.074	0.433	0.980	0.080

Each row summarizes 100 alternative-hypothesis replicates. Coverage is for a 95% interval. Empirical power is the frequency of posterior probability of benefit exceeding 0.95 (PSCL: lower 95% confidence limit above the threshold). Gaussian decisions use a threshold of 0.5 and survival decisions use a ratio threshold of 1. RMSE is the square root of the mean squared error. NP denotes no historical pooling (trial controls only); CP denotes complete pooling; PP denotes source-adjusted pooling using a source indicator. PP is distinct from the hierarchical comparator. lrcBART requests ESS 100; the MAP comparator in Table 2 also requests 100. Full configurations and Monte Carlo standard errors appear in the appendix. The hierarchical comparator uses discrepancy-variance prior scale $s_{\tau}^2 = 0.25$.

Table S20: Single-arm performance for Gaussian mean differences and median-survival ratios.

Method	Bias	RMSE	Coverage	Power
<i>Gaussian, Sc1</i>				
lrcBART	-0.015	0.156	0.950	0.890
BART-CP	-0.014	0.153	0.940	0.940
LM-CP	-0.029	0.270	0.960	0.600
hier. LM	-0.025	0.272	1.000	0.090
<i>Gaussian, Sc2</i>				
lrcBART	0.062	0.177	0.980	0.910
BART-CP	0.056	0.176	0.930	0.950
LM-CP	0.290	0.395	0.850	0.910
hier. LM	0.357	0.444	1.000	0.500
<i>Survival, Sc1</i>				
lrcBART	-0.020	0.178	0.990	0.590
BART-CP	-0.011	0.180	0.970	0.730
AFT-CP	0.055	0.373	0.970	0.370
hier. AFT	0.097	0.388	1.000	0.020
<i>Survival, Sc2</i>				
lrcBART	0.070	0.216	0.980	0.730
BART-CP	0.069	0.216	0.990	0.770
AFT-CP	0.676	0.859	0.820	0.810
hier. AFT	0.872	1.042	1.000	0.370

Each row summarizes 100 alternative-hypothesis replicates. Coverage is for a 95% interval; empirical power is the frequency of posterior probability of benefit exceeding 0.95, using a threshold of 0.5 for Gaussian mean differences and 1 for survival ratios. All single-arm lrcBART primary rows use fixed $w = 1$, $H_f = 50$, $H_g = 5$ and requested ESS 100. The hierarchical discrepancy variance has prior $\text{IG}(3/2, 3s_\tau^2/2)$. In single-arm studies, CP uses all historical controls with full likelihood weight and no source discrepancy; there are no observed trial controls to pool. Full configurations and Monte Carlo standard errors appear in the appendix. The single-arm hierarchical comparator uses discrepancy-variance prior scale $s_\tau^2 = 0.05$.

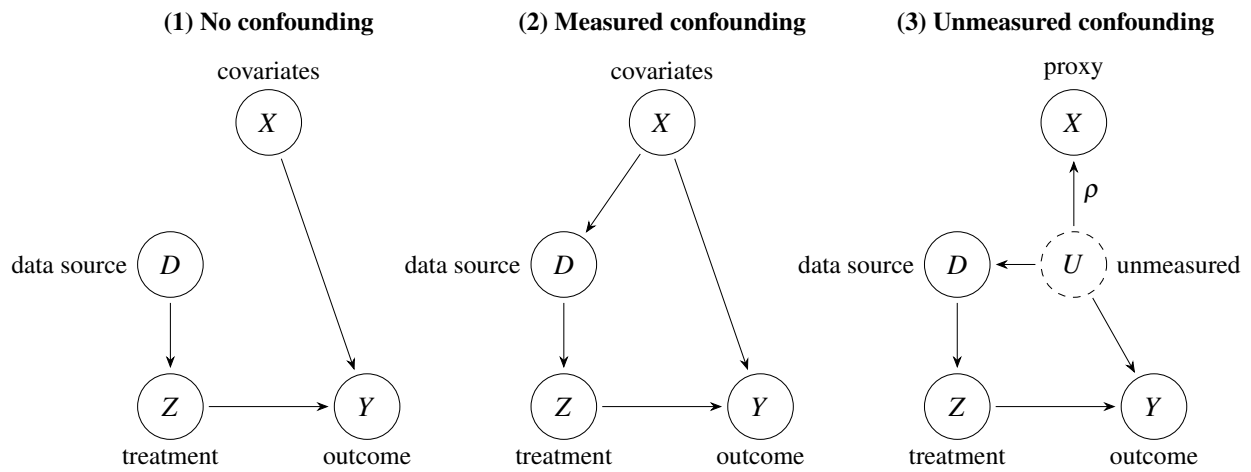


Figure S3: Directed acyclic graphs (DAGs) for the source-selection mechanisms in Sc1–Sc3. In this schematic, D denotes data source (trial or historical), Z treatment assignment, Y an individual outcome, X the measured predictors relevant to the depicted paths and U unmeasured prognostic variables; participant and group indices are suppressed. Sc1: measured covariates predict the outcome but do not influence source selection. Sc2: measured covariates influence both source selection and outcome. Sc3: unmeasured variables influence source selection and outcome, and generate observed proxies through $U \rightarrow X$, with proxy association indexed by ρ . Other measured predictors are omitted from the schematic. The $D \rightarrow Z$ edge reflects that historical observations are controls, whereas treatment is assigned within the trial. These diagrams illustrate the simulation mechanisms, not an established causal structure for the clinical application.

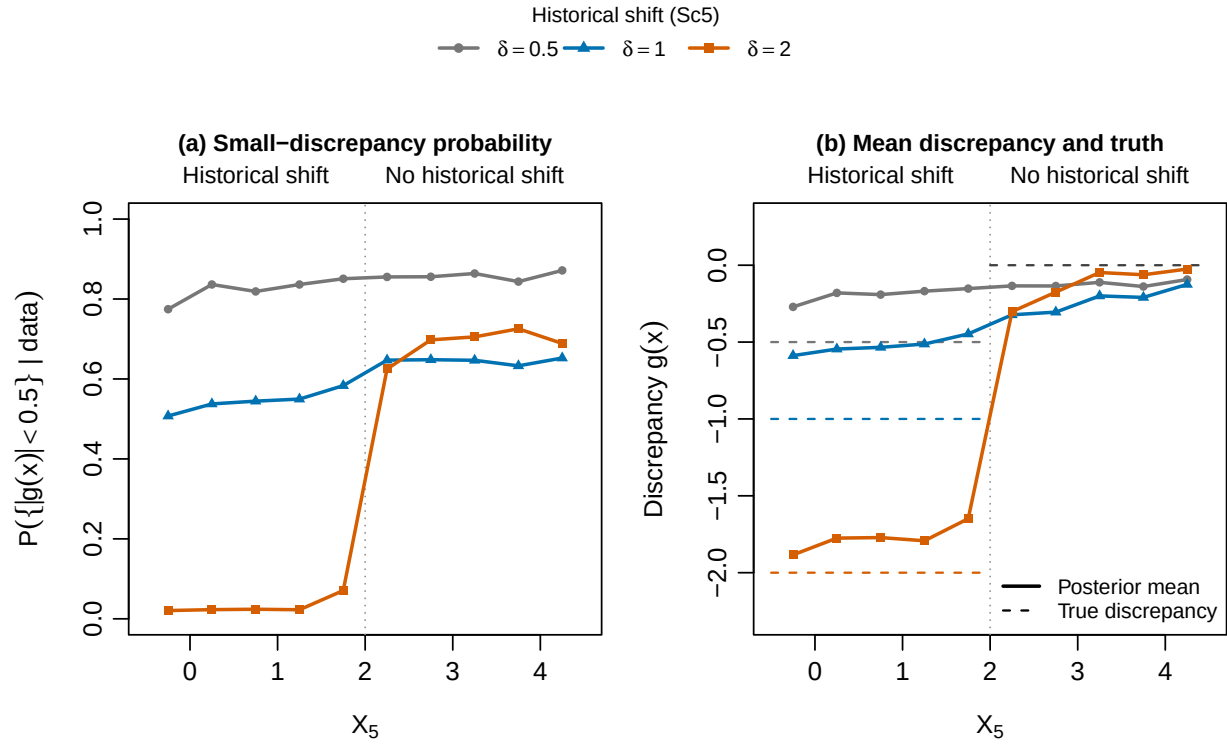


Figure S4: Separate-treatment model: regional historical shifts for Gaussian outcomes (Sc5). Historical outcomes are shifted upward by $\delta = 0.5, 1, 2$ where $X_5 \leq 2$, and are unshifted otherwise. (a) Posterior probability of a small discrepancy, $P\{|g(x)| < 0.5 \mid \text{data}\}$. (b) Posterior mean discrepancy; dashed lines show the true trial-minus-historical discrepancy, $-\delta$ in the shifted region and zero elsewhere. Solid curves summarize pooled trial-control profiles within X_5 bins of width 0.5 across 40 replicates; they are not uncertainty intervals. The dotted line marks the shift boundary. Primary lrcBART uses $H_f = 50$, $H_g = 5$, $w \sim \text{Beta}(1, 1)$ and requested prior ESS 100.

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